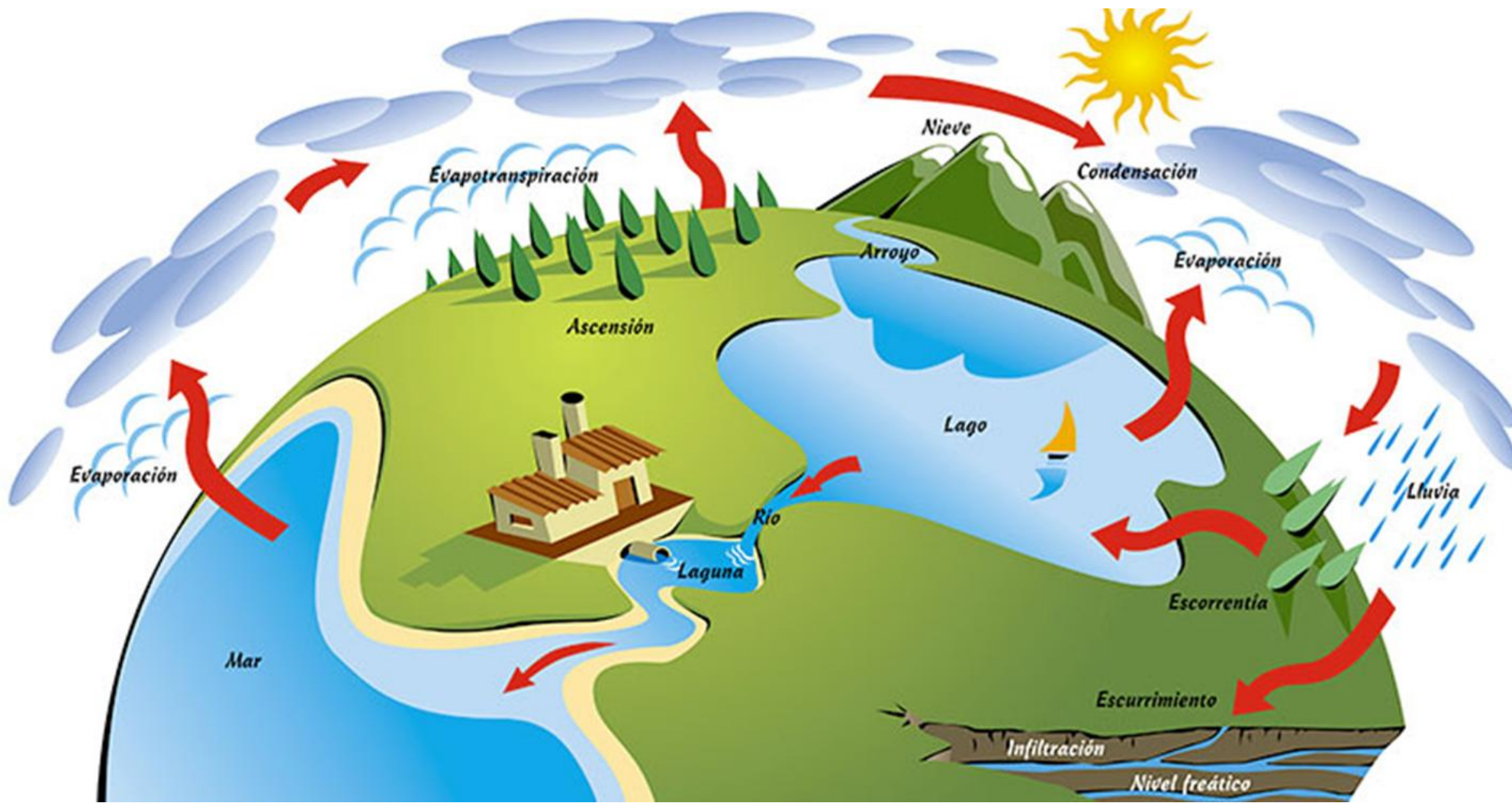




Unidad 1. Introducción y Balance hídrico





¿Qué es la hidrología?

La hidrología (del griego ὕδωρ, hýdōr, "agua", y λόγος, lógos, "estudio") es una **rama de las ciencias de la Tierra que se enfoca en analizar el agua, su presencia, distribución, movimiento y características físicas, químicas y mecánicas**, tanto en océanos como en la atmósfera y la superficie terrestre. Entre sus campos de estudio se encuentran las precipitaciones, escorrentía superficial, humedad del suelo, evapotranspiración y el balance de masas glaciares, entre otros.

Por otro lado, el análisis específico del agua subterránea es abordado por la hidrogeología.



¿Qué es la hidrogeología?

La hidrogeología es la rama de la hidrología que se enfoca en el estudio de las aguas subterráneas.

Estudio de la fase terrestre subterránea del ciclo hidrológico, sus aspectos físicos (**hidrogeología**), sus aspectos dinámicos (**hidrodinámica**), sus aspectos químicos (**hidroquímica**) y su relación con los seres vivos.

- **Distribución del agua subterránea.**
- **Movimiento del agua a través de diferentes tipos de suelos y rocas.**
- **Interacción del agua con el medio ambiente geológico**



Conexiones interdisciplinarias de la hidrología

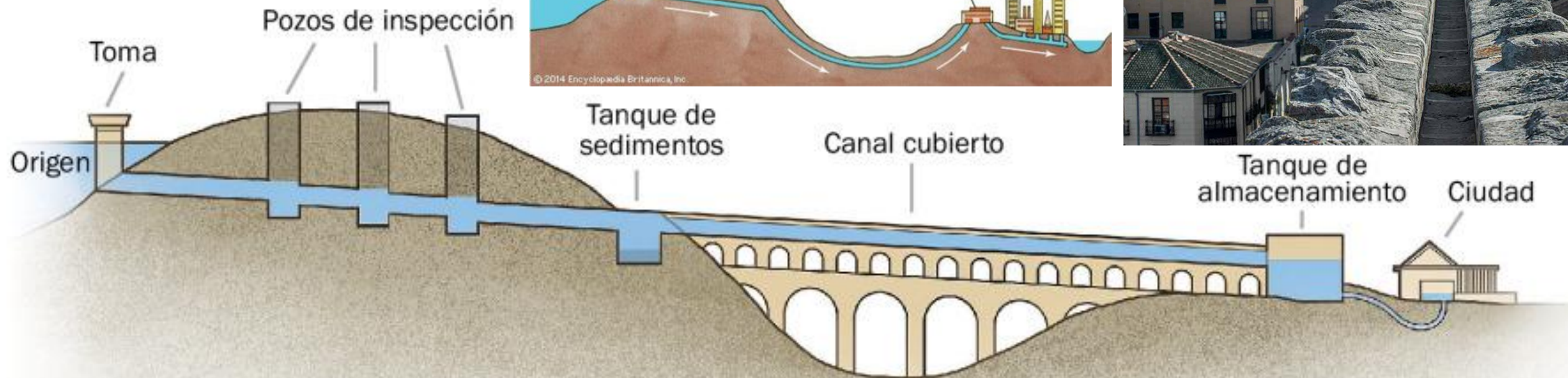
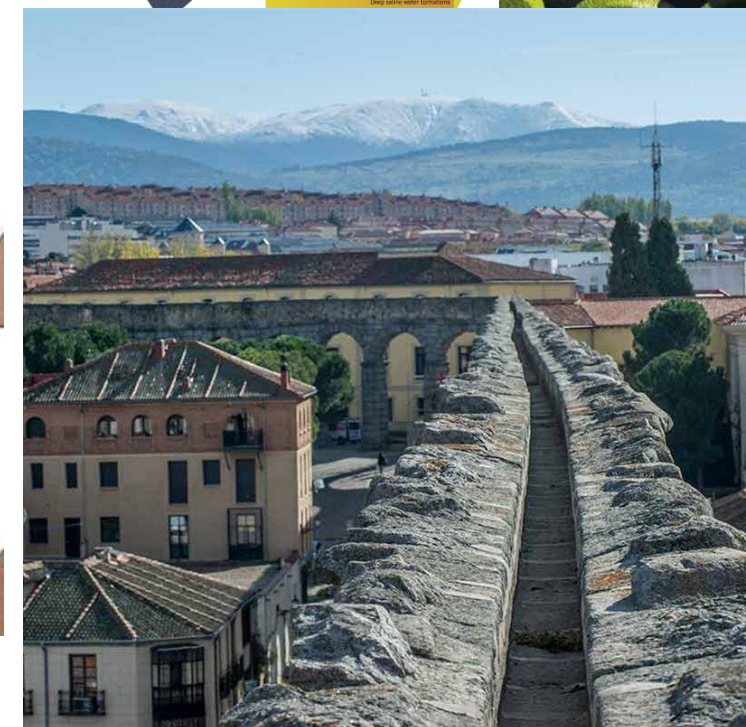
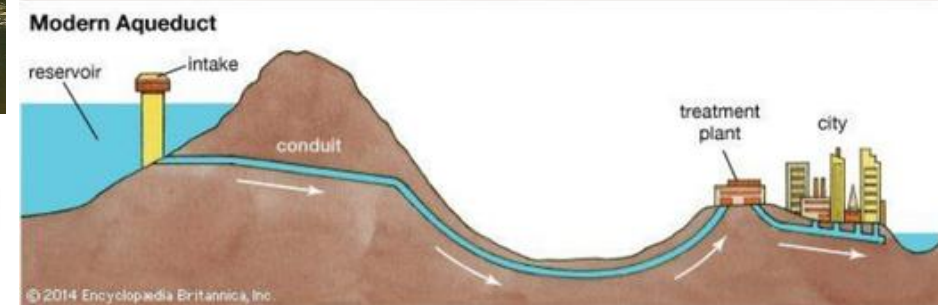
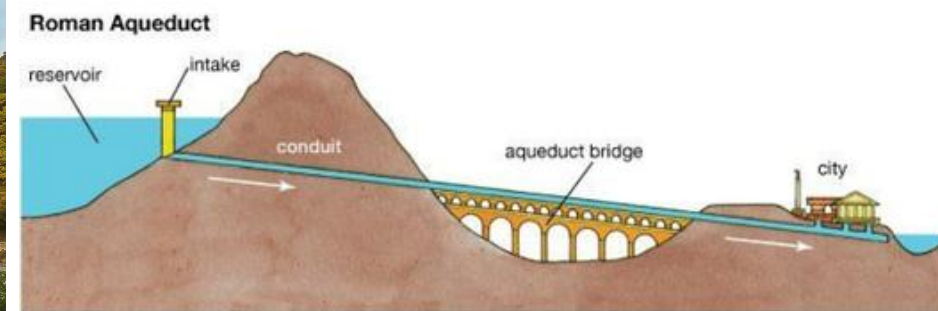
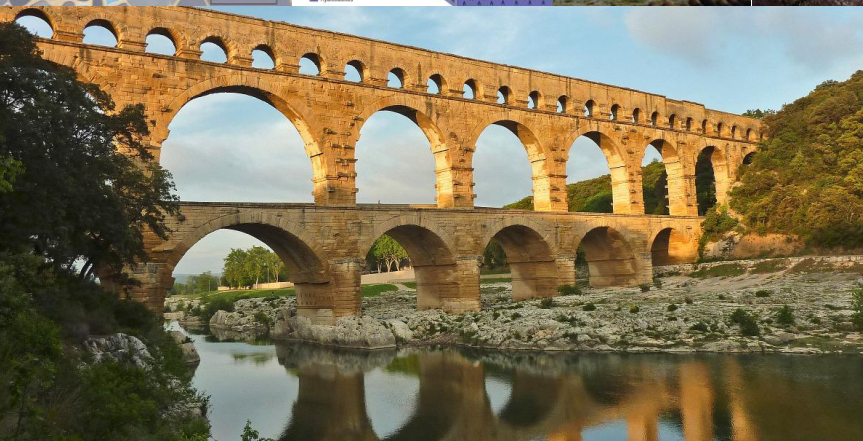
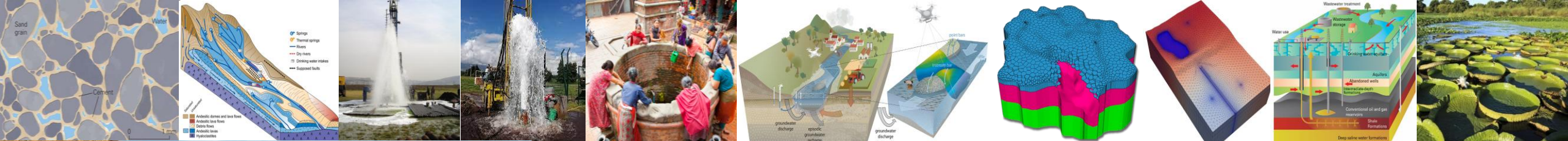
Integra conocimientos de diversas disciplinas científicas:

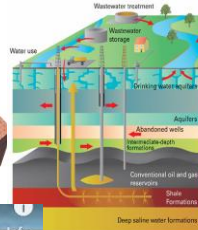
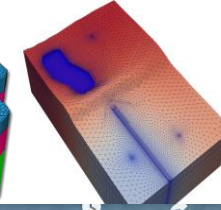
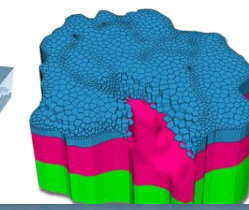
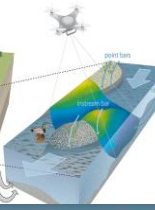
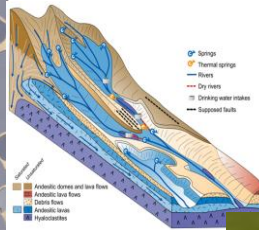
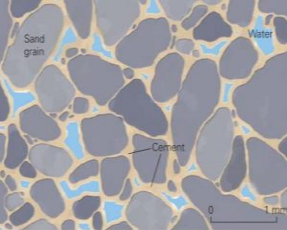
- **Principios de física** (mecánica de fluidos, termodinámica, balance energético) para comprender el movimiento y almacenamiento del agua.
- **Conceptos de química** (calidad del agua, ciclos biogeoquímicos, transporte de contaminantes) para evaluar la salud y la contaminación del agua.
- **Principios de biología** (ecología acuática, microbiología, dinámica de ecosistemas) para comprender el papel del agua en la vida.
- **Conocimientos de geología** (hidrogeología, geomorfología, sedimentología) para estudiar las interacciones del agua con la Tierra.
- **Ciencias atmosféricas** (meteorología, climatología, teledetección) para investigar el papel del agua en el tiempo y el clima.



Desarrollo Histórico de la Hidrología









Babilonia, Persia, Egipto...

Qanats- foggaras

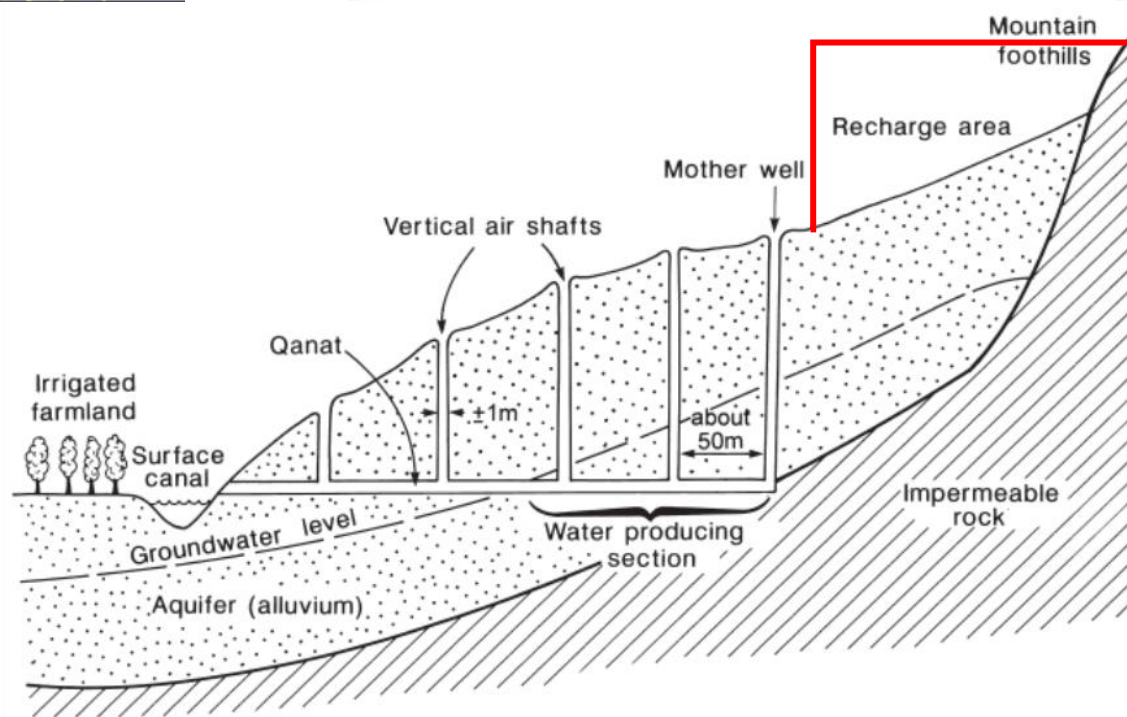
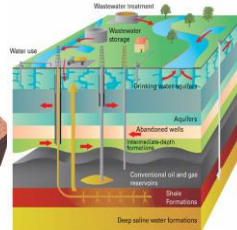
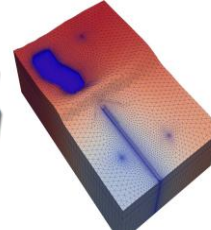
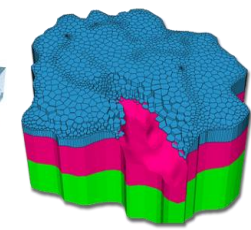
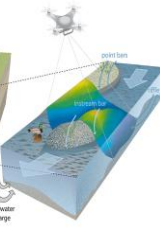
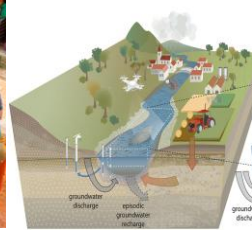
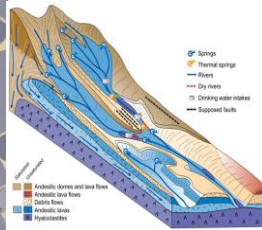
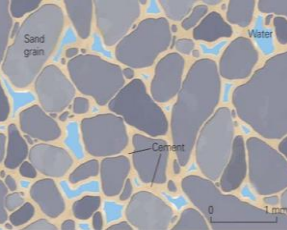
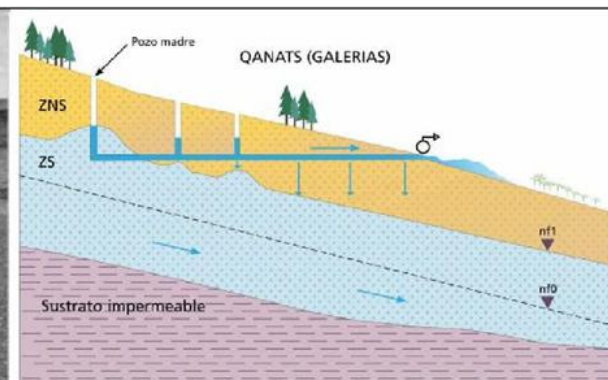
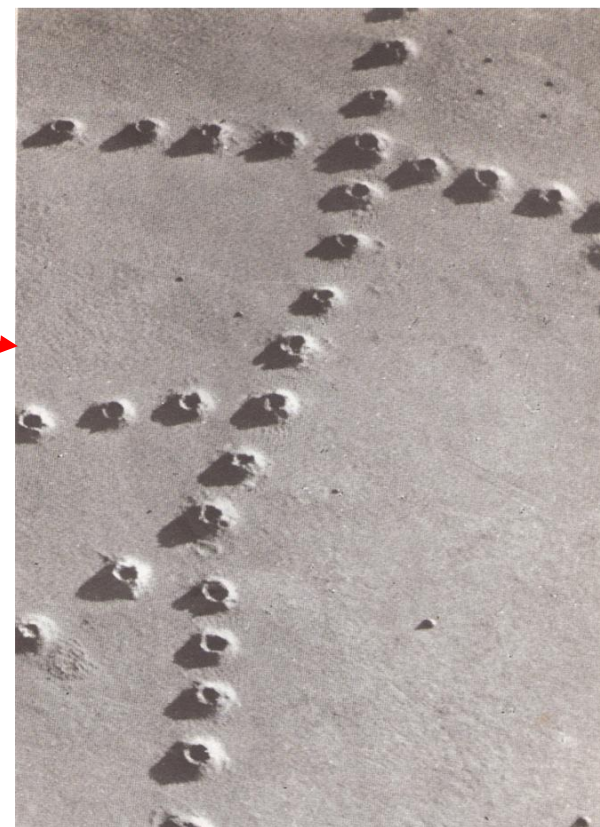


Fig. 1.3 Longitudinal section of a qanat. Based on Beaumont (1968) and Biswas (1972).



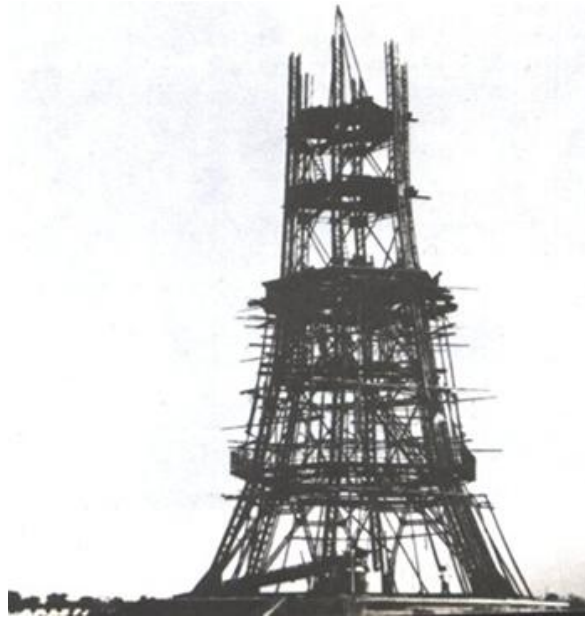
Water flow in the foggara

a) through rainfall



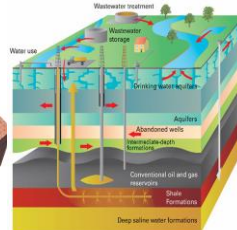
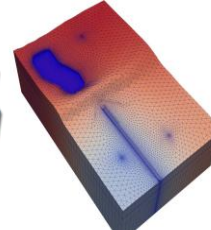
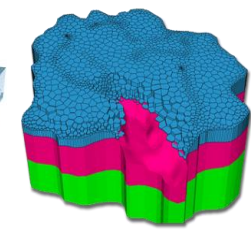
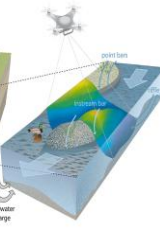
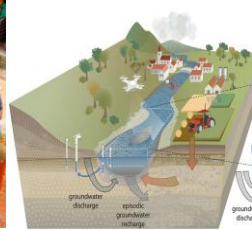
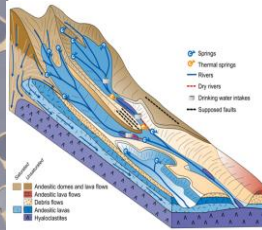
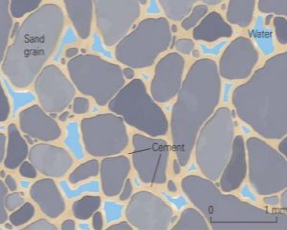


Los primeros en aprovechar las aguas superficiales y subterráneas fueron los griegos y romanos, quienes ya reconocían la importancia del agua y desarrollaban métodos para su utilización y manejo.



China Pozos y sondeos

Los pozos de bambú típicamente se construían perforando manualmente el suelo blando y utilizando secciones de bambú para encamisar y sostener las paredes del pozo. Esto prevenía el colapso de las paredes del pozo y permitía un acceso más seguro y duradero al agua subterránea. Estos pozos no solo se utilizaban para extraer agua para uso doméstico y agrícola, sino también en la creación de sistemas de irrigación innovadores.



Viernes 21 de diciembre 2012

En Alemania descubren cuatro pozos de agua

Dichos pozos habrían estado abiertos entre 5206 - 5098 a.J.C.

Twitter 0 Me gusta 0 0

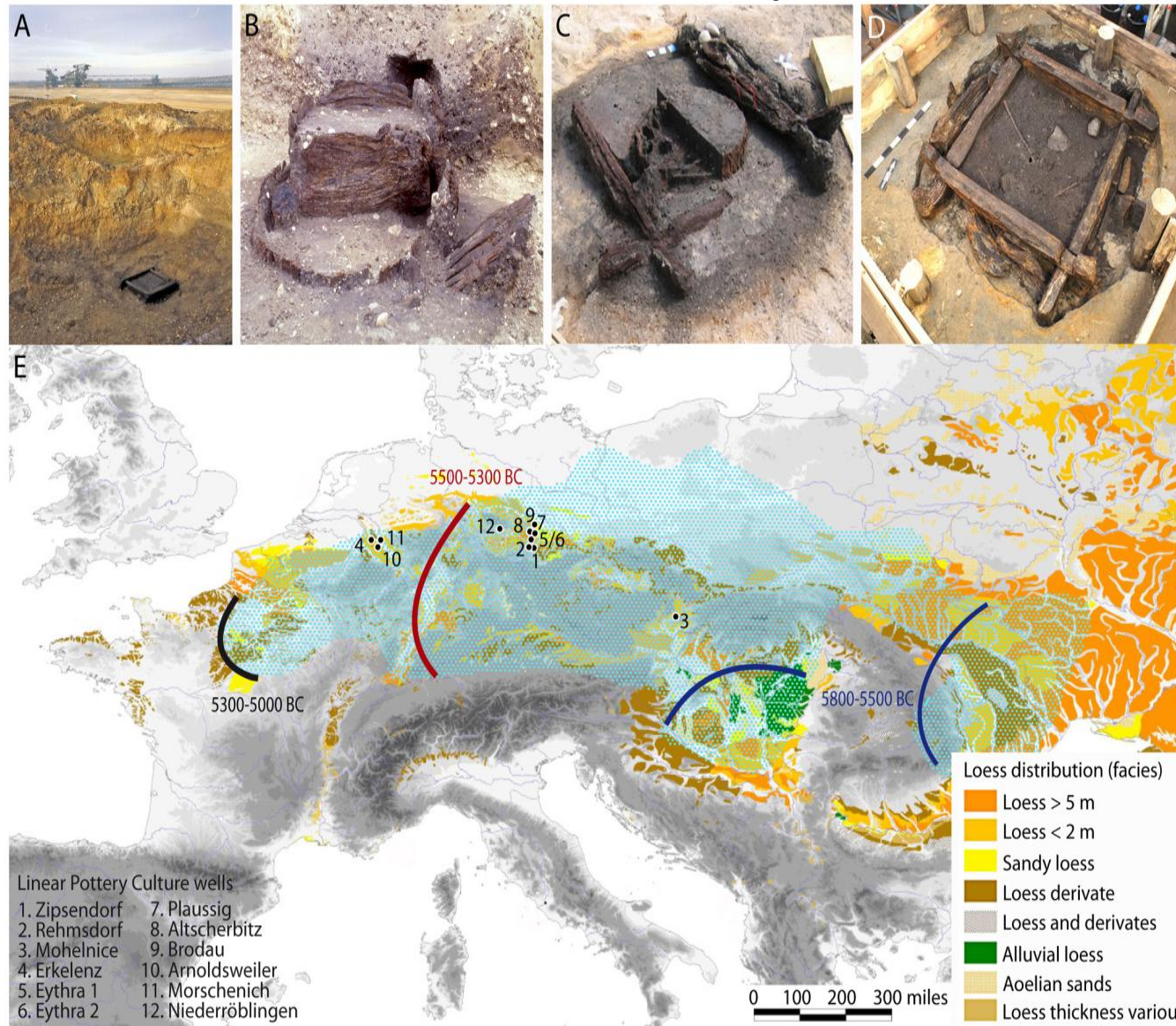


Foto:Generacion.com

Hoy se dio a conocer que arqueólogos de la Universidad de Friburgo, Suiza, hallaron en el este de Alemania antiguos pozos desconocidos hasta ahora en territorio de Europa.

Son cuatro pozos de hasta siete metros de profundidad. Se encuentran cerca de Leipzig, Sajonia. Fueron abiertos entre 5206 - 5098 a.J.C., supuestamente para abastecer de agua un pequeño poblado. La madera de sus paredes se ha conservado bien bajo tierra en un medio sin aire.

"El hallazgo testimonia que el nivel de vida y de desarrollo de la gente de aquella época era mucho más alto de lo que creíamos antes", dijo Willy Tegel, jefe del grupo que descubrió los pozos.

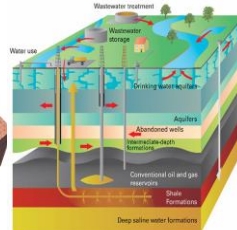
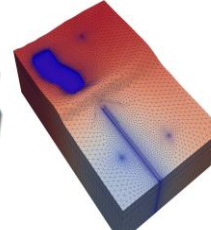
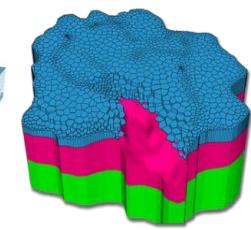
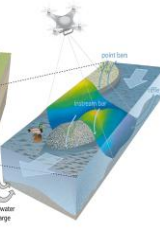
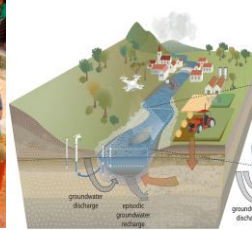
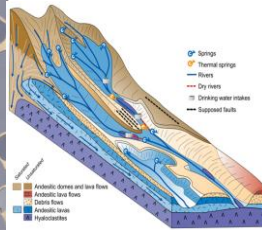
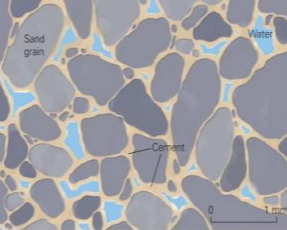




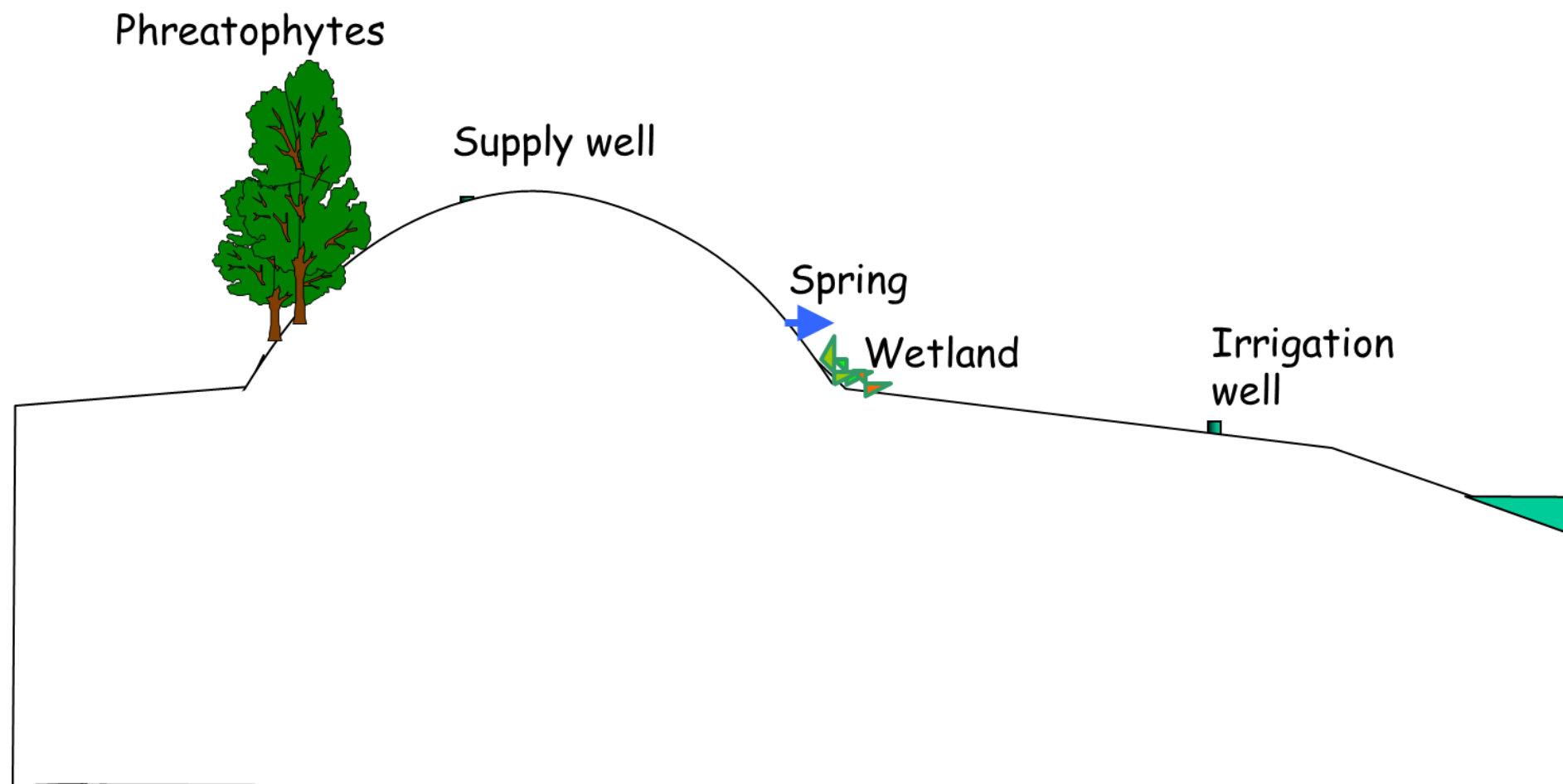
Groundwater and the landscape

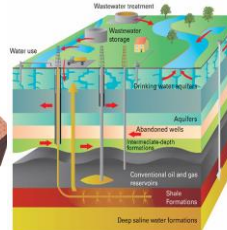
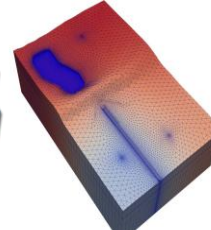
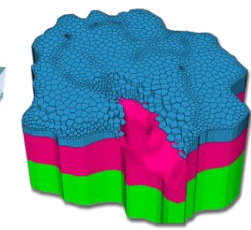
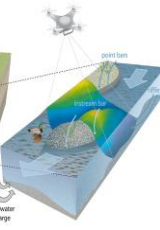
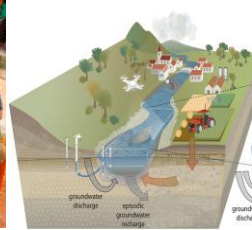
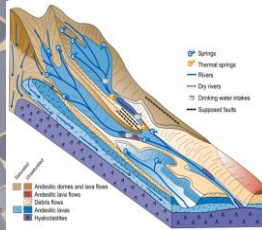
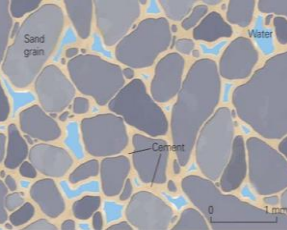


http://www.thewaterpage.com/photo_gallery_4.htm

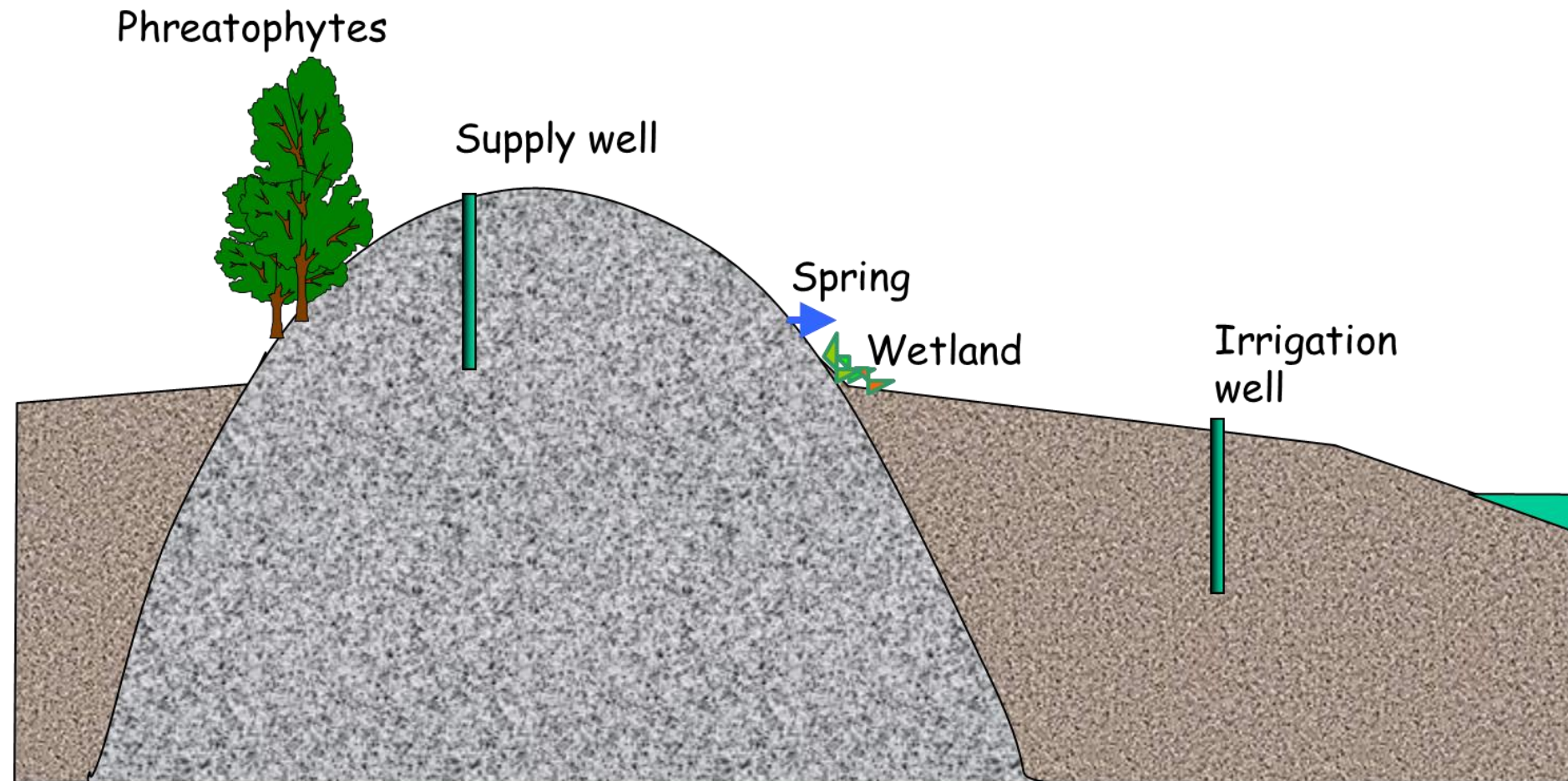


Landscape for a lay person



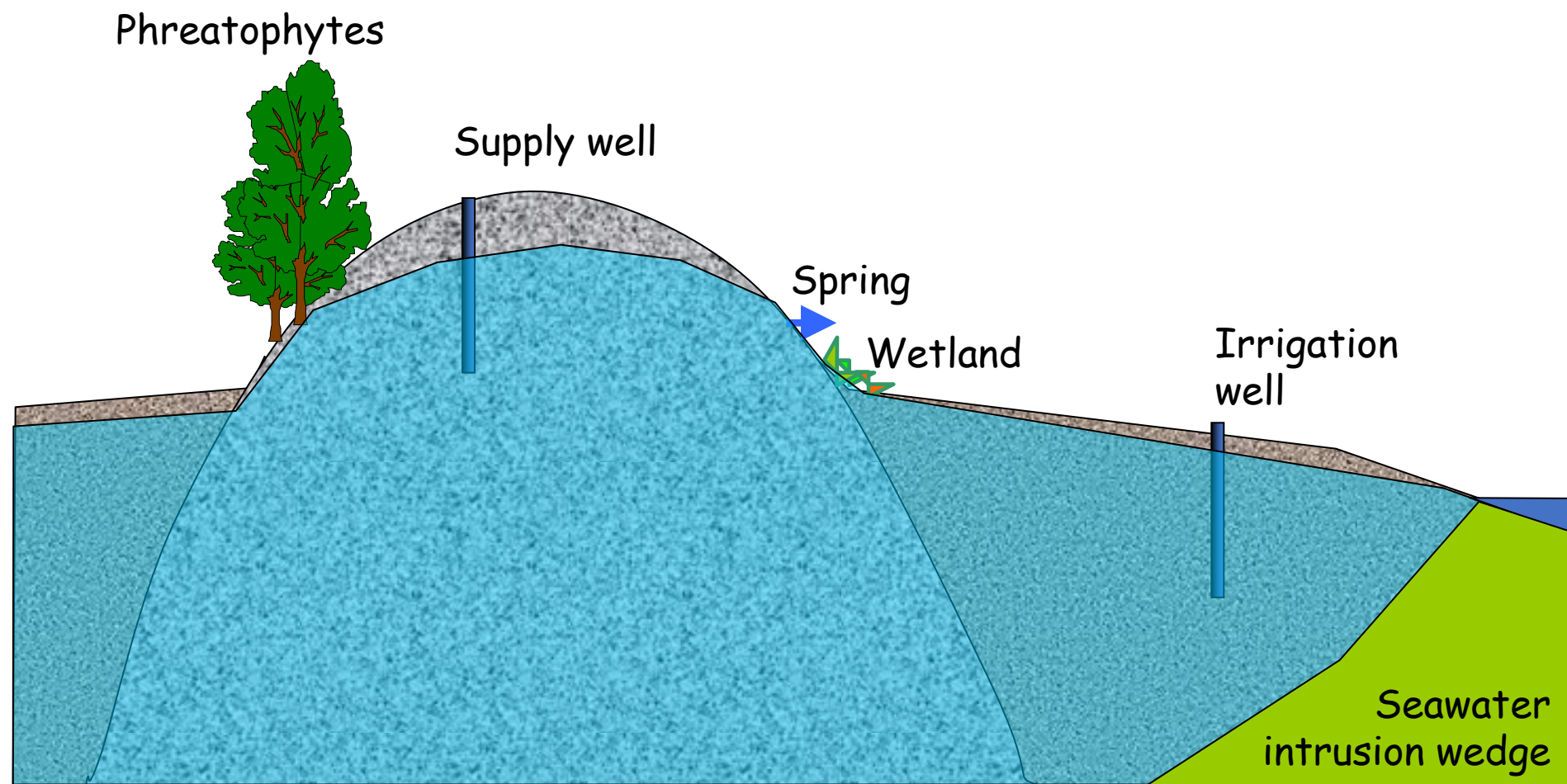


Landscape for a geologist



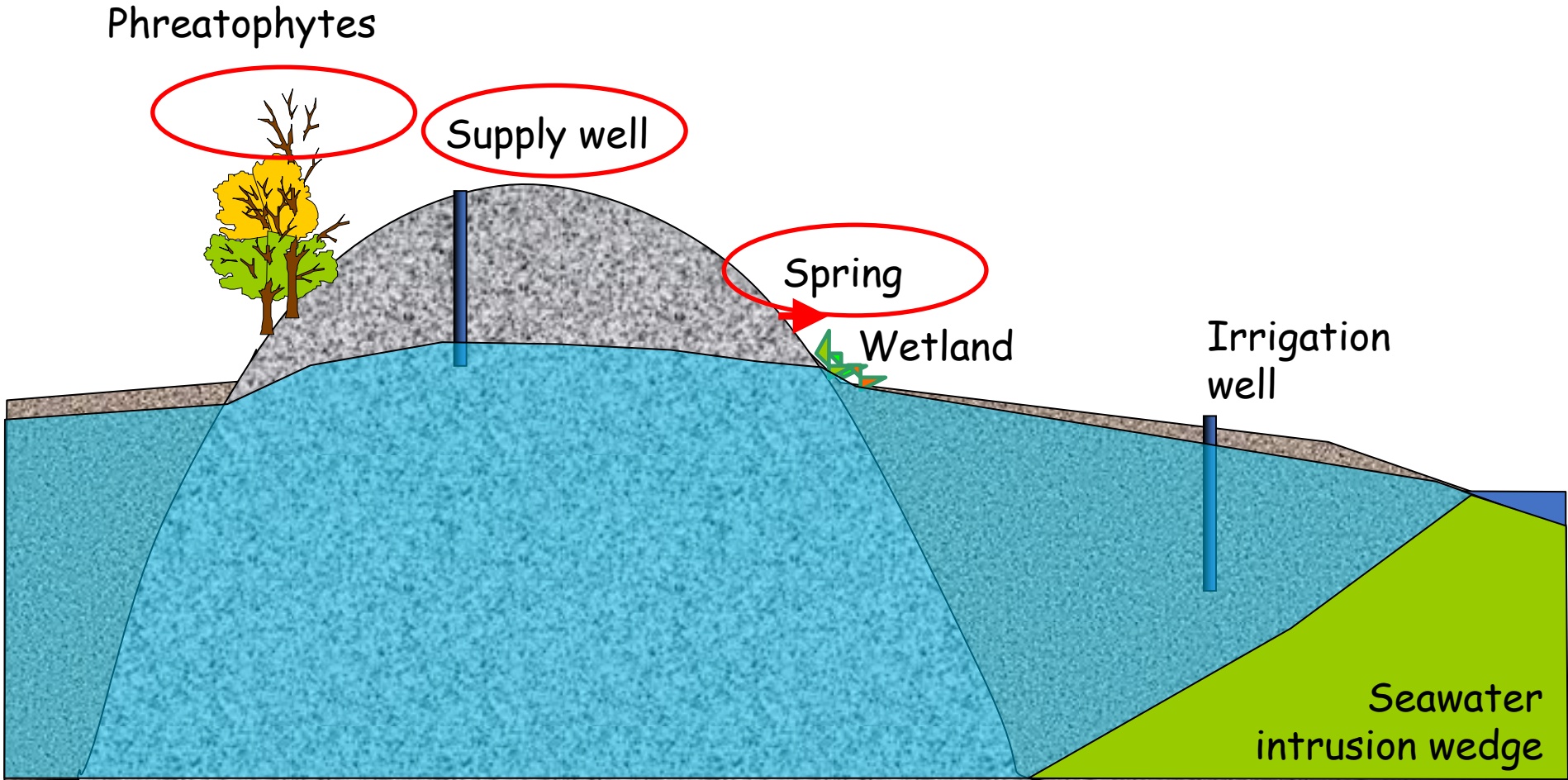


Landscape and the subsurface (as seen by hydrogeologists)





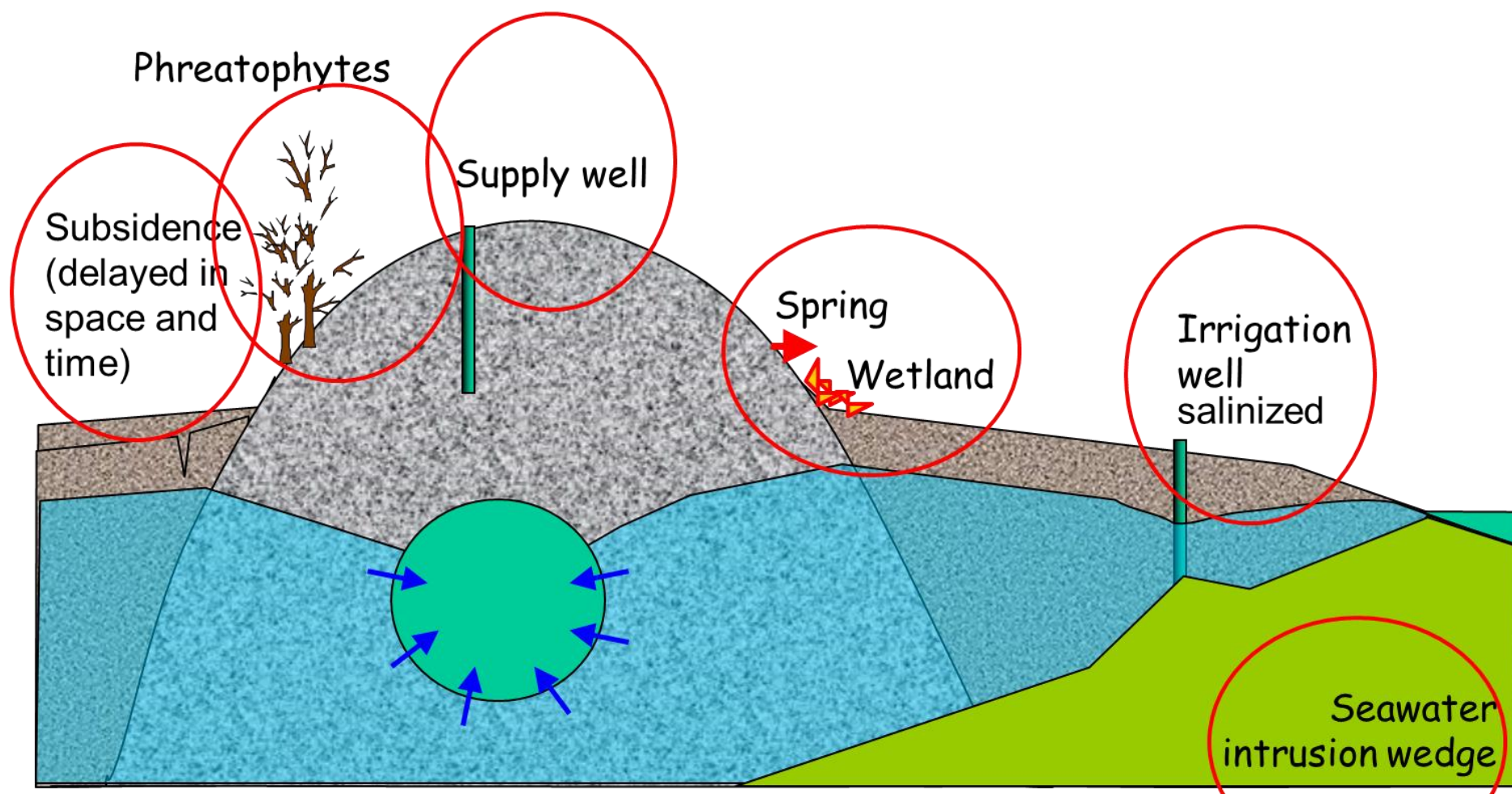
Landscape and the subsurface after long term heavy pumping





Landscape and the subsurface after long term very heavy extractions

La extracción de grandes volúmenes de agua subterránea reduce la **presión de poros** lo que provoca la compactación de sedimentos y la subsidencia del terreno, siendo irreversible en acuíferos no consolidados debido a la pérdida de espacio poroso. En acuíferos confinados, esta presión también se conoce como **presión piezométrica**, que impulsa el flujo del agua subterránea.





Typical subsidence pictures

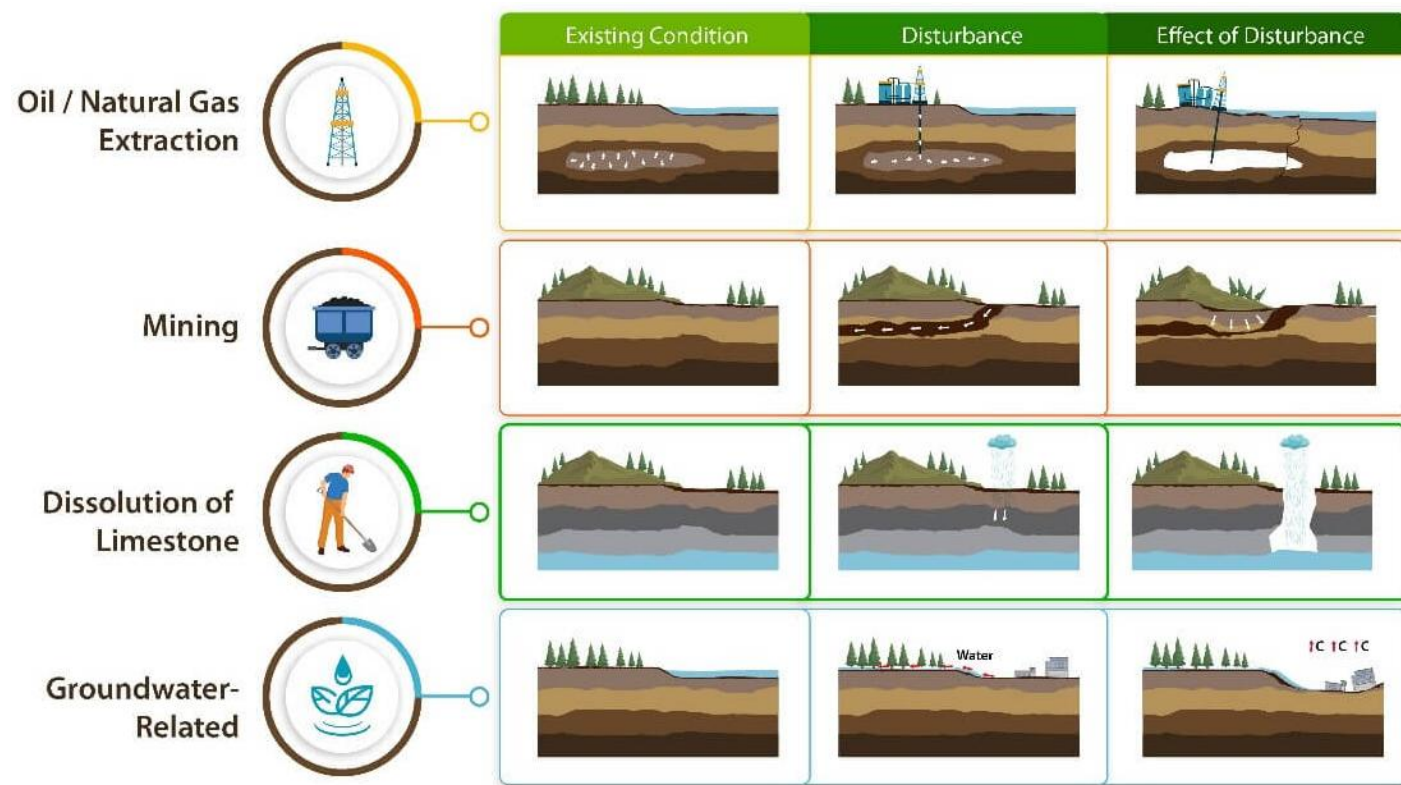
El Valle de San Joaquín tiene todos los ingredientes necesarios para el hundimiento: grandes cantidades de arcilla, agua subterránea a presión que sostiene la arcilla y grandes volúmenes de bombeo de agua subterránea que reduce drásticamente la capacidad de la arcilla para sostener el suelo sobre ella.

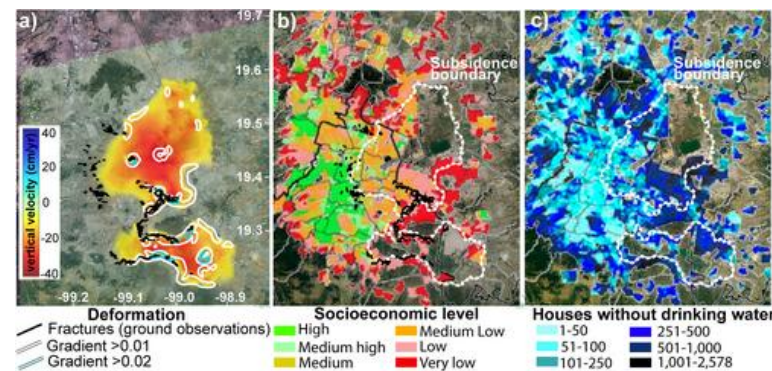
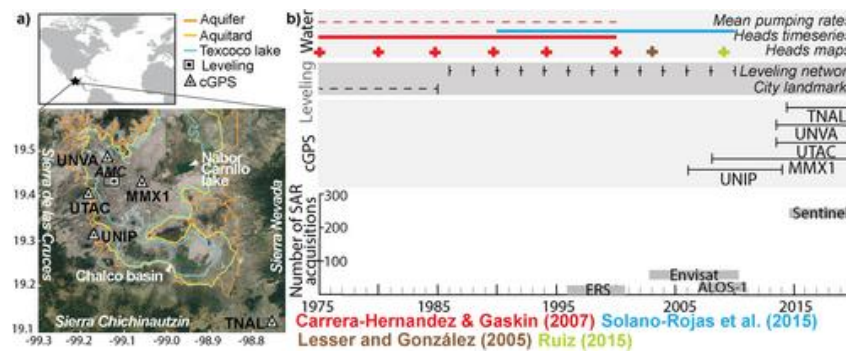
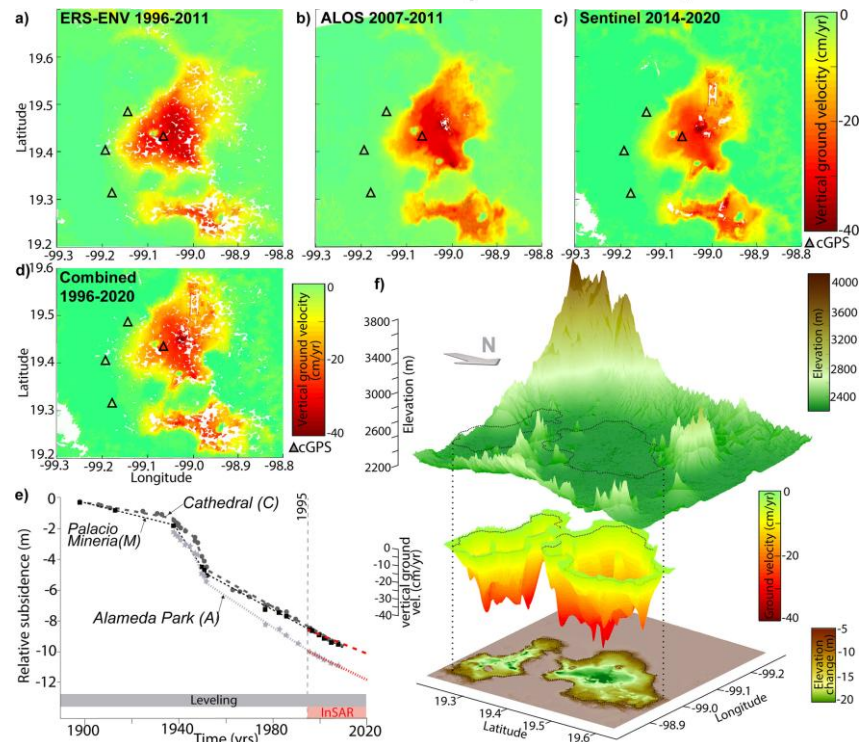
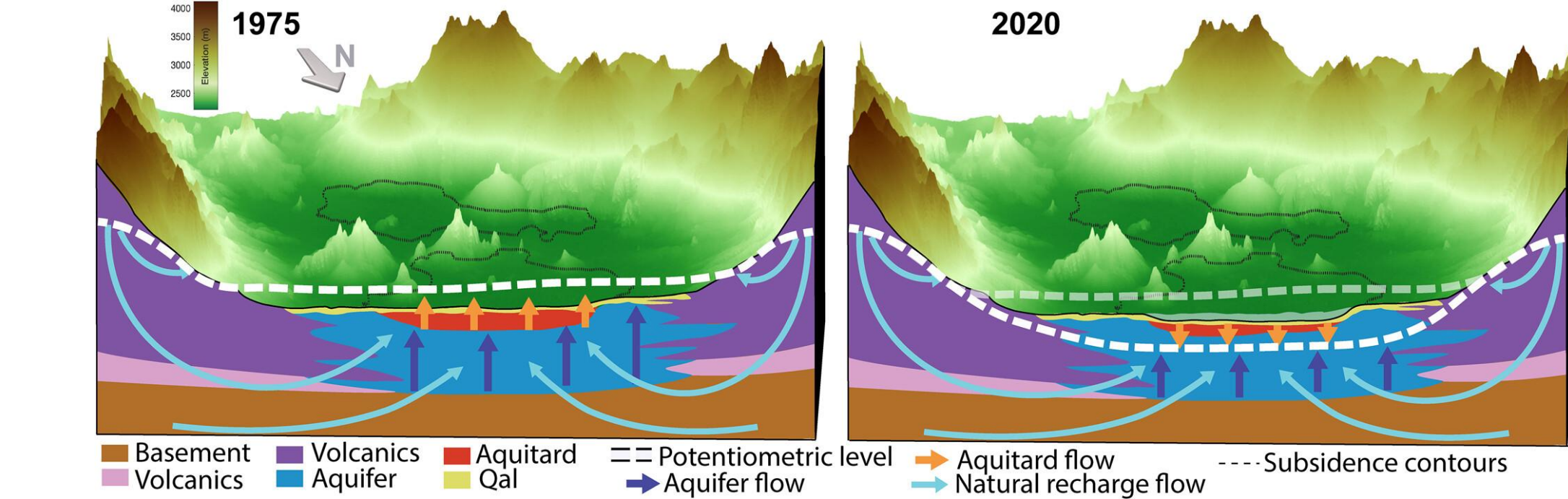
Un investigador se encuentra junto a un poste en el Valle de San Joaquín, al suroeste de Mendota, California. Los carteles muestran la altitud aproximada de la superficie terrestre en 1925, 1955 y 1977. La superficie terrestre disminuyó unos nueve metros entre 1925 y 1977 debido a la compactación del sistema acuífero. Dominio público/USGS



Edge of Tucson (Arizona) basin.

San Joaquín Valley (California)
9 m settlement between 1925 y 1977



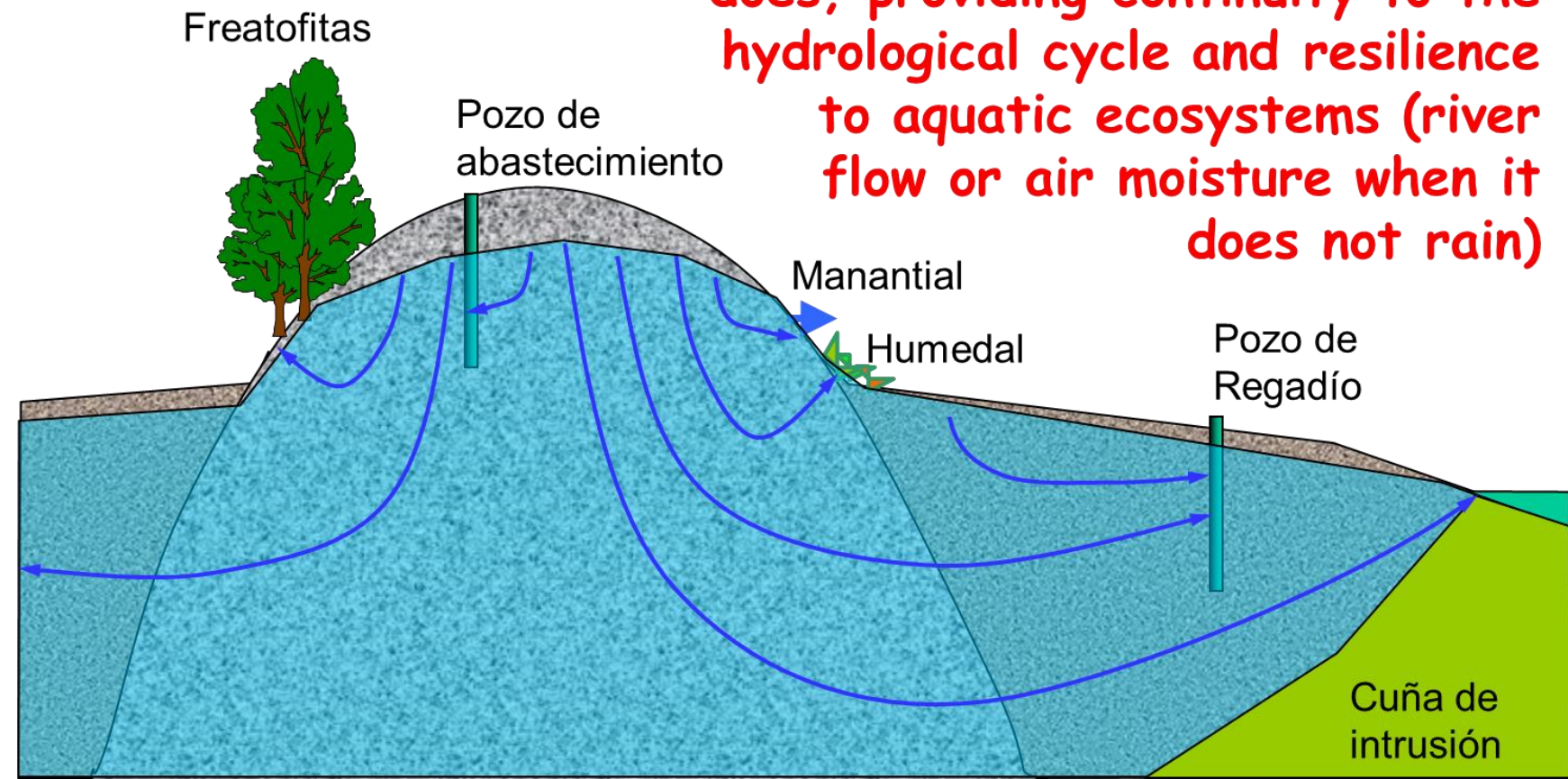


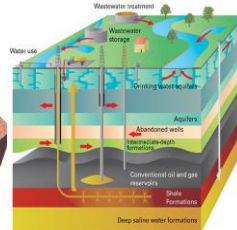
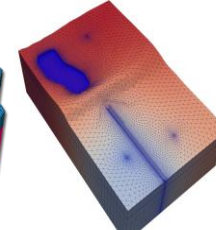
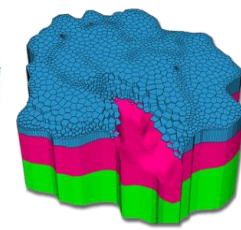
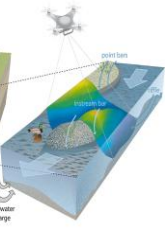
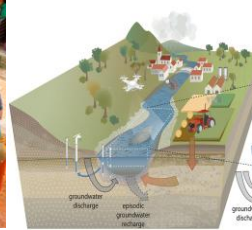
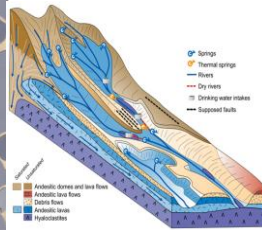
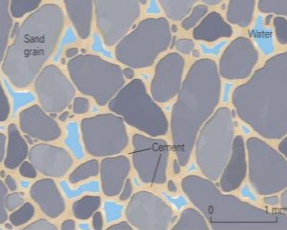
Chaussard, E., Havazli, E., Fattahi, H., Cabral-Cano, E., & Solano-Rojas, D. (2021). Over a century of sinking in Mexico City: No hope for significant elevation and storage capacity recovery. *Journal of Geophysical Research: Solid Earth*, 126(4), e2020JB020648.



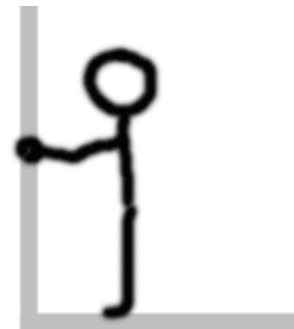
For a hydrologist, the landscape depends on water, which flows!

Grounwater flows slowly, but it does, providing continuity to the hydrological cycle and resilience to aquatic ecosystems (river flow or air moisture when it does not rain)





Whenever you see luxurious vegetation think of "spoiled" plants and GW





¿Qué es el balance hídrico?

El concepto de balance hídrico se deriva del concepto de balance de materia, es decir, que es el equilibrio entre todos los recursos hídricos que ingresan al sistema y los que salen del mismo, en un intervalo de tiempo determinado.

$$I(t) - Q(t) = 0$$

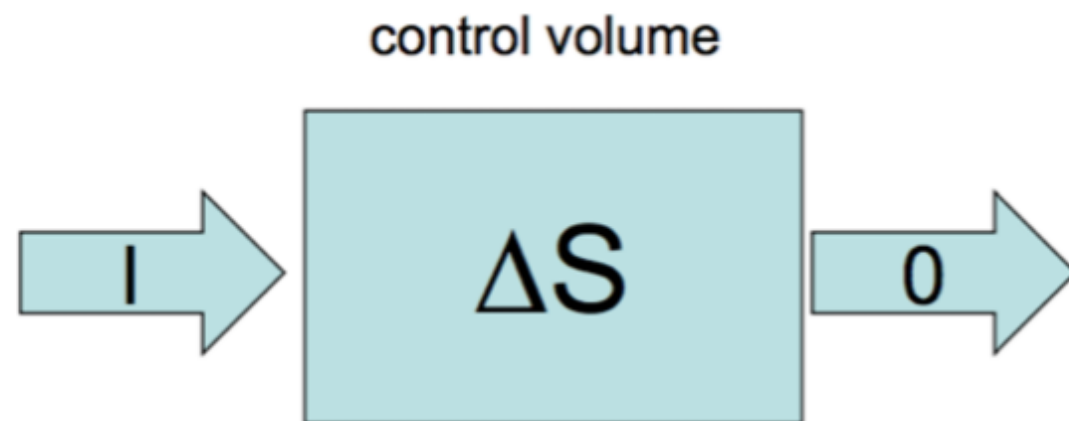
$$I(t) - Q(t) = \frac{dS}{dt}$$

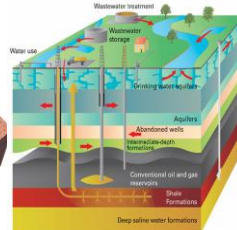
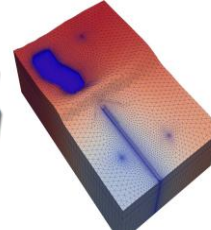
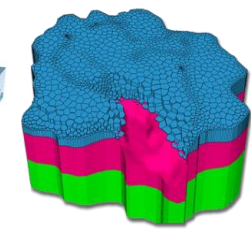
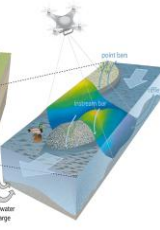
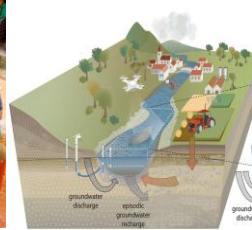
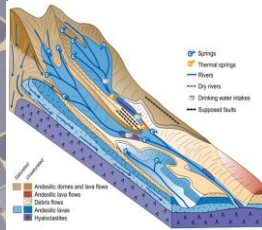
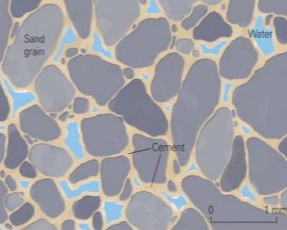
Donde:

$I(t)$: Entradas al sistema (PCP, SNOW)

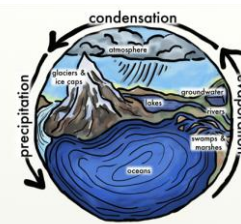
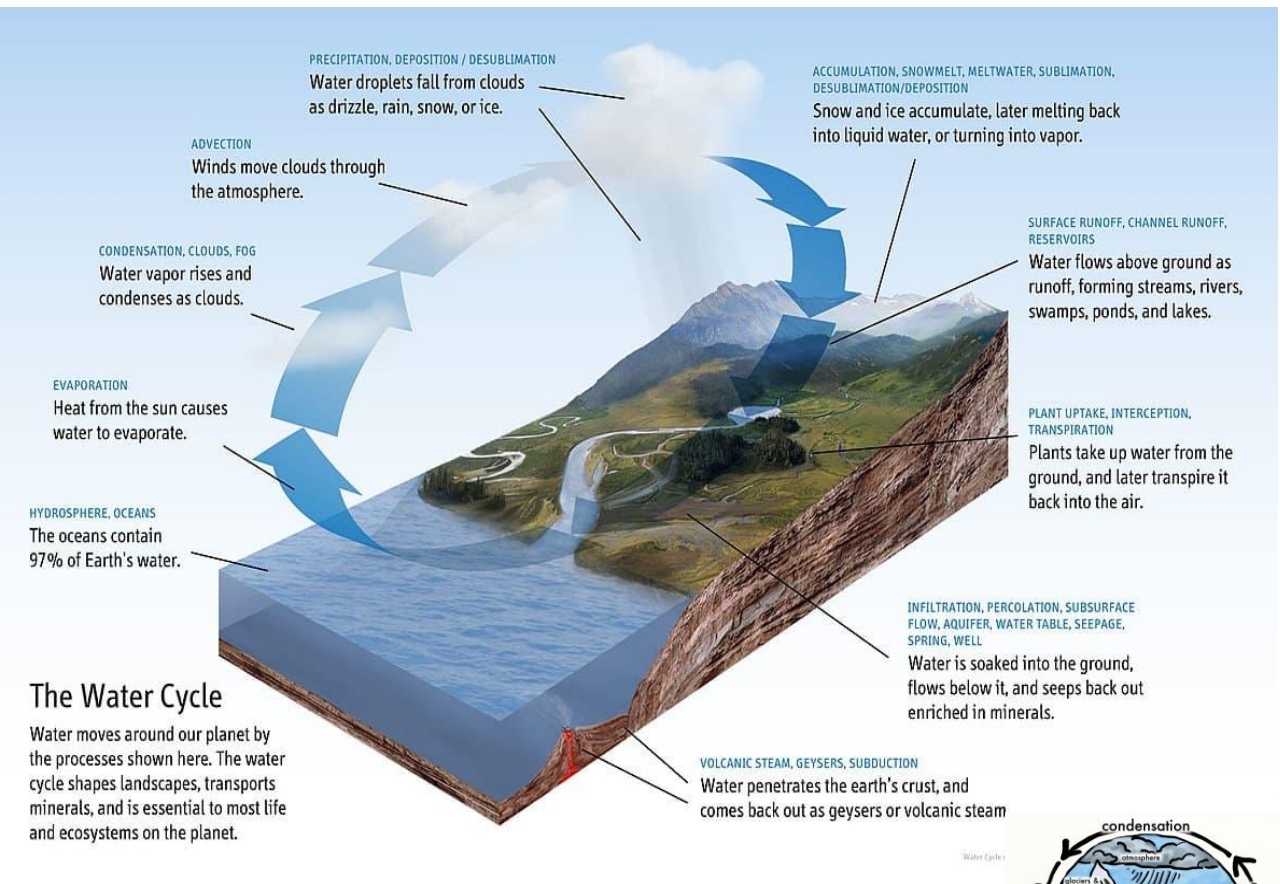
$Q(t)$: Salidas del Sistema (ET, SURQ, LATQ, BSFLOW)

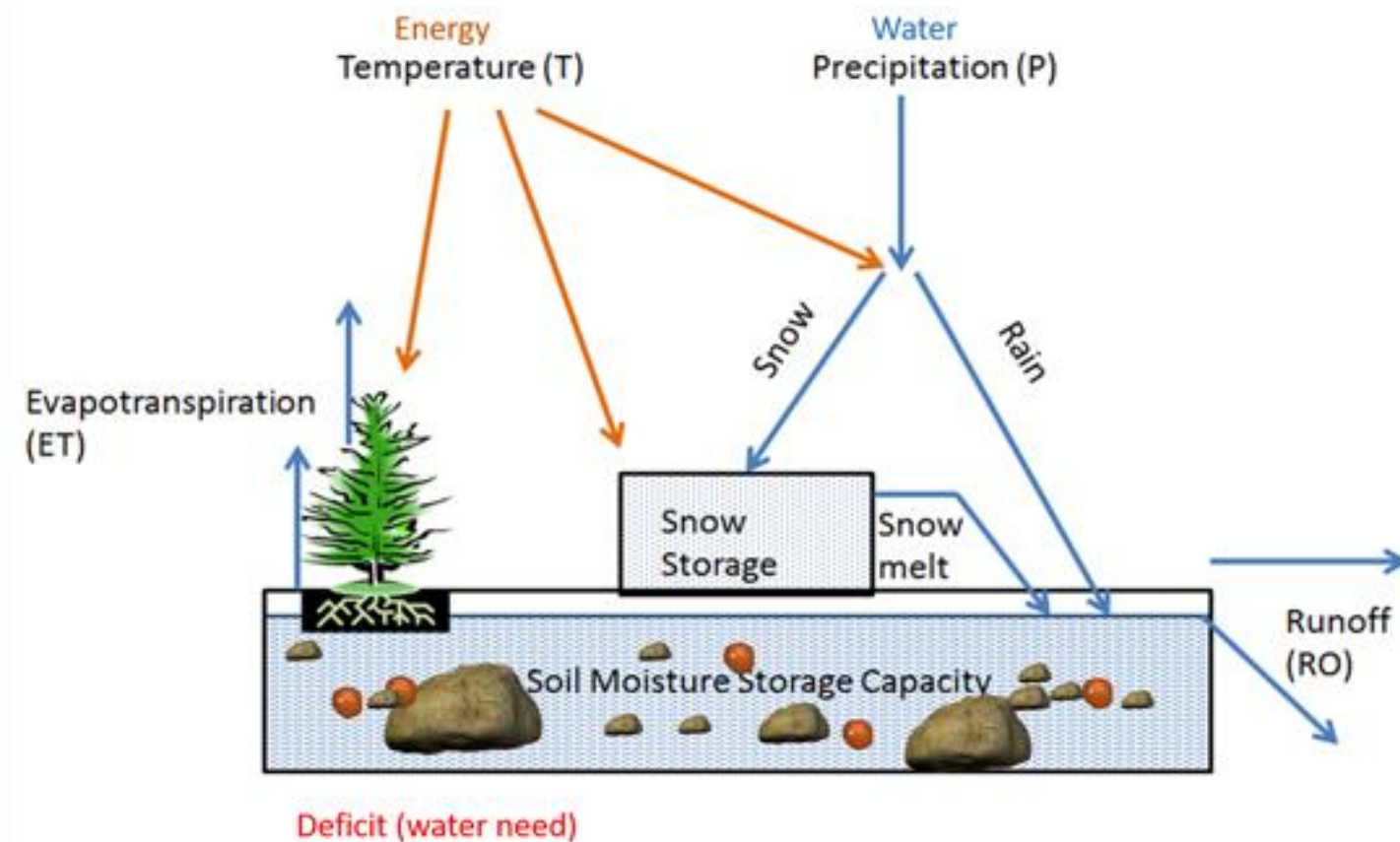
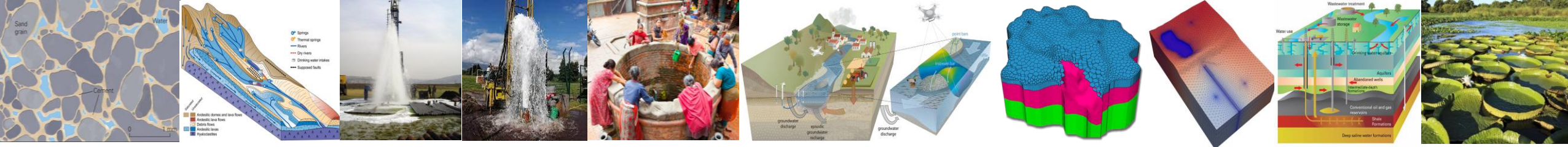
$\frac{dS}{dt}$: Variación del almacenamiento (HMD, GWH)





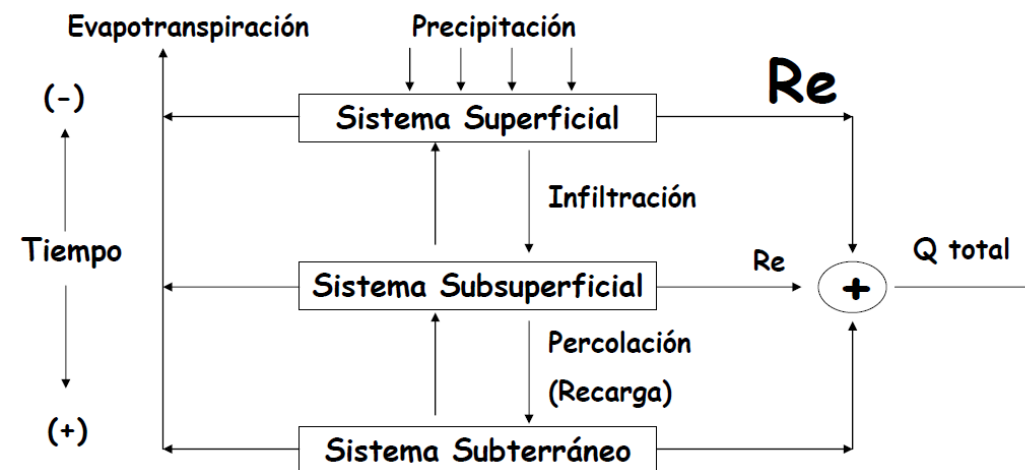
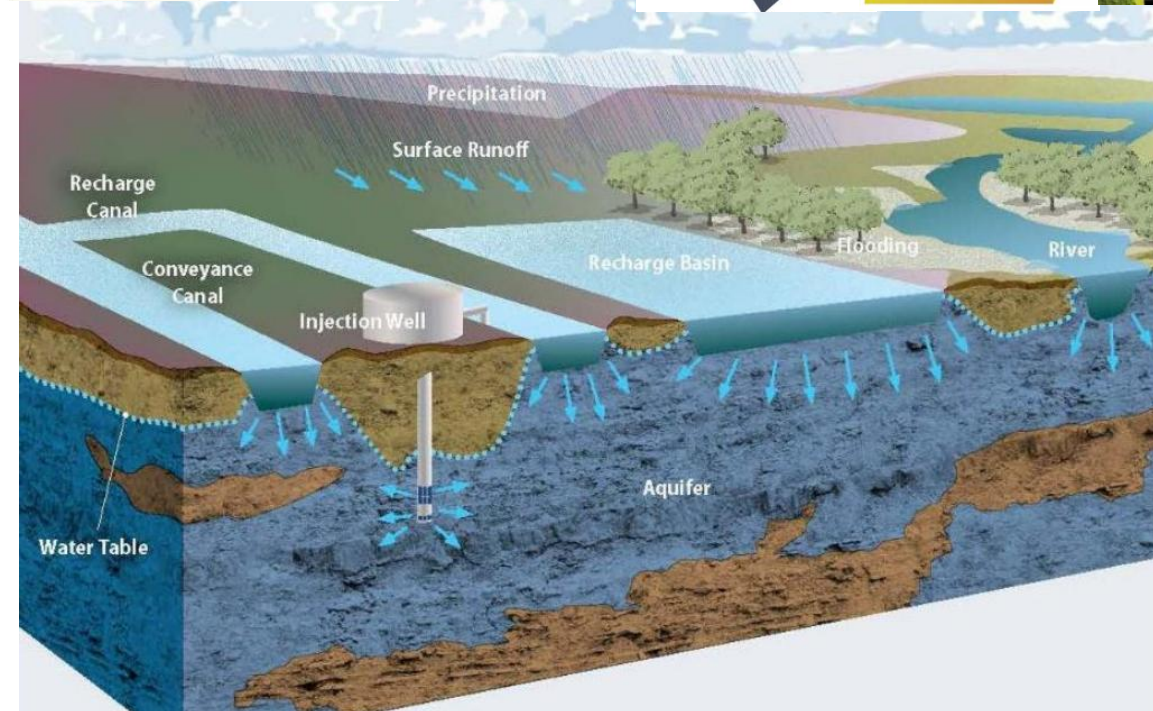
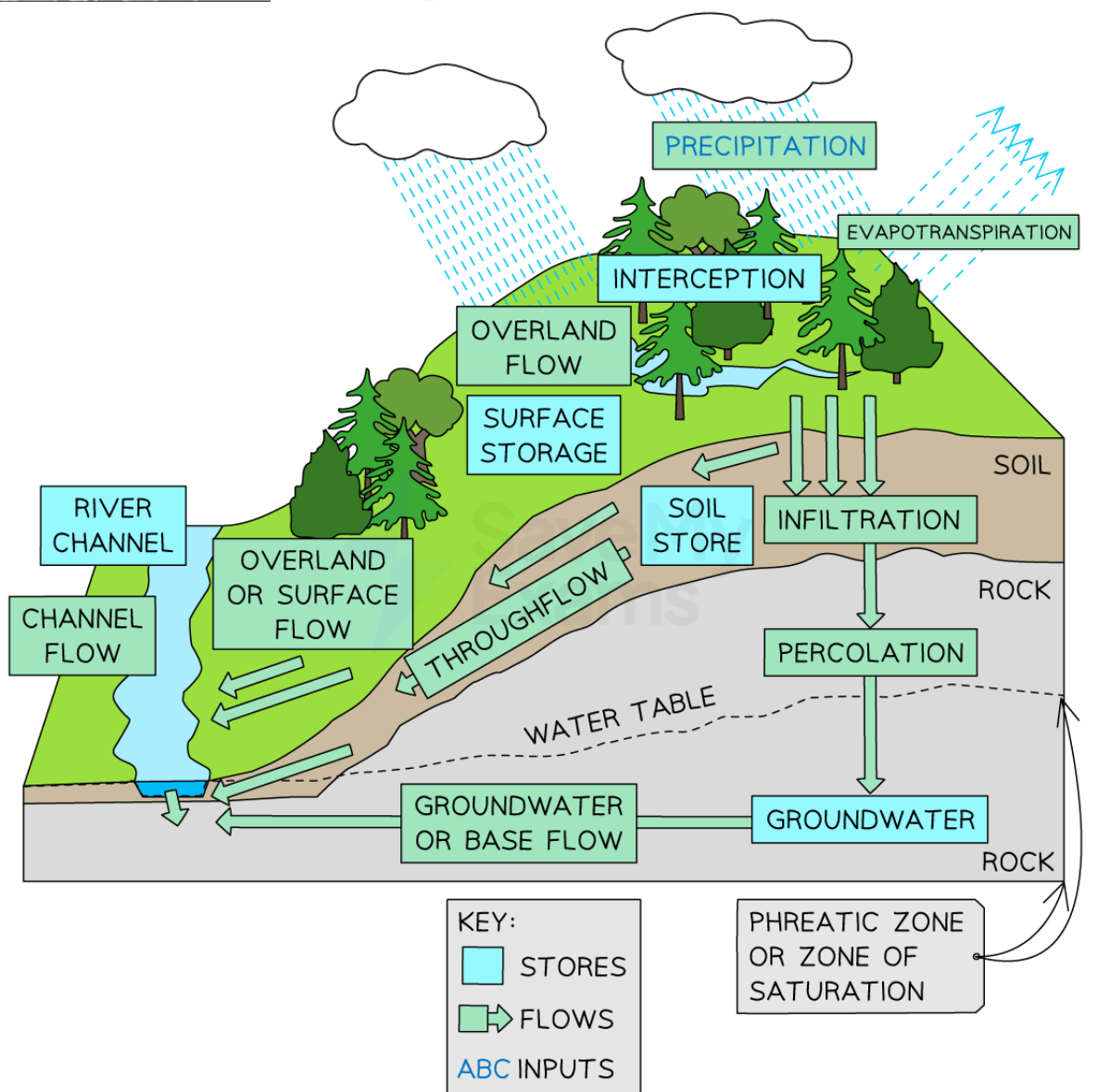
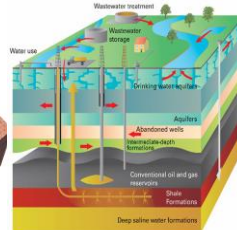
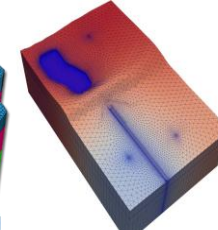
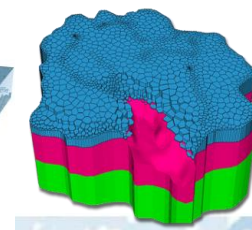
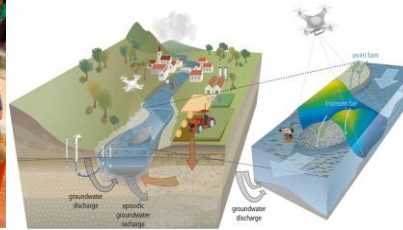
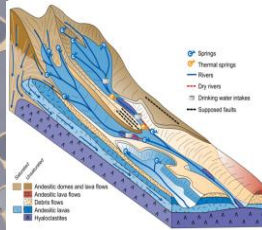
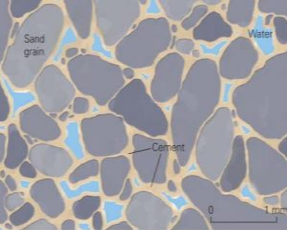
Movimiento del agua en la naturaleza

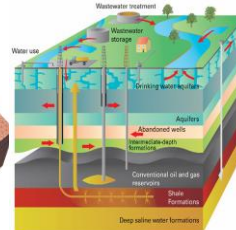
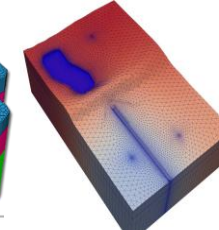
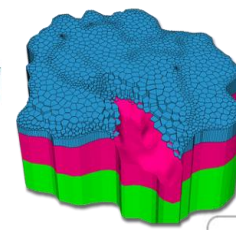
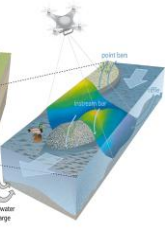
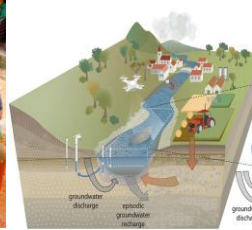
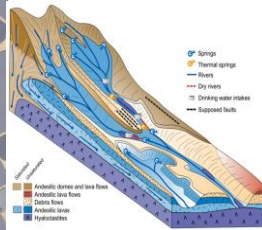
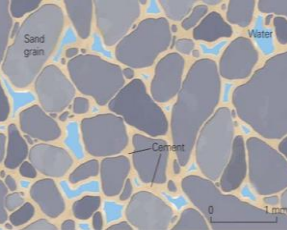




El calor sensible provoca un cambio de temperatura en una sustancia.

El calor latente causa un cambio de estado (fase) de la sustancia, pero sin alterar su temperatura





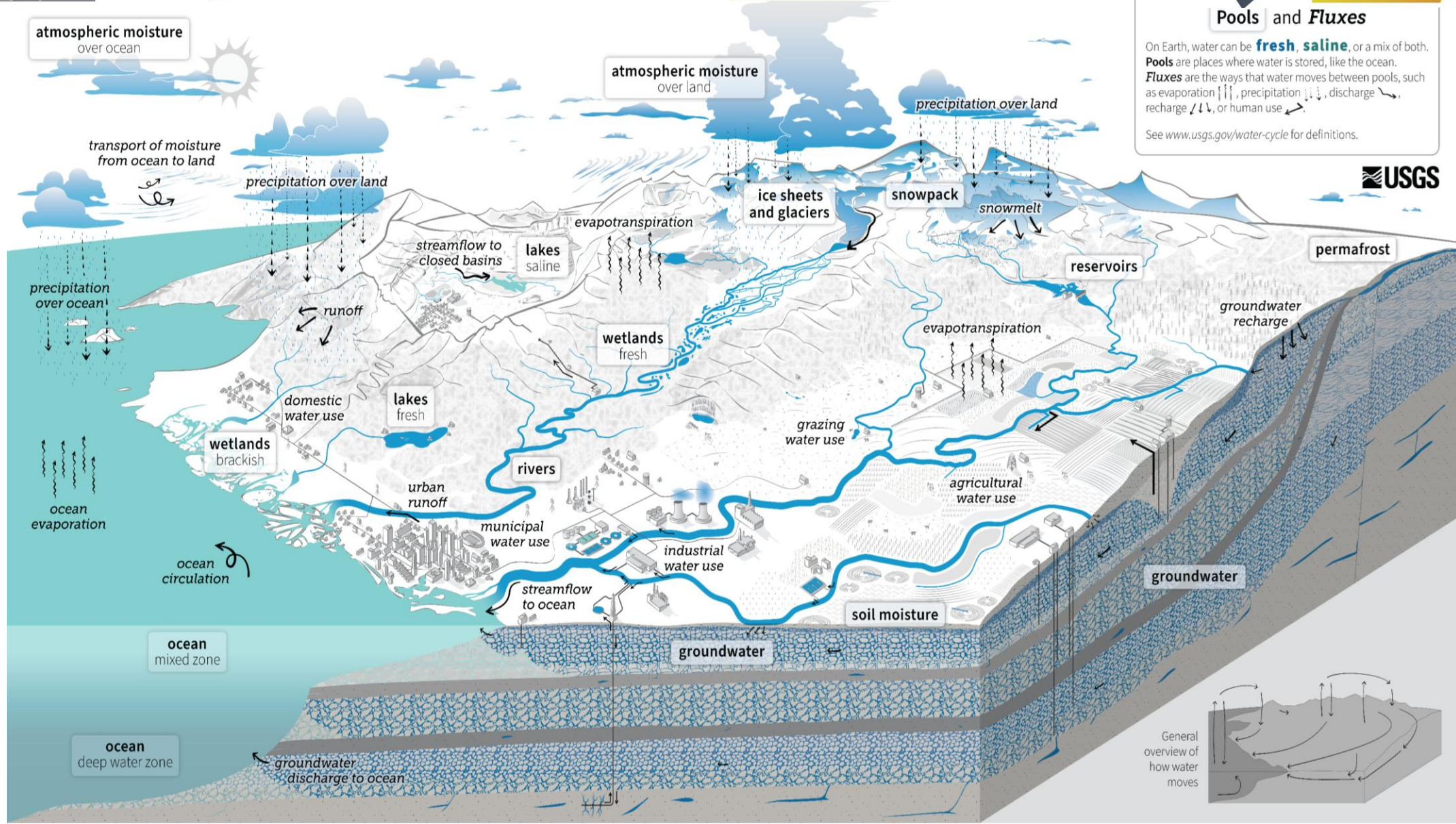
atmospheric moisture
over ocean

atmospheric moisture
over land

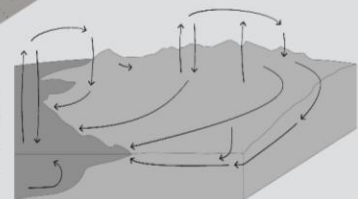
Pools and Fluxes

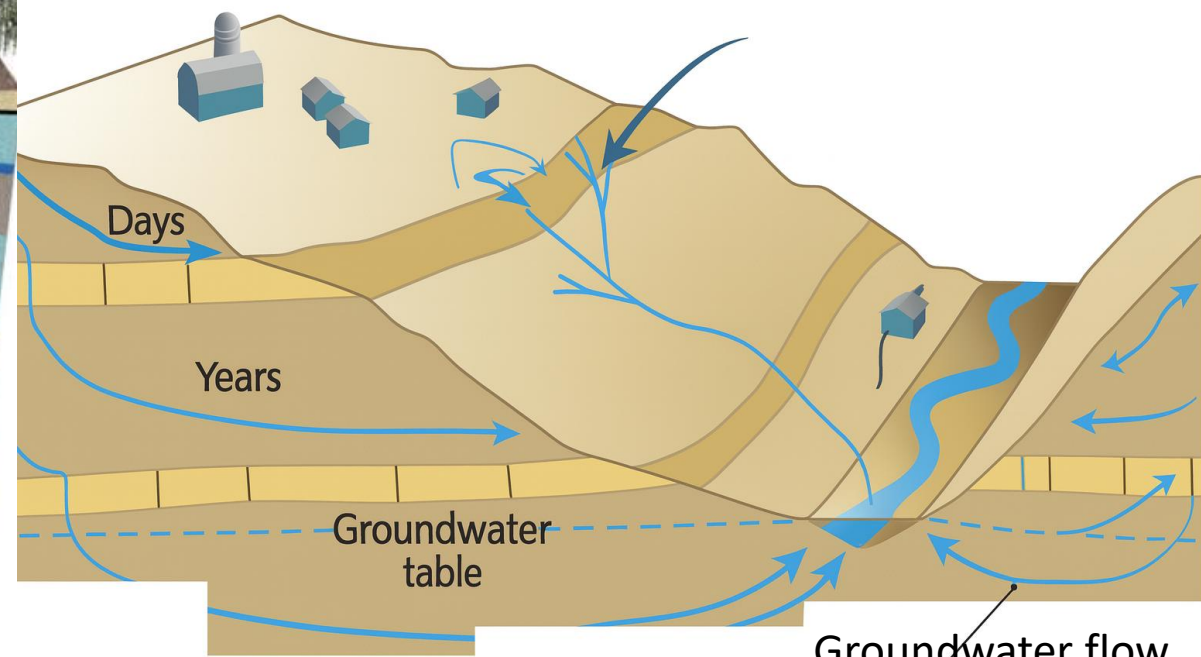
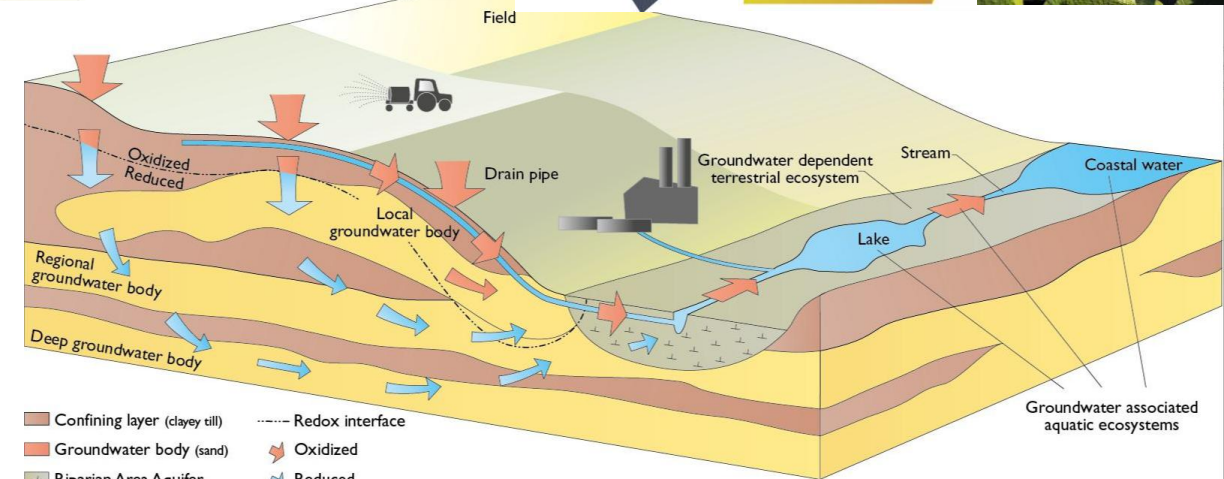
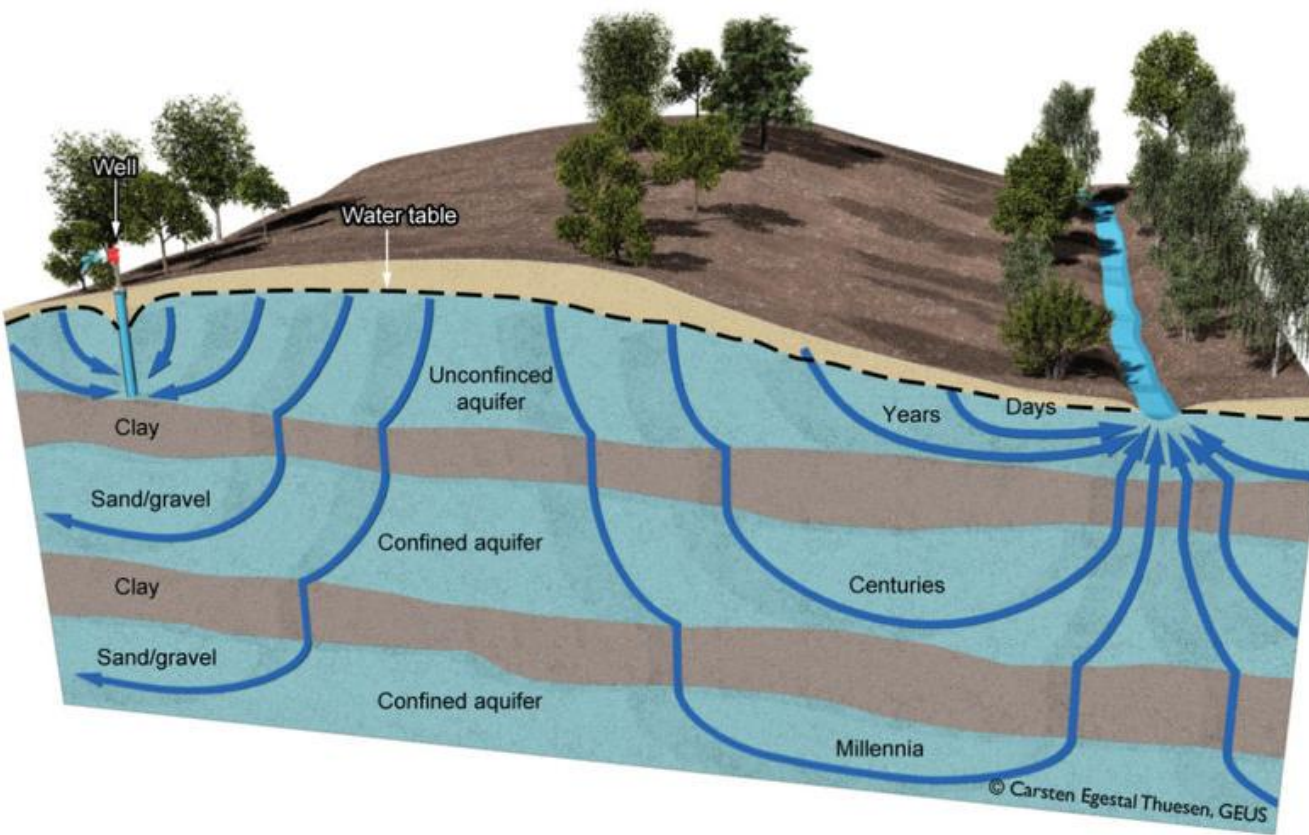
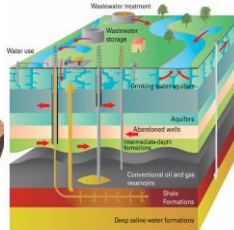
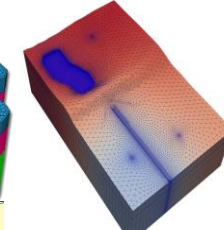
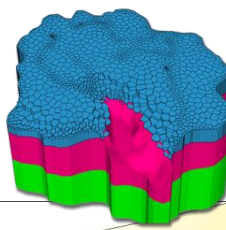
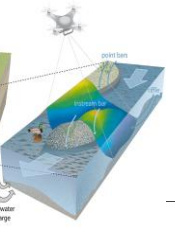
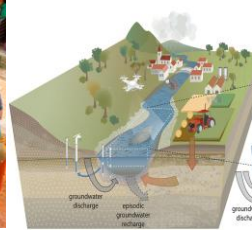
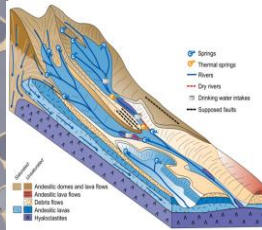
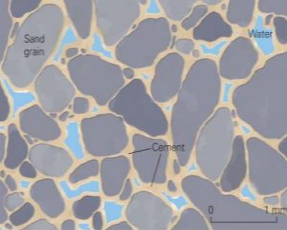
On Earth, water can be **fresh, saline**, or a mix of both. **Pools** are places where water is stored, like the ocean. **Fluxes** are the ways that water moves between pools, such as evaporation $\uparrow\uparrow\uparrow$, precipitation $\downarrow\downarrow\downarrow$, discharge $\rightarrow$, recharge $\nearrow$, or human use $\hookrightarrow$.

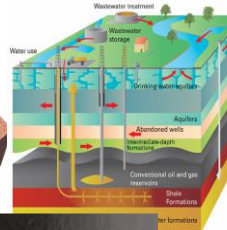
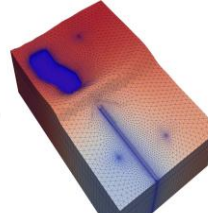
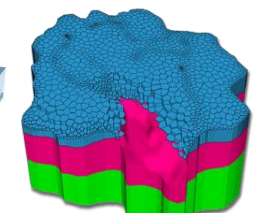
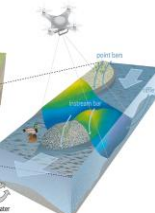
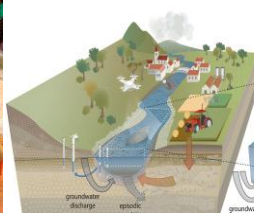
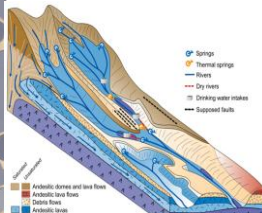
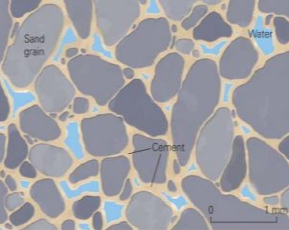
See www.usgs.gov/water-cycle for definitions.



General
overview of
how water
moves



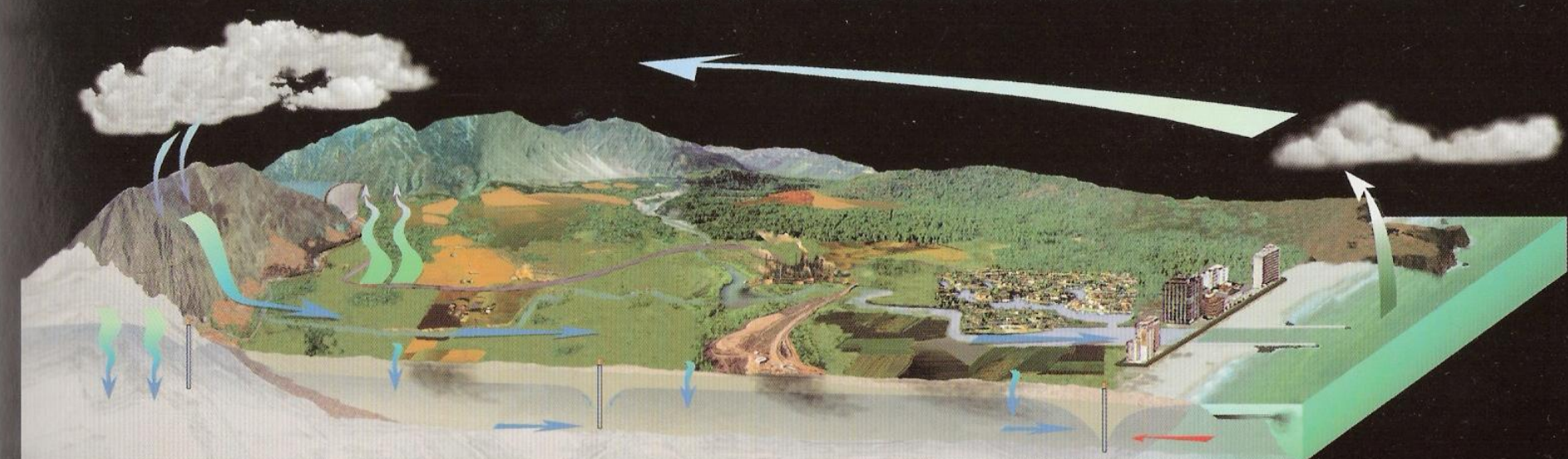


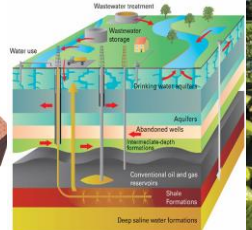
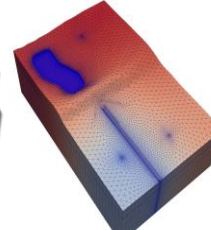
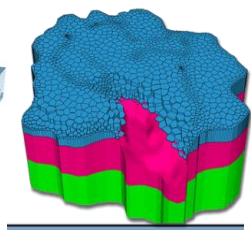
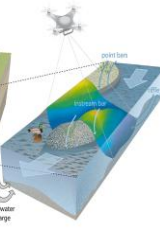
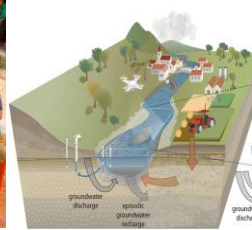
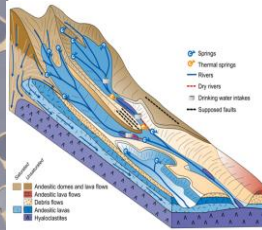
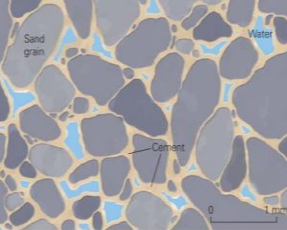


CICLO HIDROLÓGICO EN ESTADO NATURAL

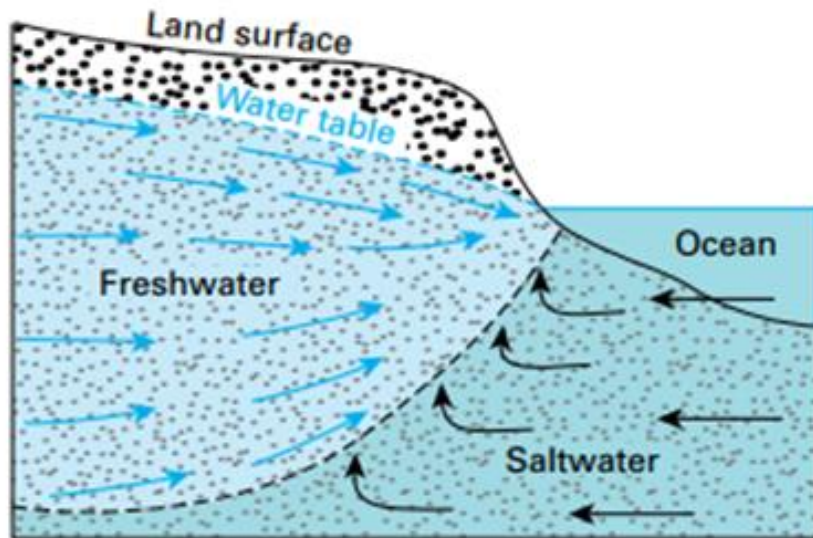


CICLO HIDROLÓGICO AFECTADO POR EL HOMBRE

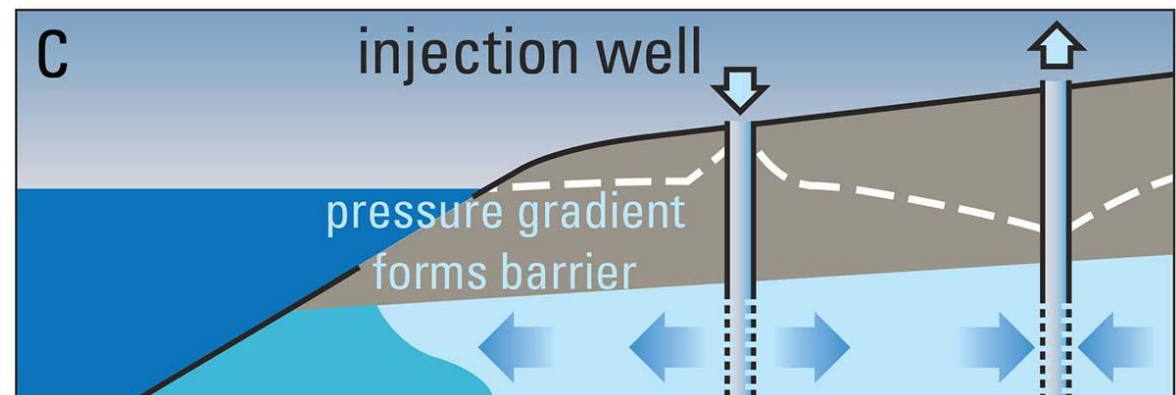
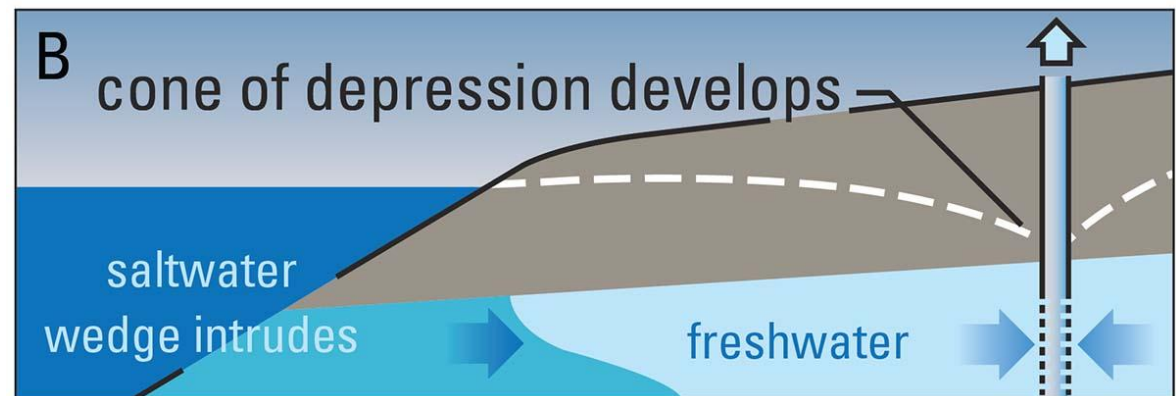
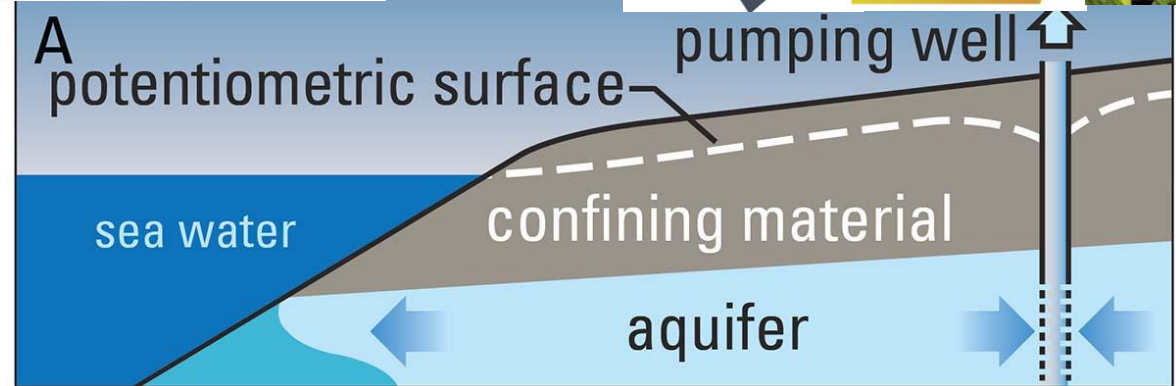
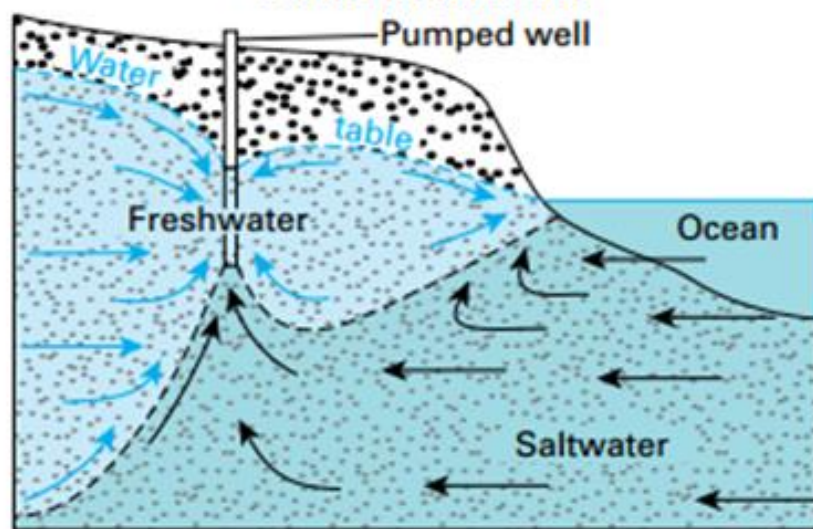




Natural Conditions

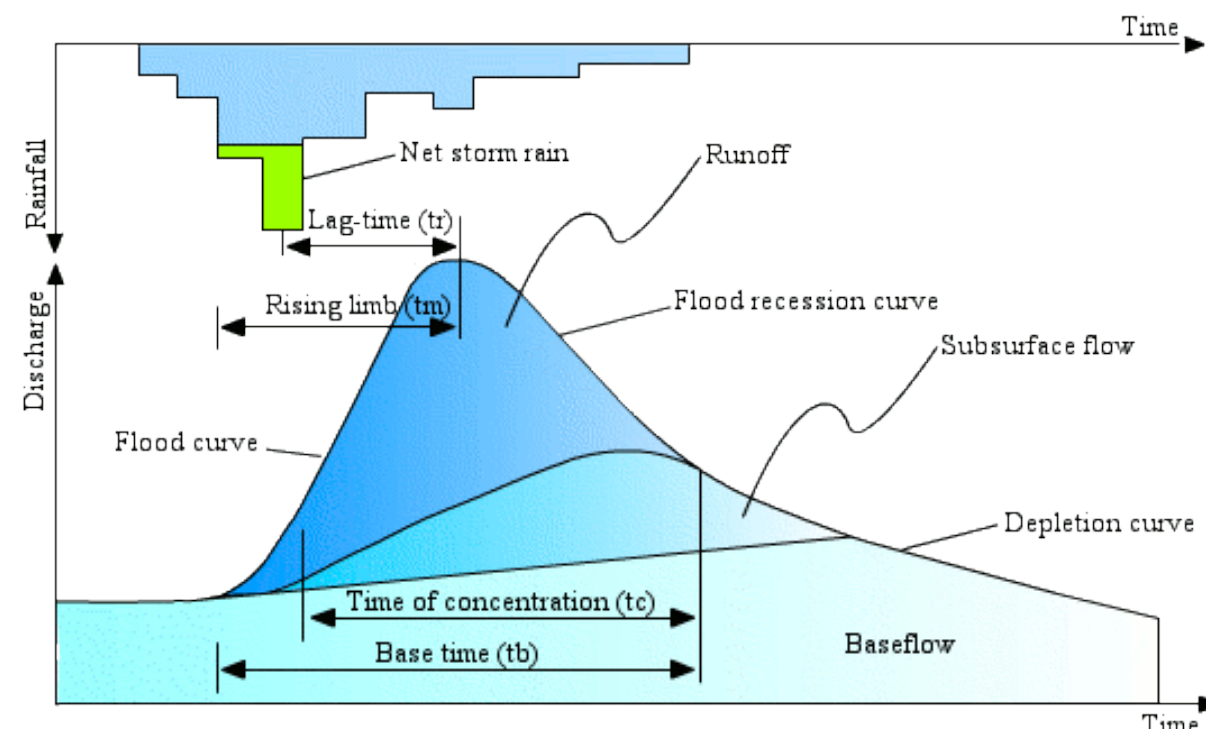
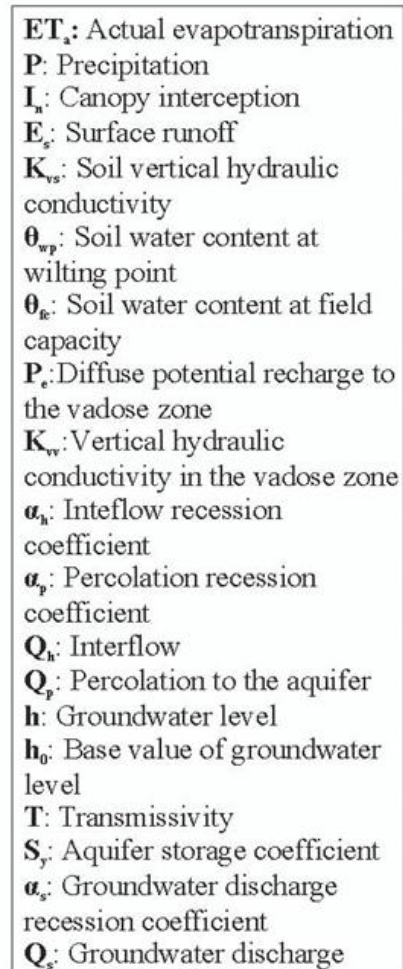


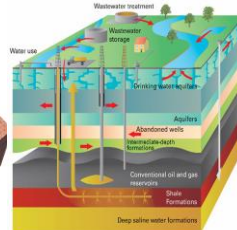
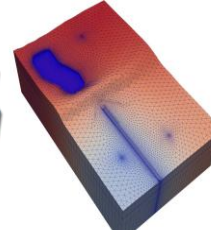
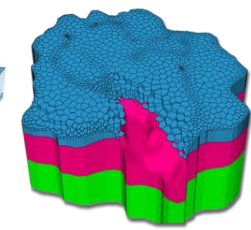
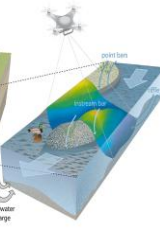
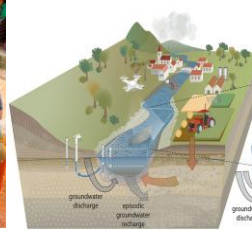
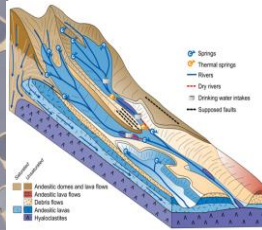
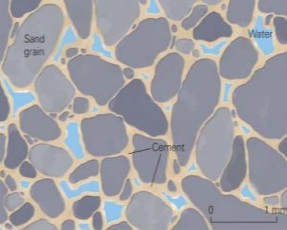
Salt-Water Intrusion



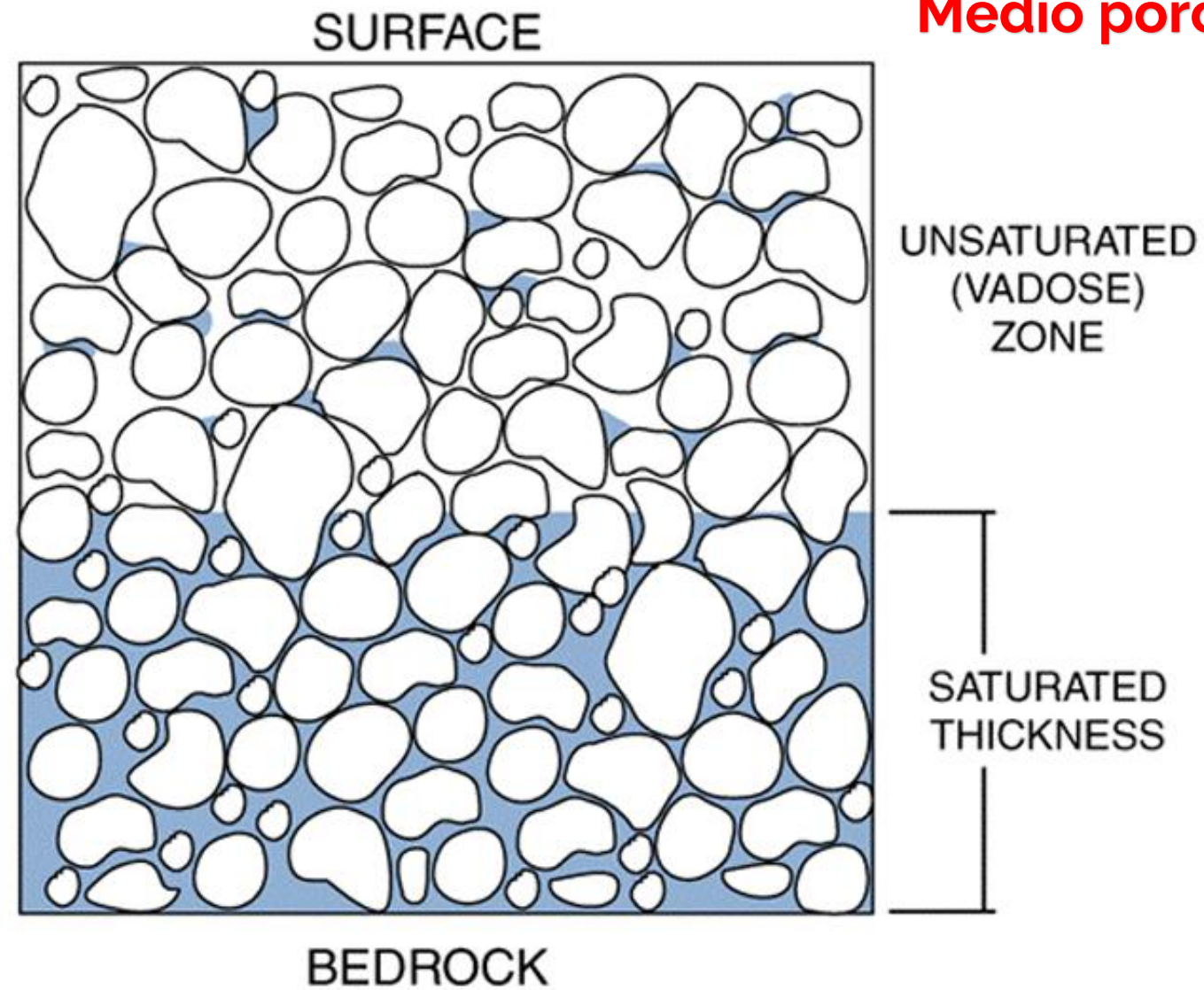


1. Infiltración o Percolación
2. Esgurrimiento superficial y subsuperficial
3. Evapotranspiración
4. Recarga del acuífero
5. Flujo de aguas subterráneas
6. Descarga de aguas subterráneas
7. Interacciones con cuerpos de agua superficiales



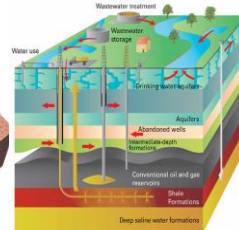
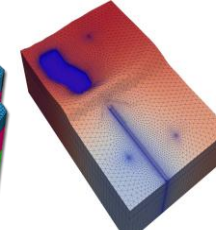
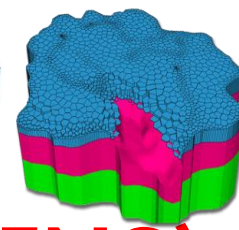
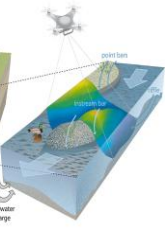
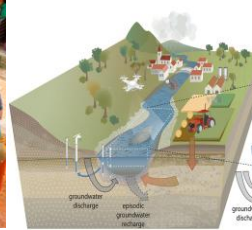
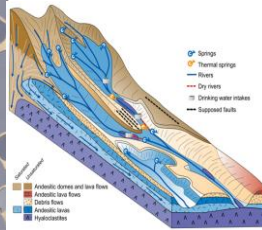
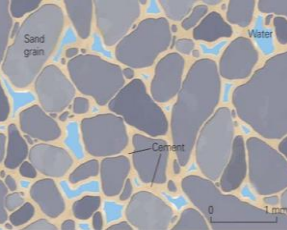


Medio poroso y distribución de fases

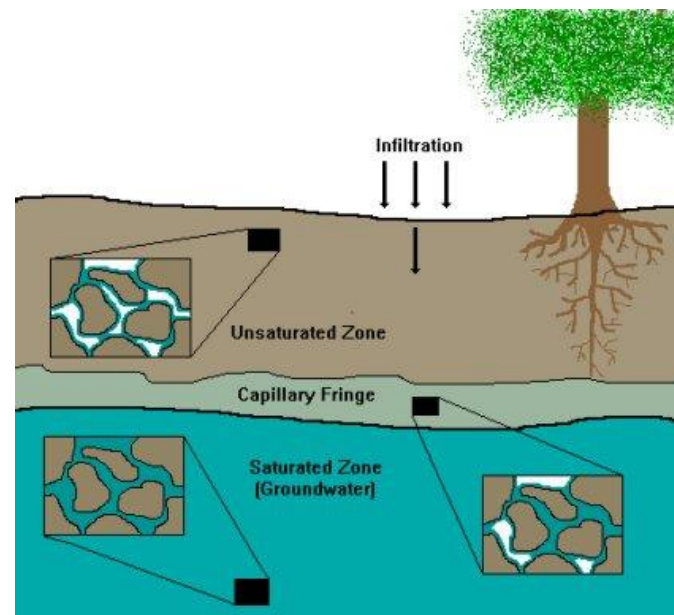
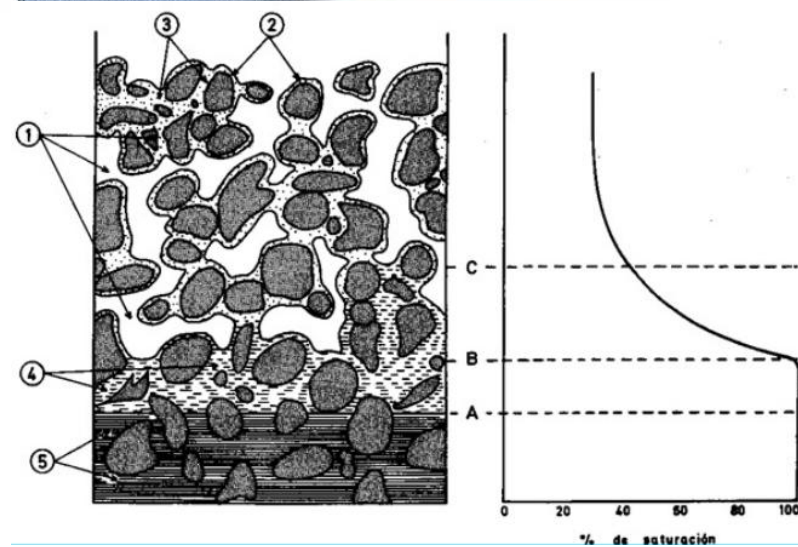
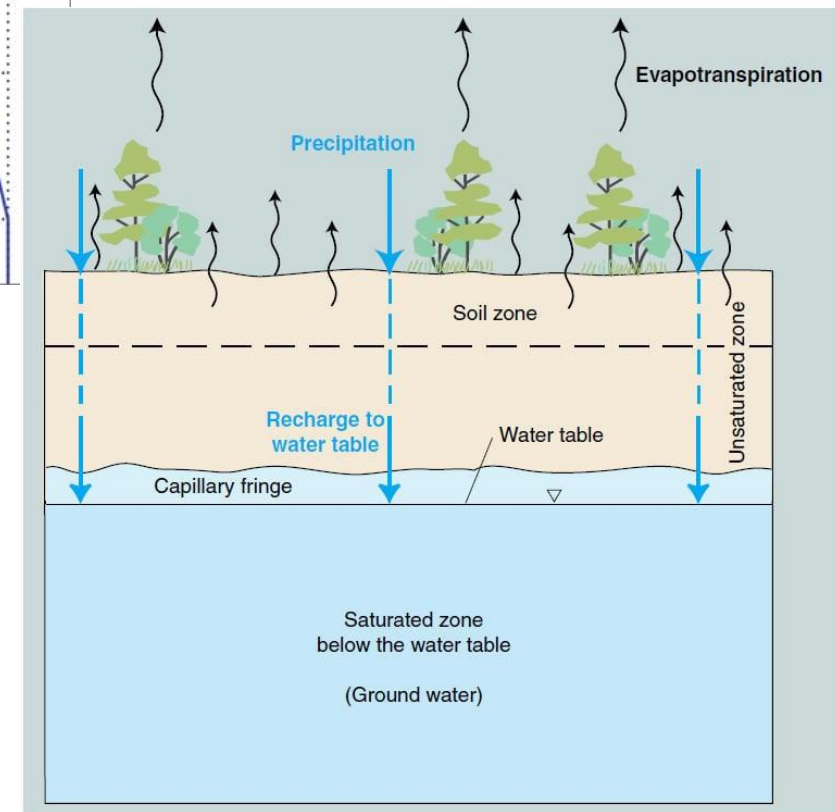
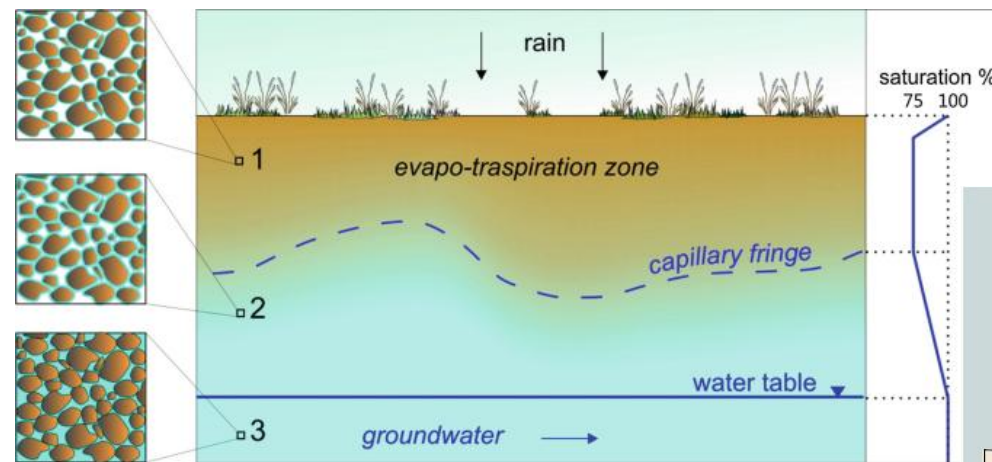
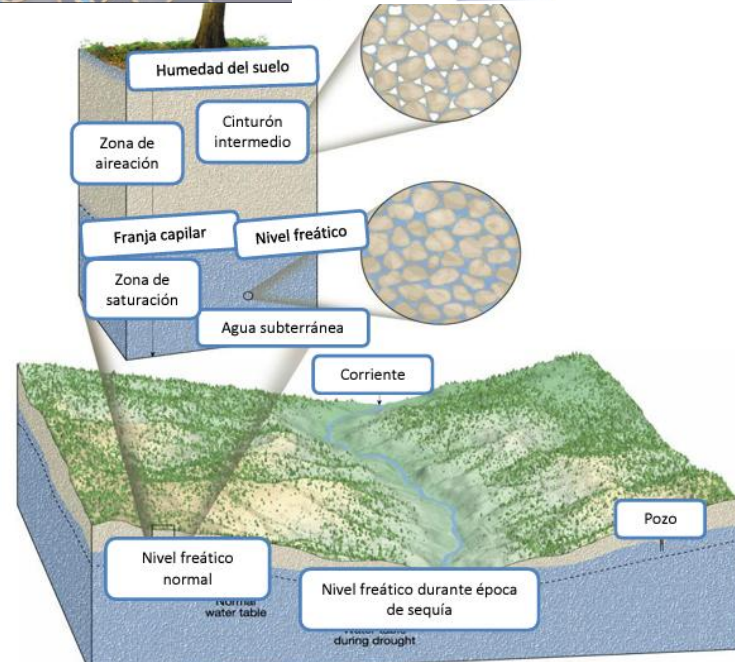


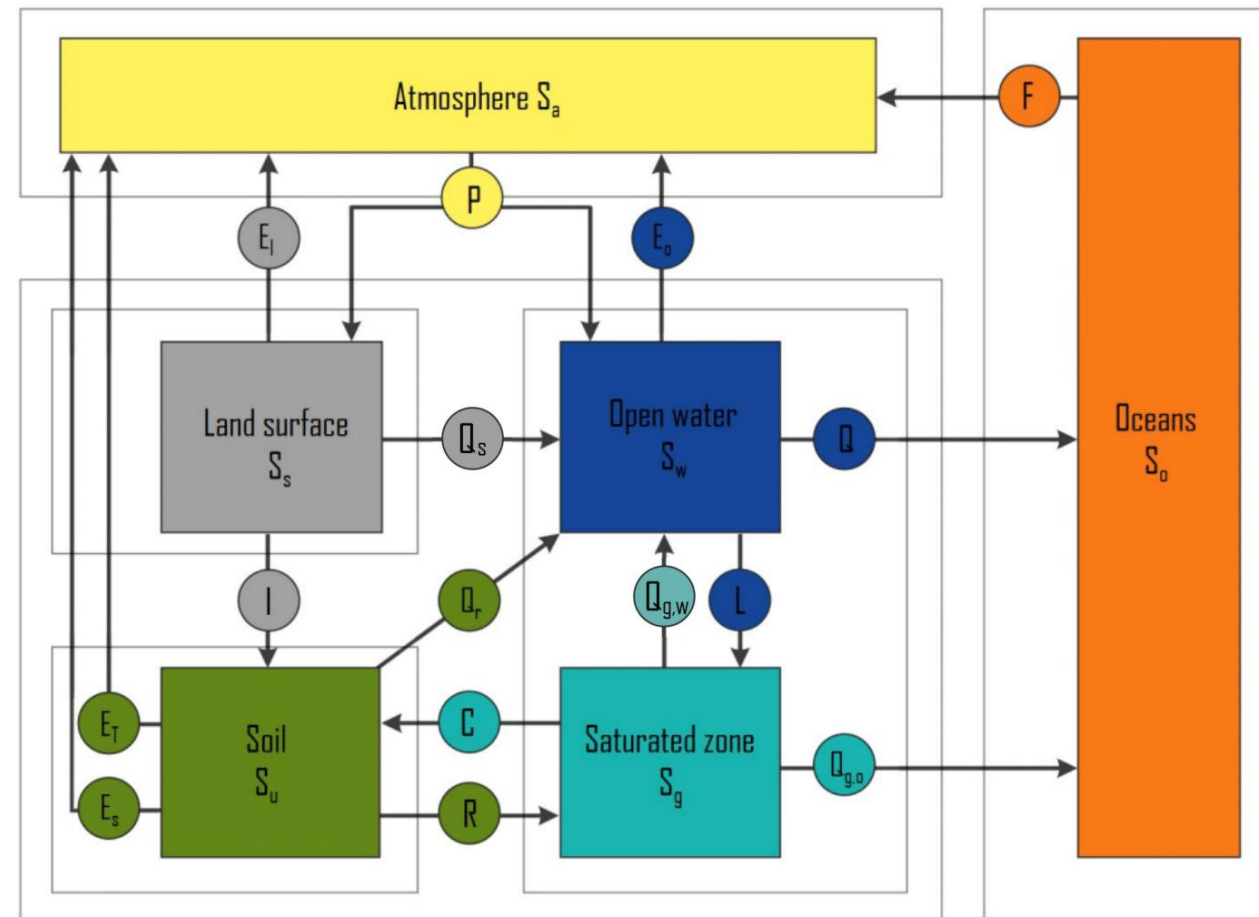
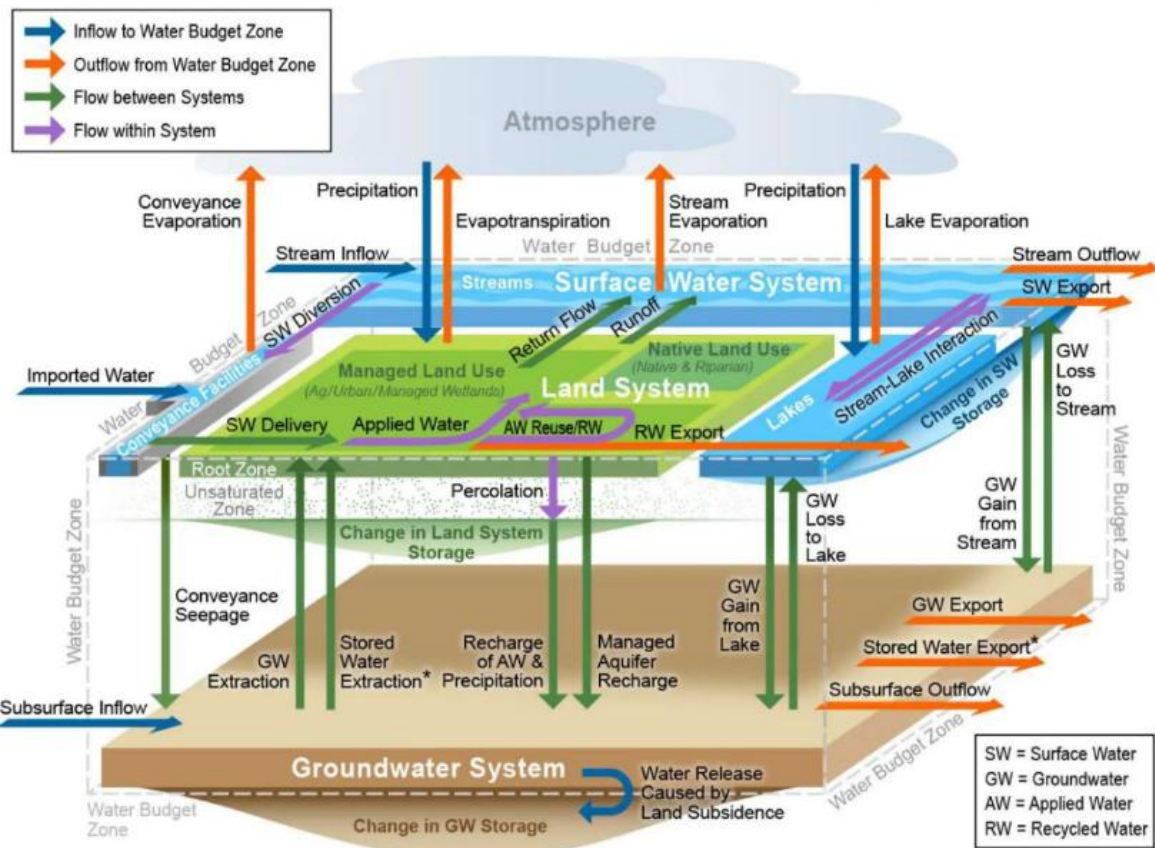
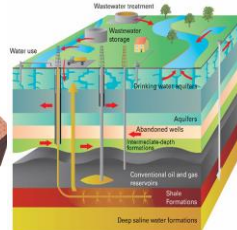
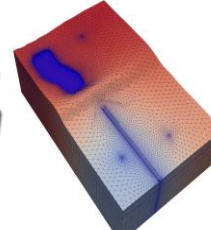
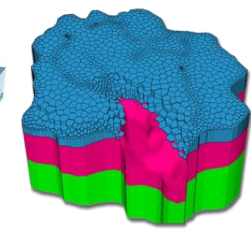
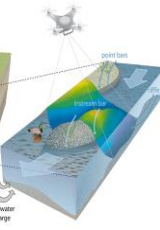
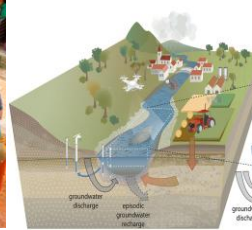
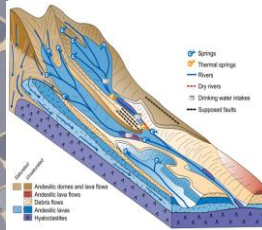
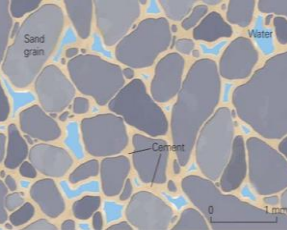
Sistema trifásico

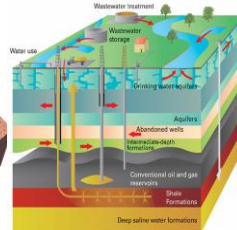
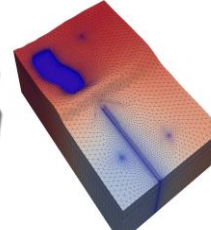
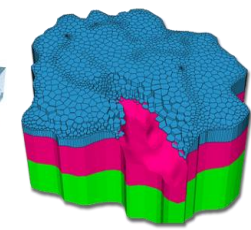
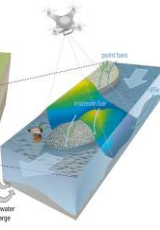
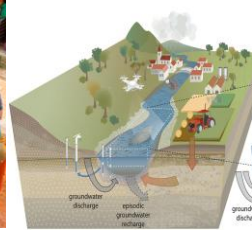
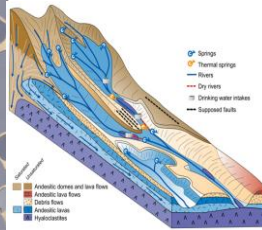
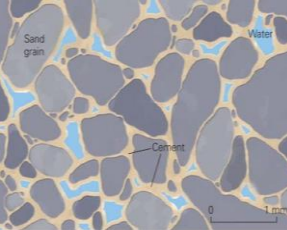
Sistema bifásico



Zona no saturada (ZNS)







$$\frac{dS}{dt} = Q_{in} - Q_{out}$$

S	Almacenamiento
Q_{in}	Flujos entrantes
Q_{out}	flujos salientes

$$A = A_s + A_w$$

A_s	Área de superficie terrestre
A_w	Área de agua superficial

$$E = E_i + E_t + E_s + E_o$$

E_i	Intercepción
E_t	Transpiración
E_s	Evaporación del suelo
E_o	Evaporación en aguas abiertas

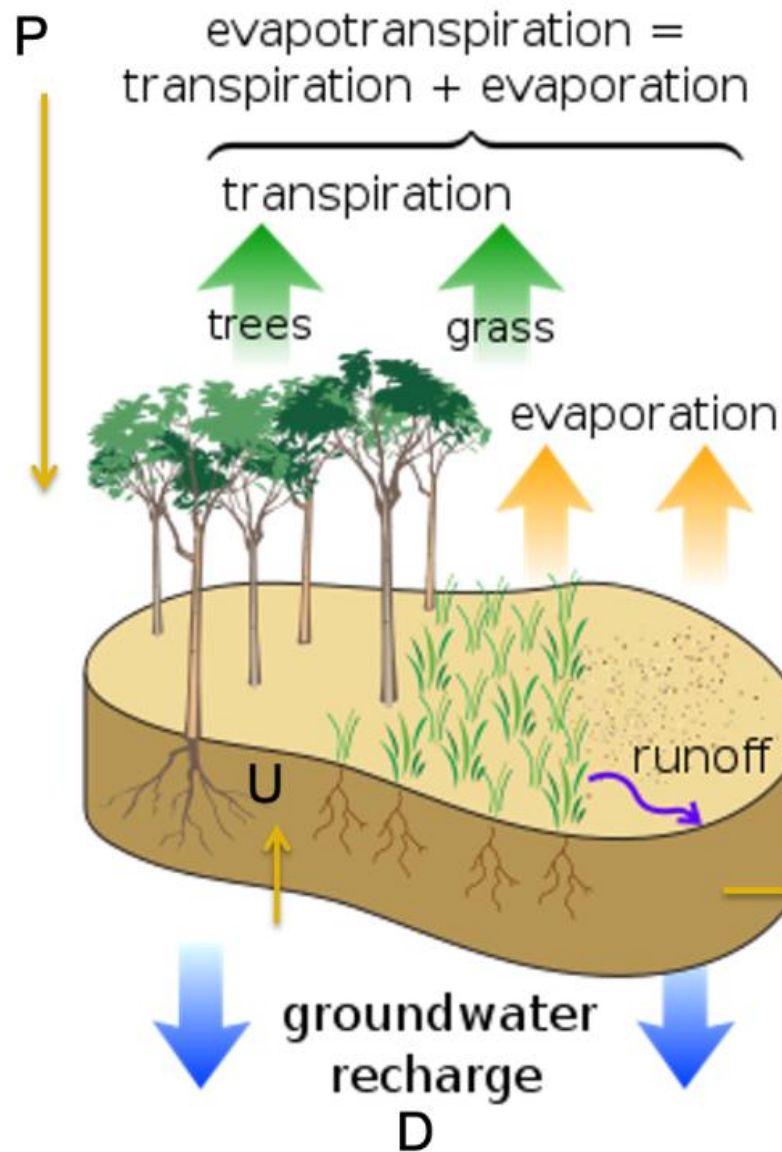
$$Q_{in} = PA$$

$$Q_{out} = EA + Q$$

P	Precipitación
A	Área
E	Evaporación
Q	Descarga del río

$$S = S_s + S_u + S_g + S_w$$

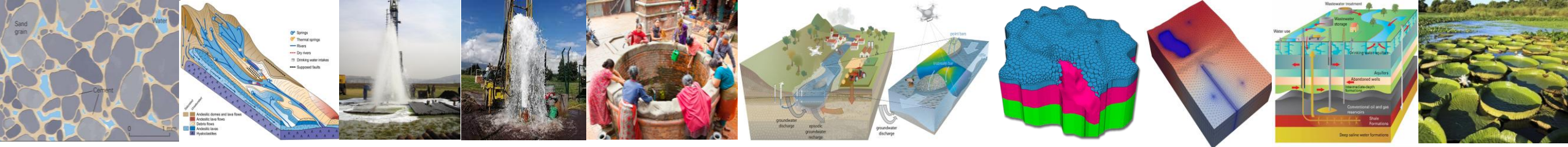
S_s	Almacenamiento en la superficie terrestre
S_u	Almacenamiento en el suelo (humedad del suelo en la zona no saturada)
S_g	Almacenamiento en zona saturada (agua subterránea)
S_w	Almacenamiento en aguas abiertas



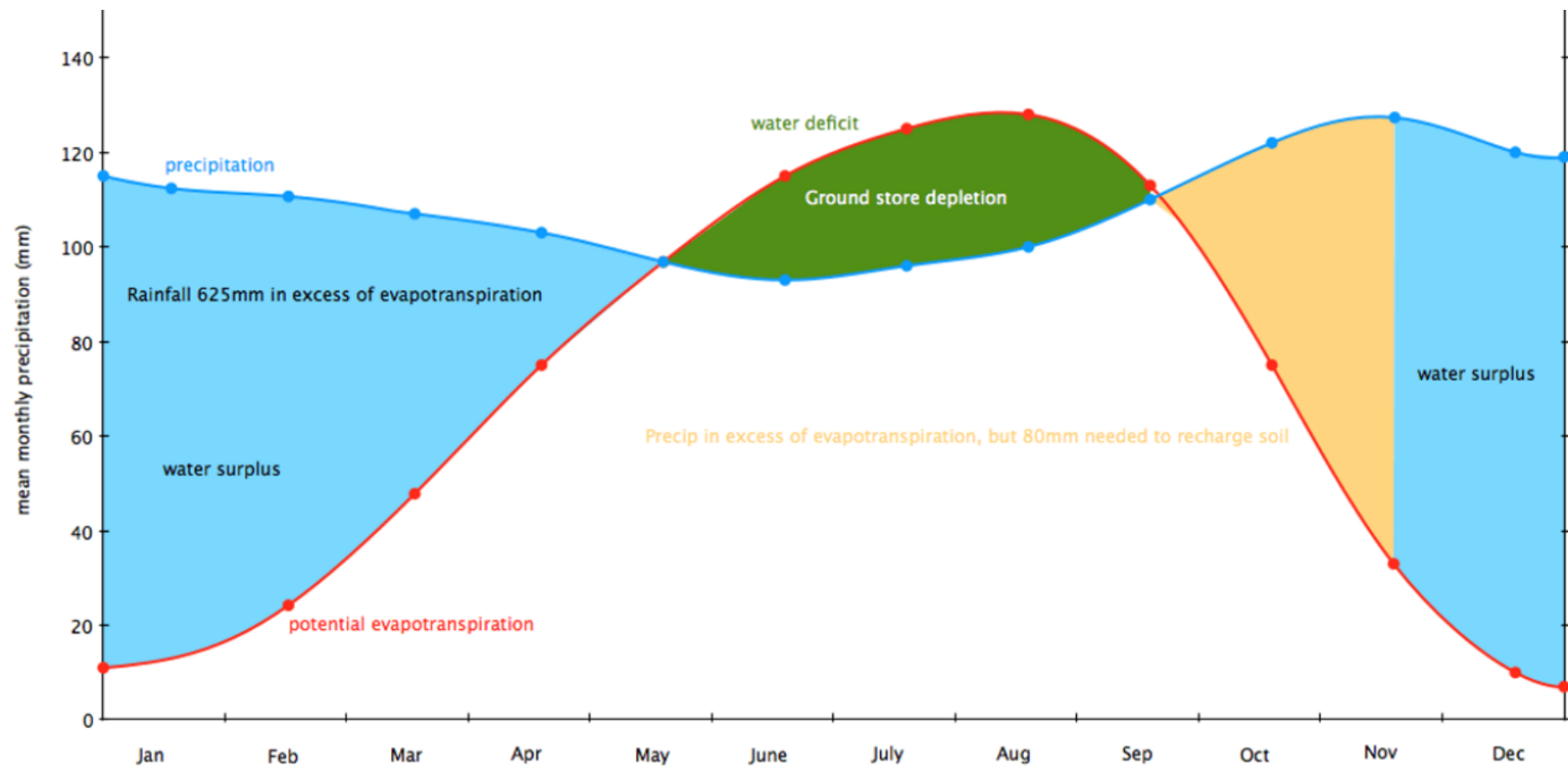
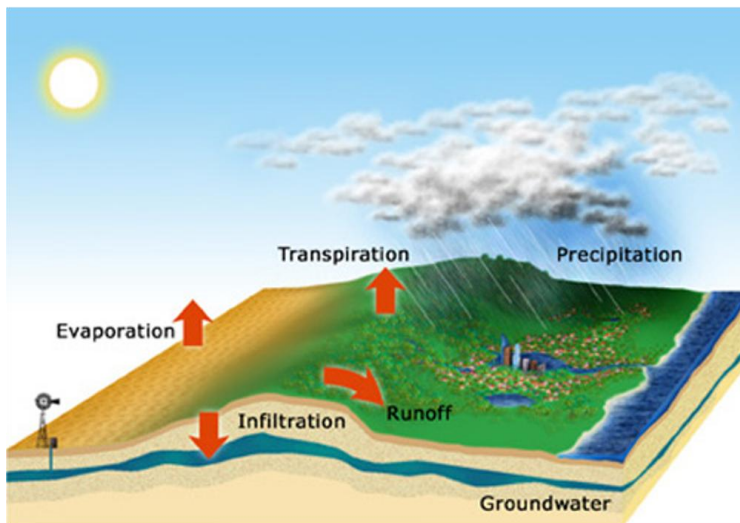
$$(P + I + U) - (ET + R + D + L) = \Delta S$$

donde:

- P = precipitación
- U = flujo capilar ascendente
- I = aportes de riego
- ET = evapotranspiración
- R = escorrentía
- D = drenaje vertical (percolación)
- L = drenaje lateral
- ΔS = cambio en el almacenamiento de agua en el suelo (almacenamiento hídrico del suelo)



Escorrentía = Precipitación – Evapotranspiración ± Cambios de almacenamiento





$$P + GW_{in} - (Q + ET + GW_{out}) = \Delta S$$

$$ET = P + GW_{in} - Q - GW_{out} - \Delta S$$

Dingman, 2002

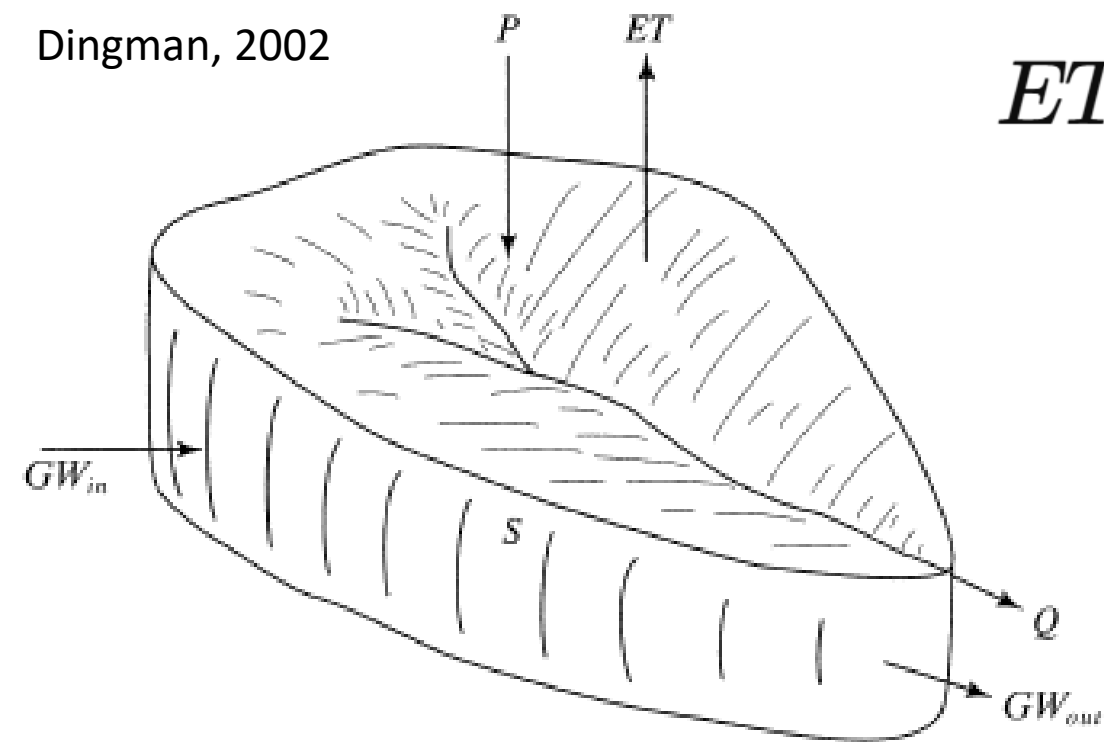


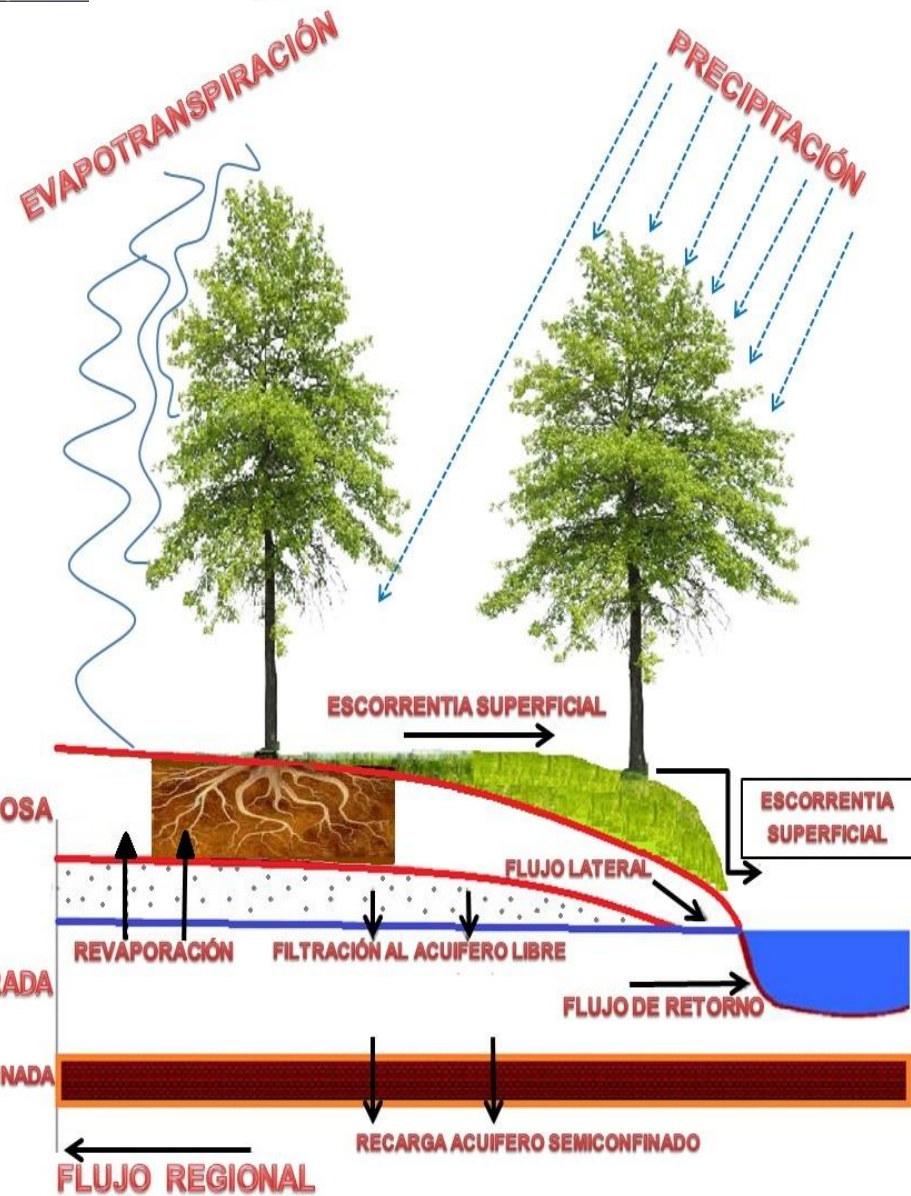
Figure 1.12 Schematic three-dimensional diagram of a watershed, showing the components of the regional water balance: P = precipitation, ET = evapotranspiration, Q = stream outflow, GW_{in} = ground-water inflow, GW_{out} = ground-water outflow.

Ecuación del Balance Hídrico

$$P + I = ET + DR + RO - \Delta W$$

Definiciones de variables

- P → Precipitación
- I → Irrigación
- ET → Evapotranspiración (suelo y planta)
- DR → Drenaje
- RO → Escorrentía superficial
- ΔW → Cambio en el almacenamiento de agua dentro del perfil del suelo (agotamiento de agua en el suelo)



Ecuación del balance hídrico

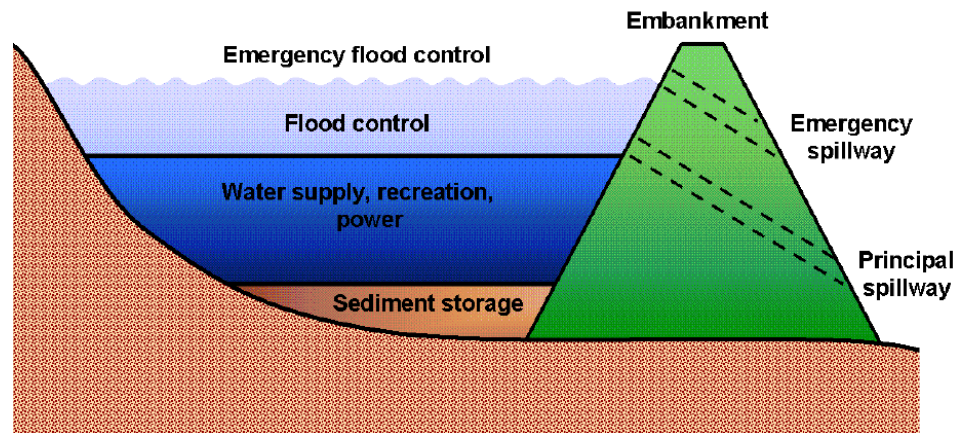
$$SW_t = SW_o + \sum (R_{day} - Q_{surf} - Ea - W_{seep} - Q_{gw})$$

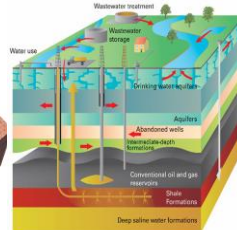
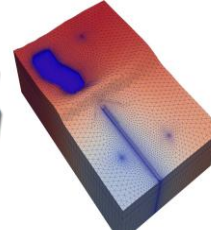
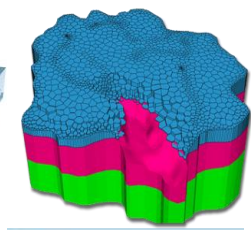
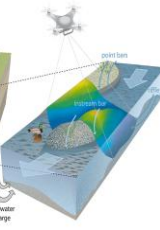
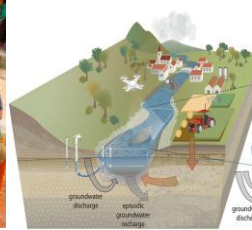
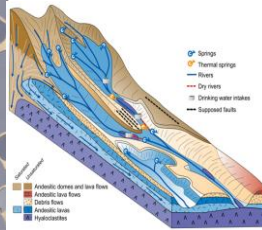
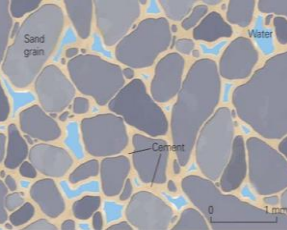
- SW_t = Contenido final de agua del suelo (soil water content) (mm)
- SW_o = Contenido Inicial de agua del suelo en el día i (mm)
- t = tiempo (días)
- R_{day} = Lluvia caída en el día i (mm)
- Q_{surf} = Esguimiento ocurrido en el día i (mm)
- Ea = Evapotranspiración en el día i (mm)
- W_{seep} = Cantidad de agua de percola del perfil de suelo a la napa freática en el día i (mm)
- Q_{gw} = Flujo subterráneo en el día i (mm).



RESERVOIRS

$$V = V_{stored} + V_{flowin} - V_{flowout} + V_{pcp} - V_{evap} - V_{seep}$$





Global Freshwater Resources Quantity and Distribution by Region

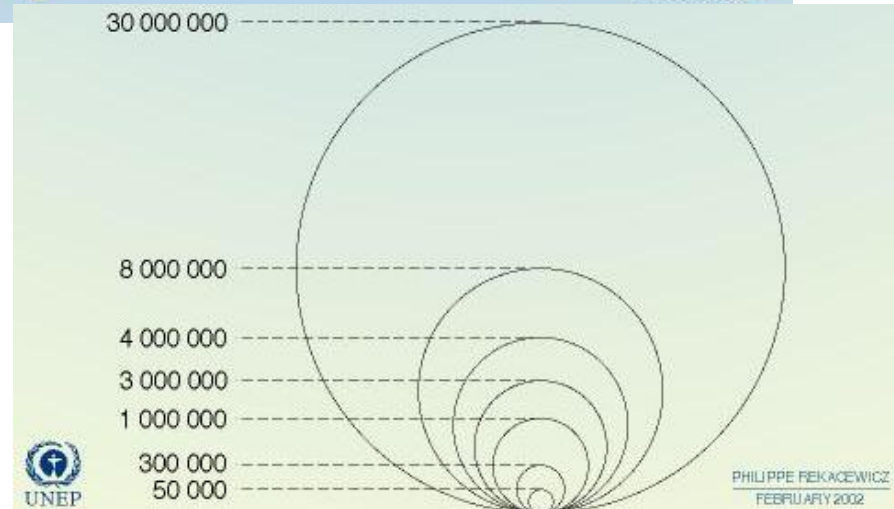
Glaciers and permanent ice caps (km³)

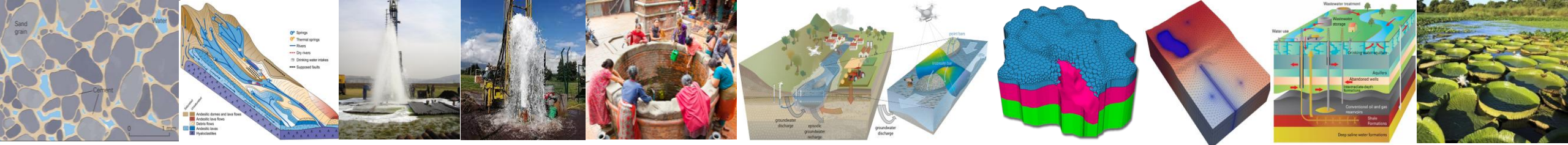


Groundwater (km³)

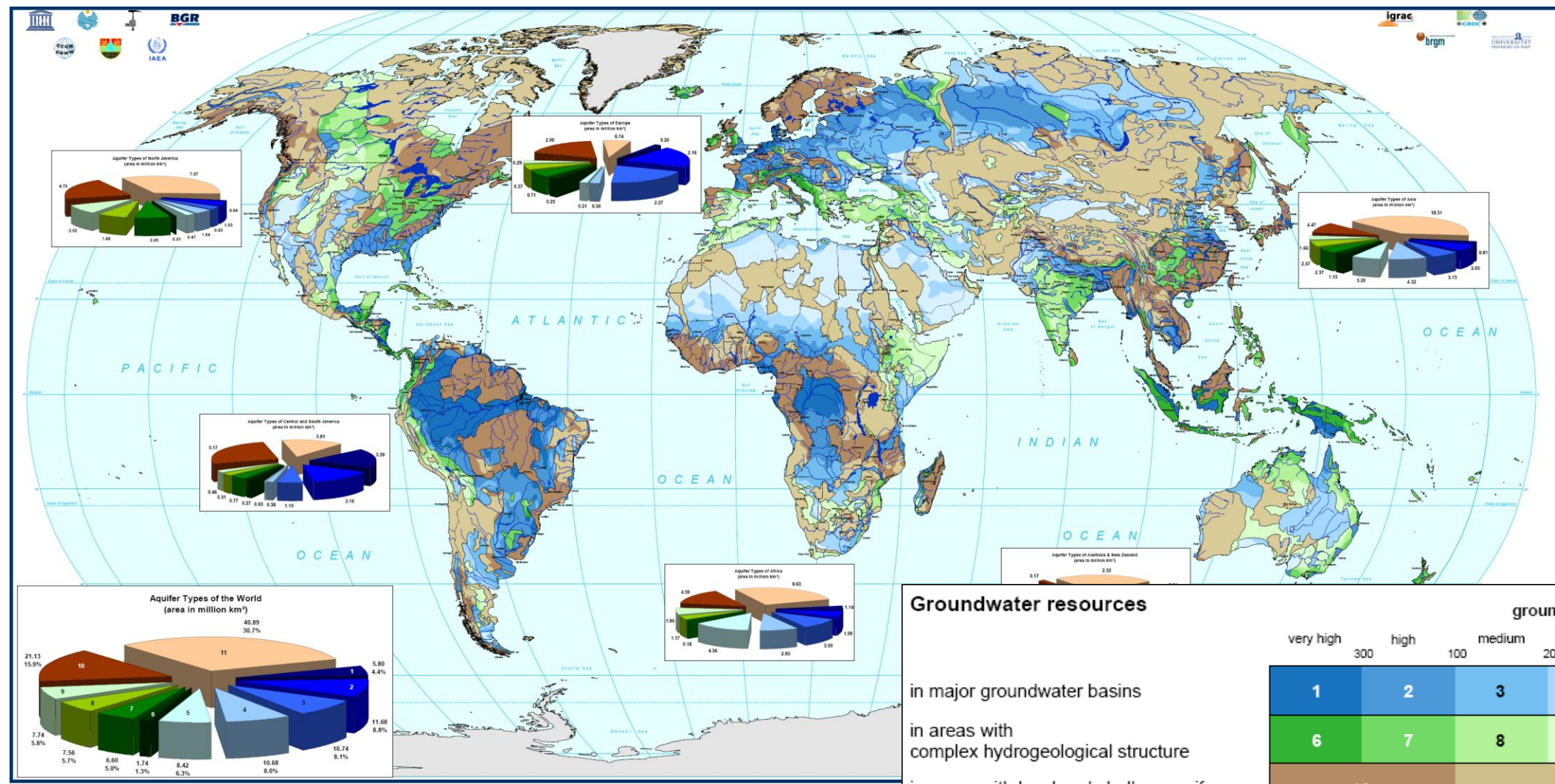


Wetlands, large lakes, reservoirs and rivers (km³)





Groundwater resources of the world

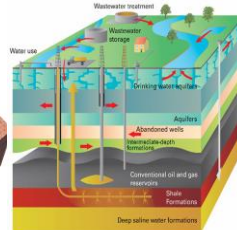
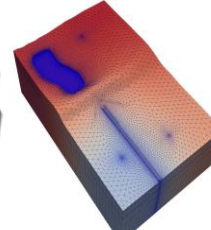
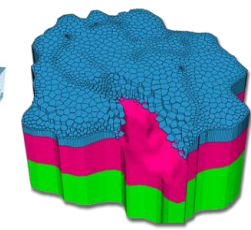
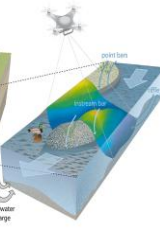
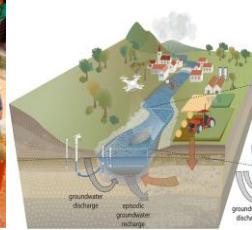
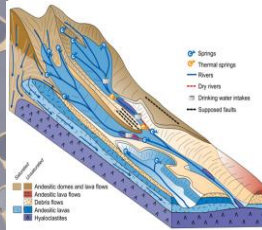
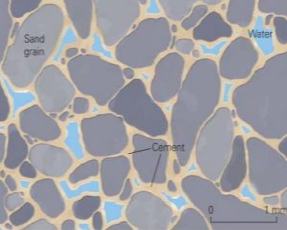


(BGR Hannover/UNESCO, 2006)

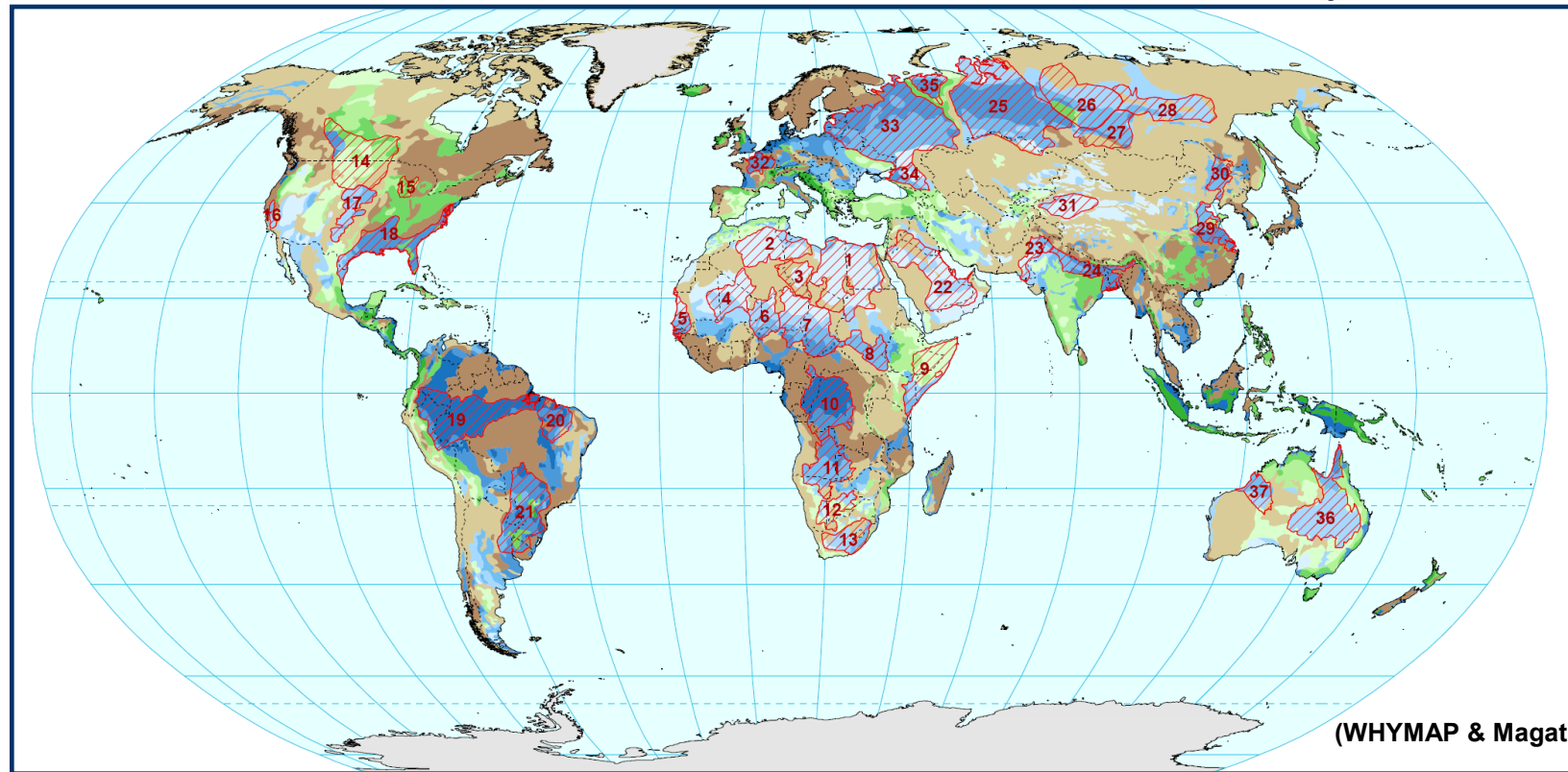
Groundwater resources

- in major groundwater basins
- in areas with complex hydrogeological structure
- in areas with local and shallow aquifers

groundwater recharge (mm/a)				
very high	high	medium	low	very low
300	100	20	2	0
1	2	3	4	5
6	7	8	9	
10			11	



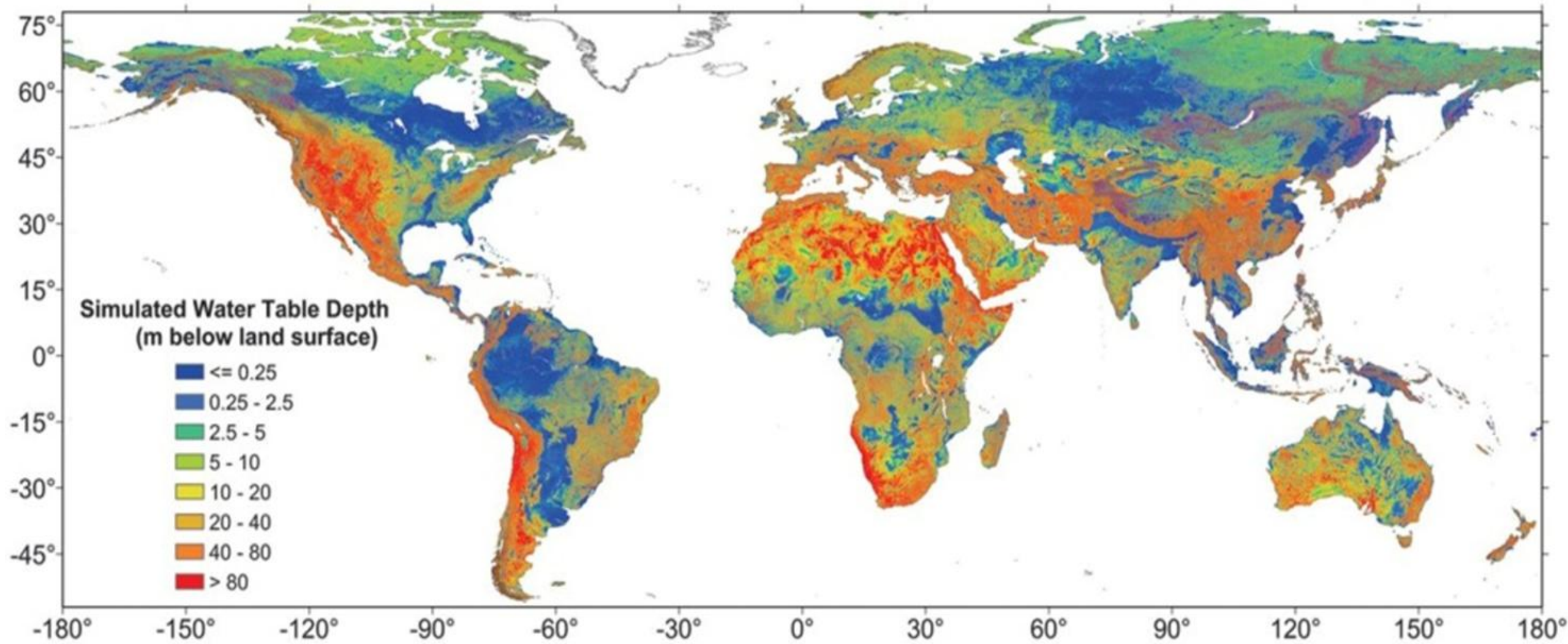
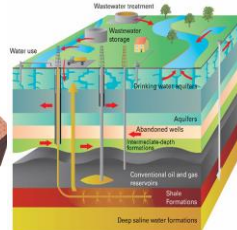
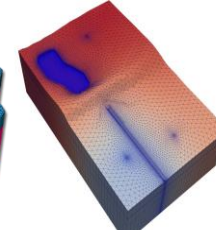
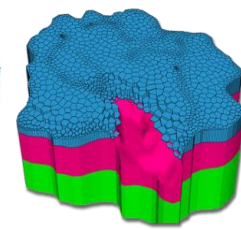
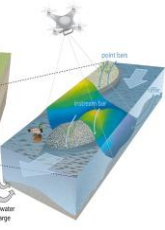
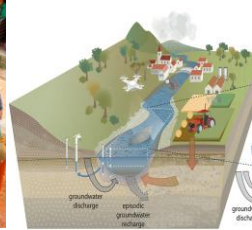
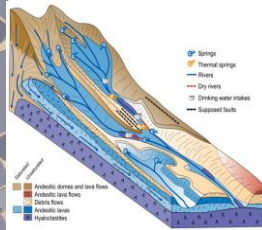
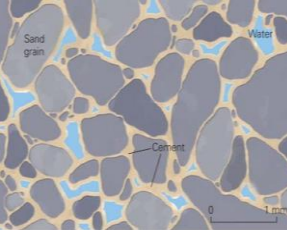
Mega-aquifers: Large aquifer systems containing very old groundwaters

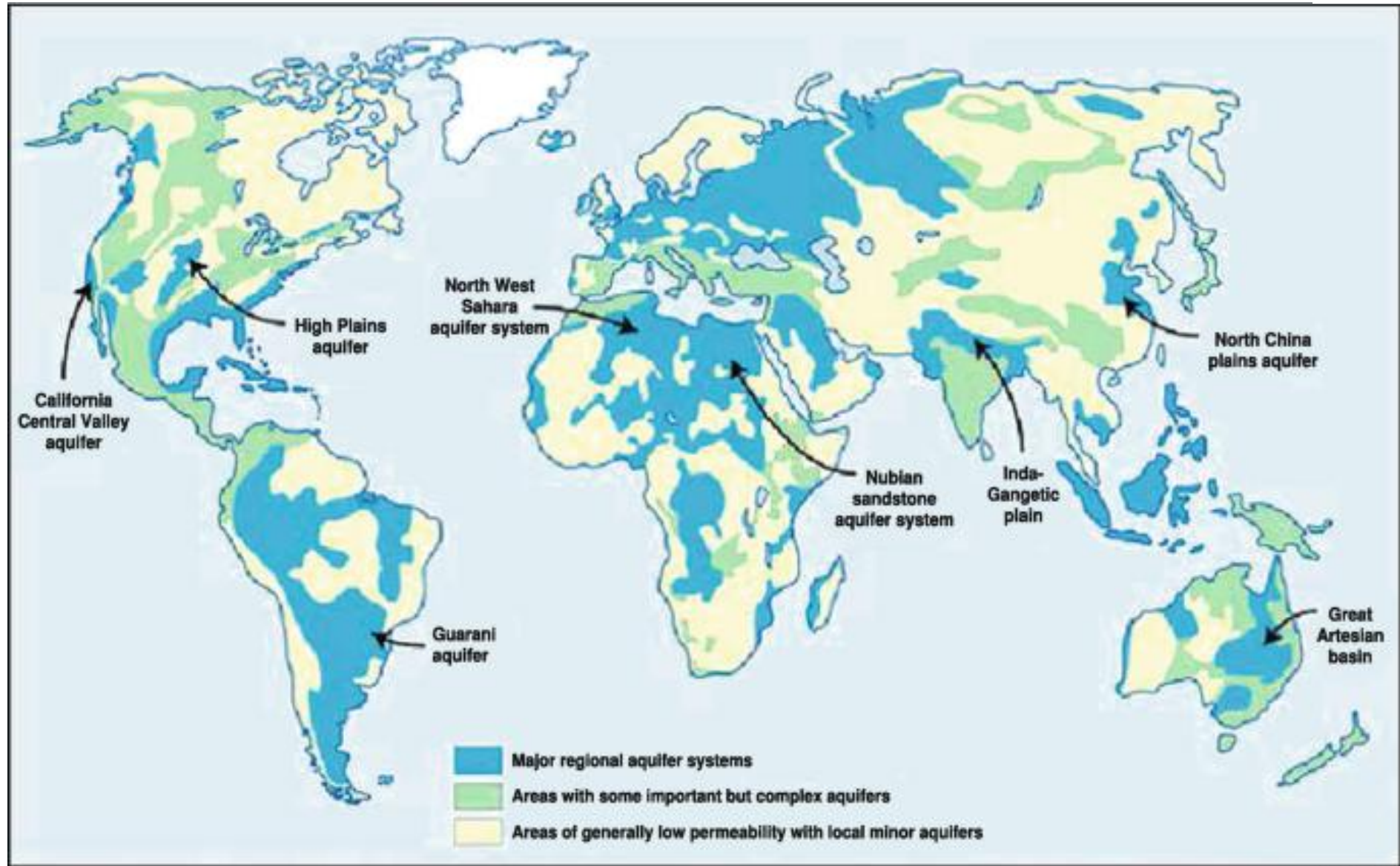


(WHYMAP & Magat, 2008)

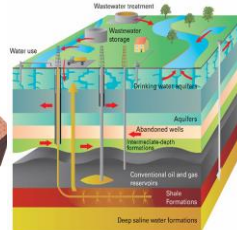
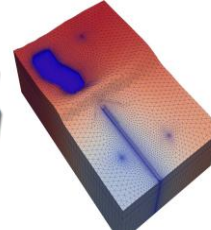
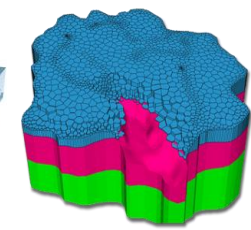
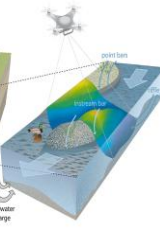
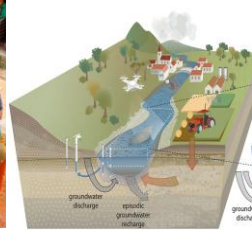
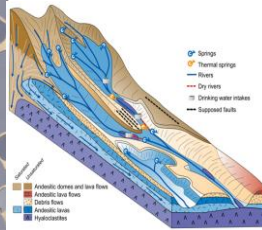
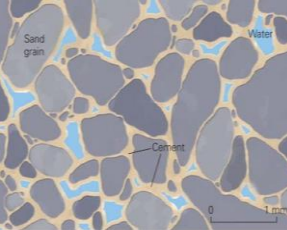
Large Aquifer Systems

- | | | | |
|--|---|--------------------------------------|----------------------------------|
| 1. Nubian Aquifer System (NAS) | 11. Northern Kalahari Basin | 21. Guarani Aquifer System | 31. Tarim Basin |
| 2. Northwest Sahara Aquifer System (NWSAS) | 12. Southeast Kalahari Basin | 22. Arabian Aquifer System | 32. Parisian Basin |
| 3. Murzuk-Djado Basin | 13. Karoo Basin | 23. Indus Basin | 33. East European Aquifer System |
| 4. Taoudeni-Tanezrouft Basin | 14. Northern Great Plains / Interior Plains Aquifer | 24. Ganges-Brahmaputra Basin | 34. North Caucasus Basin |
| 5. Senegalo-Mauritanian Basin | 15. Cambro-Ordovician Aquifer System | 25. West Siberian Artesian Basin | 35. Pechora Basin |
| 6. Iullemeden-Irhazer Aquifer System | 16. California Central Valley Aquifer System | 26. Tunguss Basin | 36. Great Artesian Basin |
| 7. Chad Basin | 17. High Plains-Ogallala Aquifer | 27. Angara-Lena Artesian Basin | 37. Canning Basin |
| 8. Sudd Basin (Umm Ruwaba Aquifer) | 18. Gulf Coastal Plains Aquifer System | 28. Yakut Basin | |
| 9. Ogaden-Juba Basin | 19. Amazonas Basin | 29. North China Plain Aquifer System | |
| 10. Congo Intracratonic Basin | 20. Maranhao Basin | 30. Songliao Basin | |

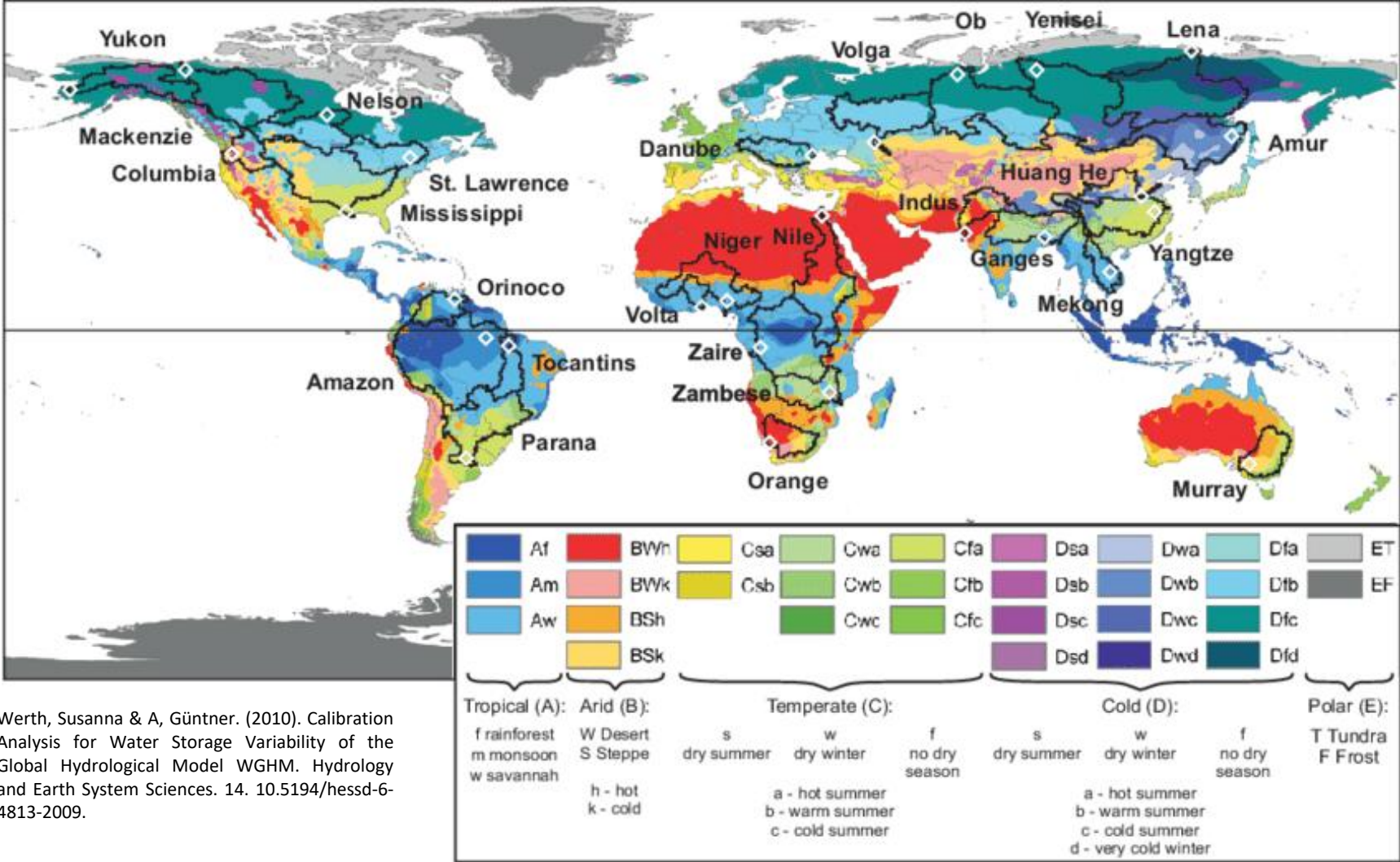




Alsharhan, A.S., Rizk, Z.E. (2020). Overview on Global Water Resources. In: Water Resources and Integrated Management of the United Arab Emirates. World Water Resources, vol 3. Springer, Cham. https://doi.org/10.1007/978-3-030-31684-6_2



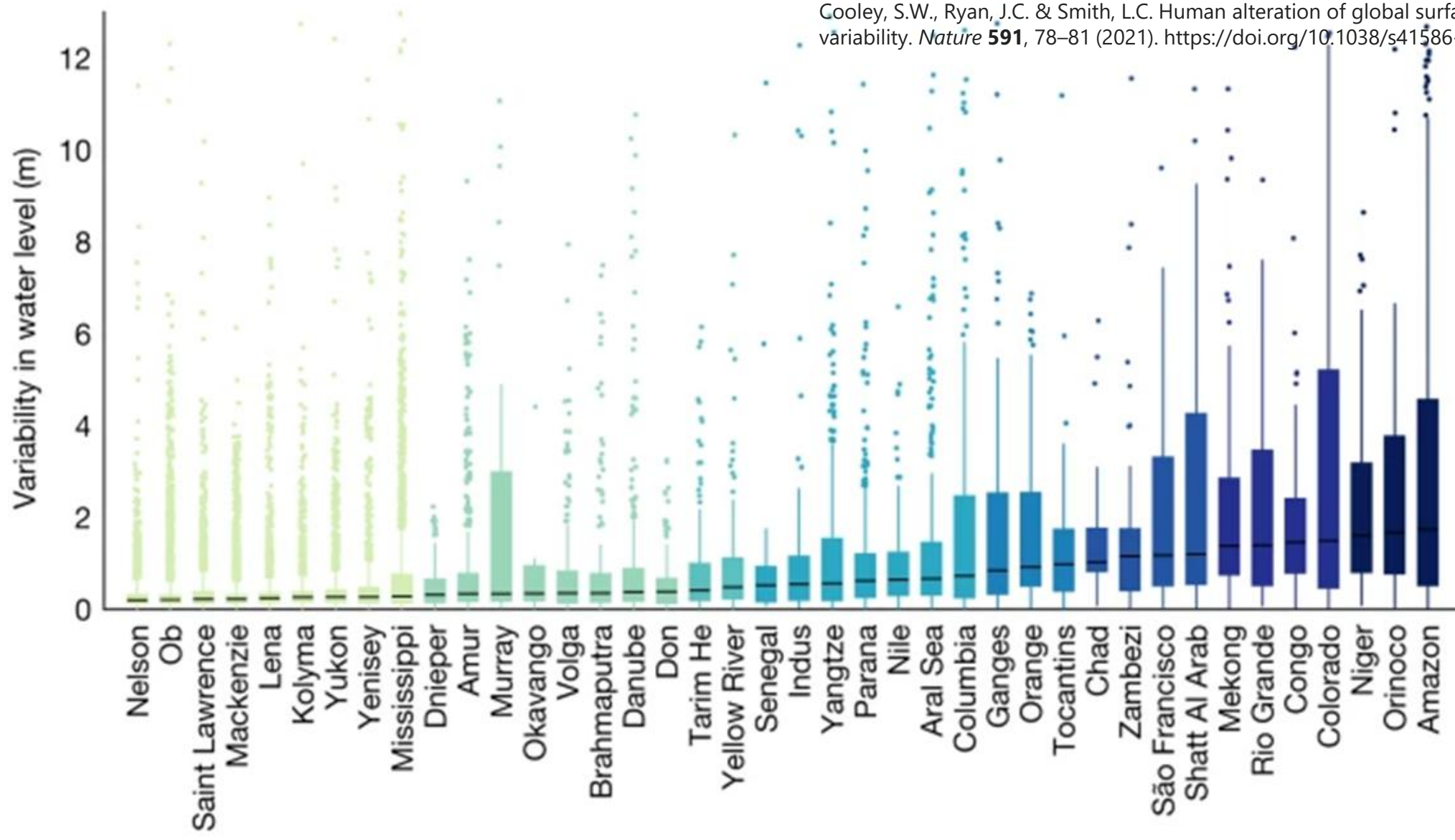
Las 28 cuencas fluviales más grandes e importantes del mundo (polígonos negros) con las zonas climáticas subyacentes de K " oppen-Geiger (para 1951-2000, por Peel et al., 2007)

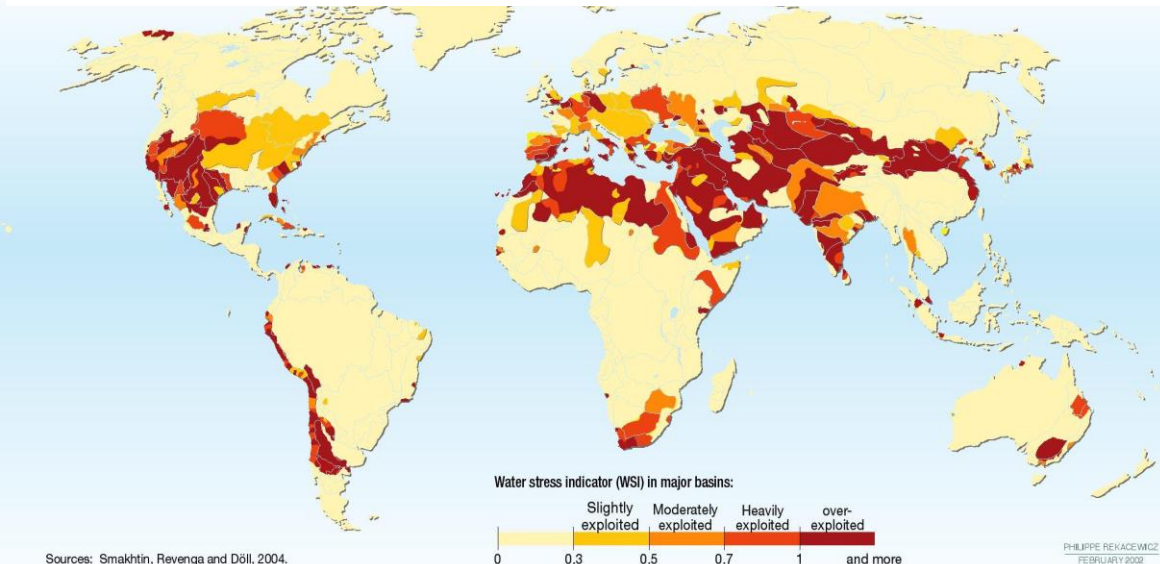
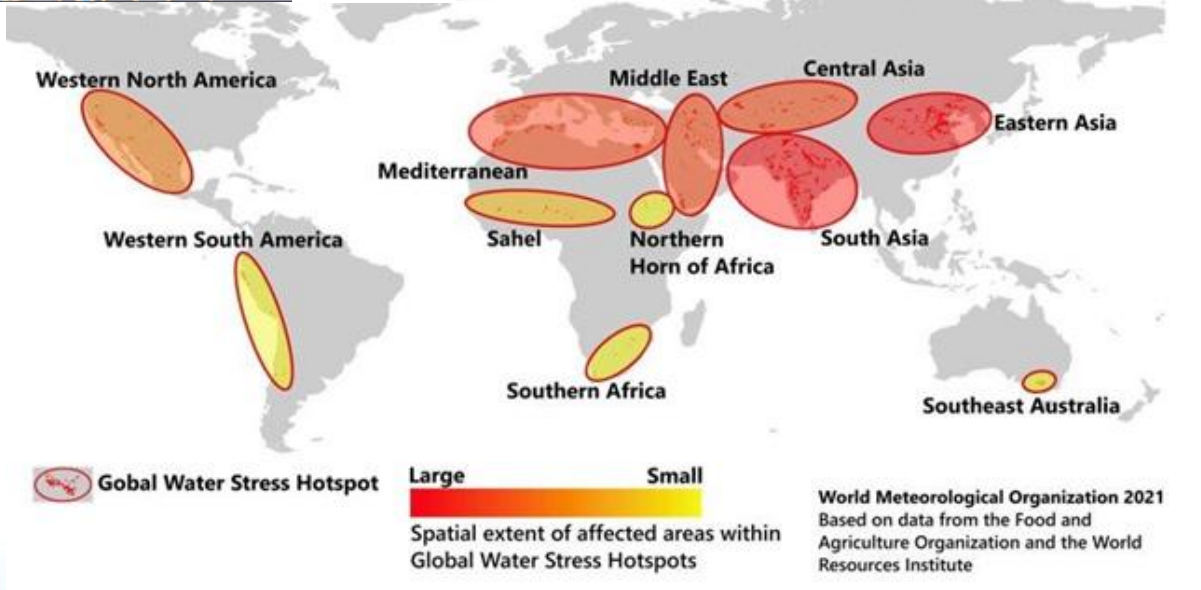


Werth, Susanna & A, Güntner. (2010). Calibration Analysis for Water Storage Variability of the Global Hydrological Model WGHM. Hydrology and Earth System Sciences. 14. 10.5194/hessd-6-4813-2009.



Cooley, S.W., Ryan, J.C. & Smith, L.C. Human alteration of global surface water storage variability. *Nature* **591**, 78–81 (2021). <https://doi.org/10.1038/s41586-021-03262-3>





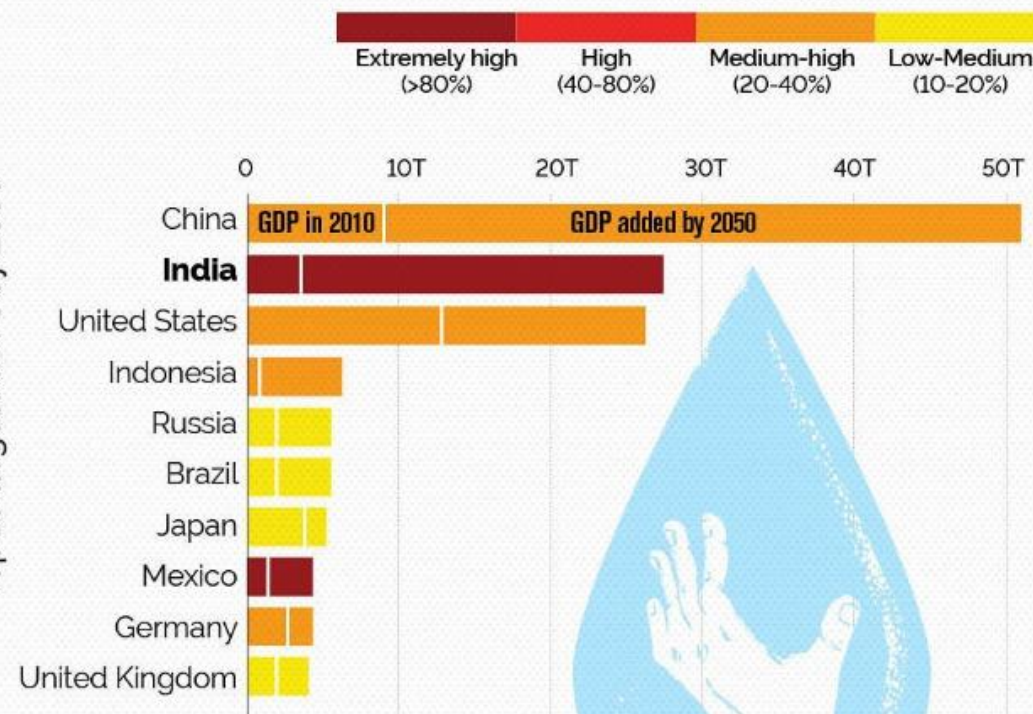
Sources: Smakhtin, Revenga and Döll, 2004.

THE ECONOMIC IMPACT OF GLOBAL WATER STRESS

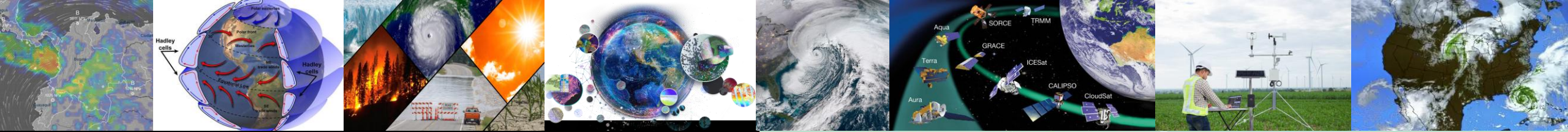
According to data from Aqueduct, 31% of global GDP — a whopping \$70 trillion — will be exposed to high water stress by 2050

PROJECTED GDP AND WATER STRESS

Top 10 largest GDPs by 2050



SOURCE: WORLD RESOURCES INSTITUTE, AQUEDUCT



WHERE THE WATER CRISIS IS AT ITS WORST



These 25 countries — housing one-quarter of the global population — are currently exposed to extremely high water stress annually.

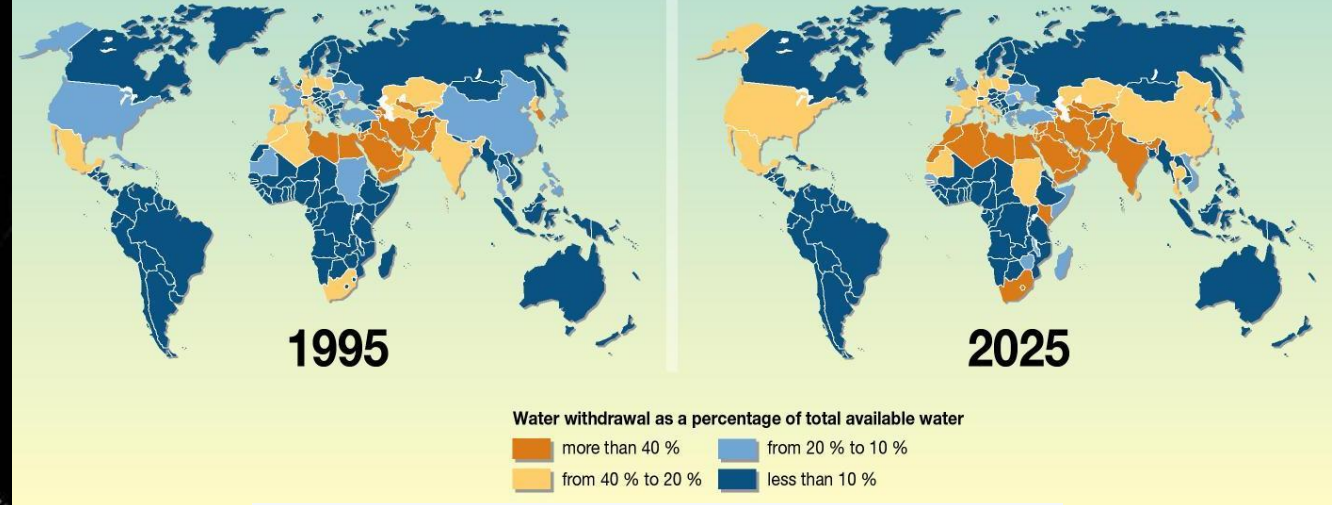
1. Bahrain
2. Cyprus
3. Kuwait
4. Lebanon
5. Oman
6. Qatar
7. United Arab Emirates
8. Saudi Arabia
9. Israel
10. Egypt
11. Libya
12. Yemen
13. Botswana
14. Iran
15. Jordan
16. Chile
17. San Marino
18. Belgium
19. Greece
20. Tunisia
21. Namibia
22. South Africa
23. Iraq
24. India
25. Syria

Network 18 creative



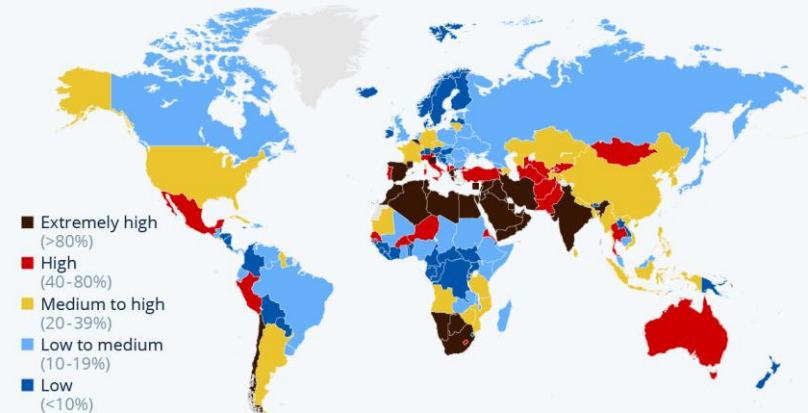
Water stress or scarcity occurs when the demand for safe, usable water in a given area exceeds the available amount during a certain period. **This map shows countries where more users are competing for limited supply of clean, usable water.**

SOURCE: WORLD RESOURCES INSTITUTE, AQUEDUCT



Where Water Stress Will Be Highest by 2050

Projected ratio of human water demand to water availability (water stress level) in 2050*

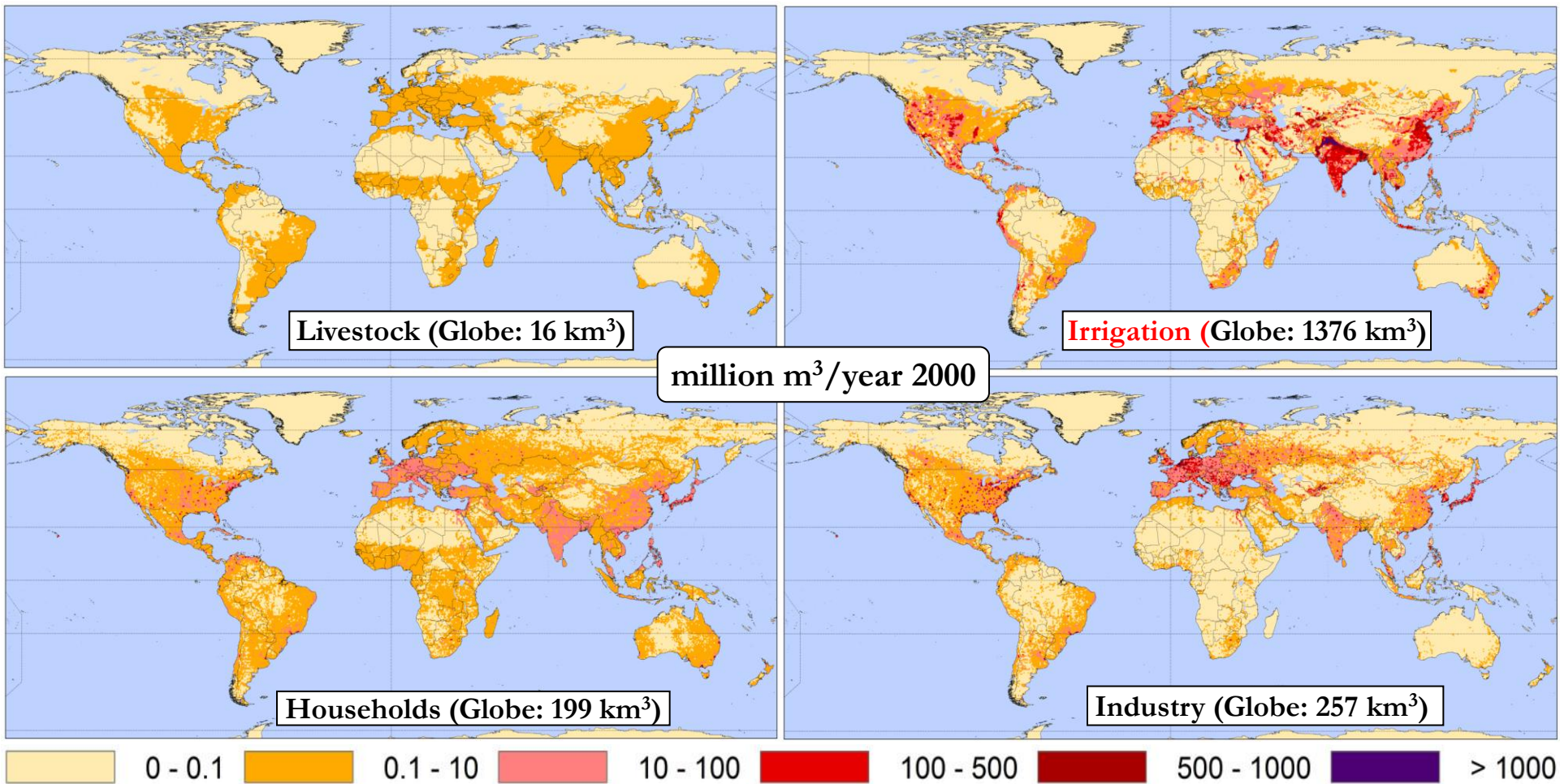


* According to "business as usual" scenario=middle-of-the-road future where temperatures increase by 2.8°C to 4.6°C by 2100

Source: World Resources Institute



Global demand for surface water and groundwater

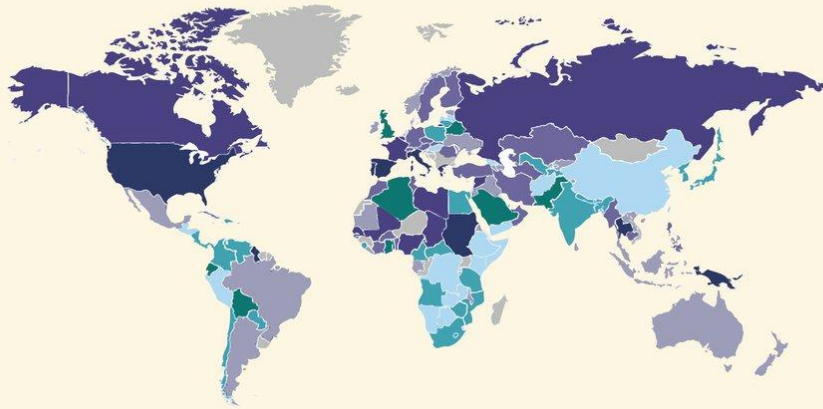


Agriculture = 74%

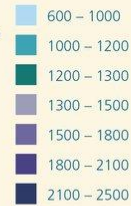
Wada et al., HESS, 2011



World Water Use

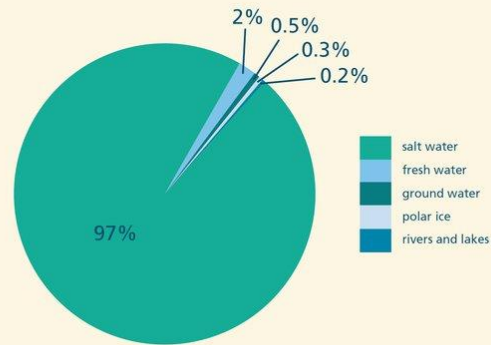


Water footprint per capita (m³)



global average water footprint is around
1,240
m³ per person

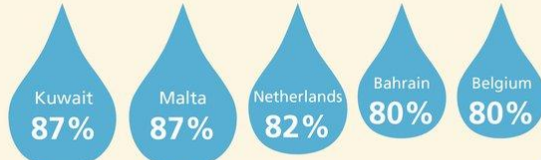
Breakdown of the world's water types



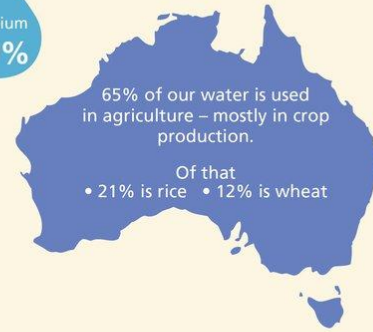
Highest water footprint per capita



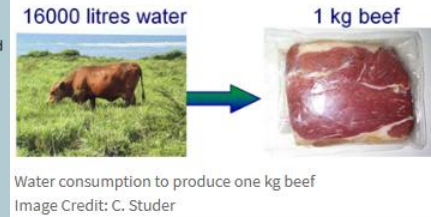
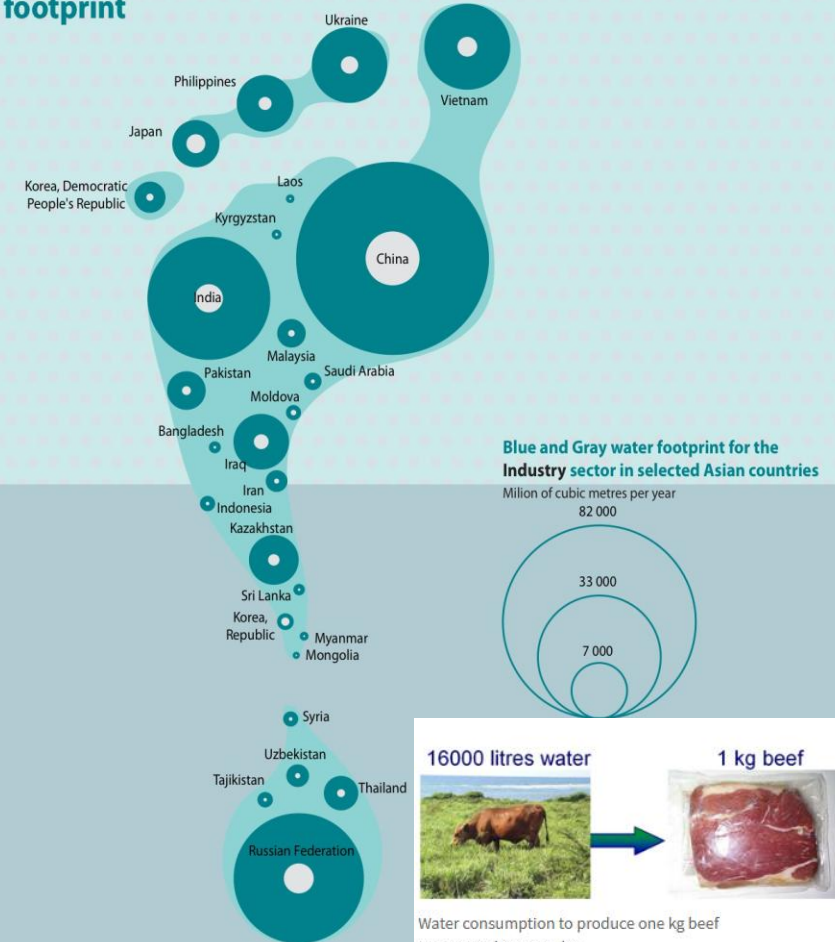
Countries most dependent on water imports



Water footprint of different foods

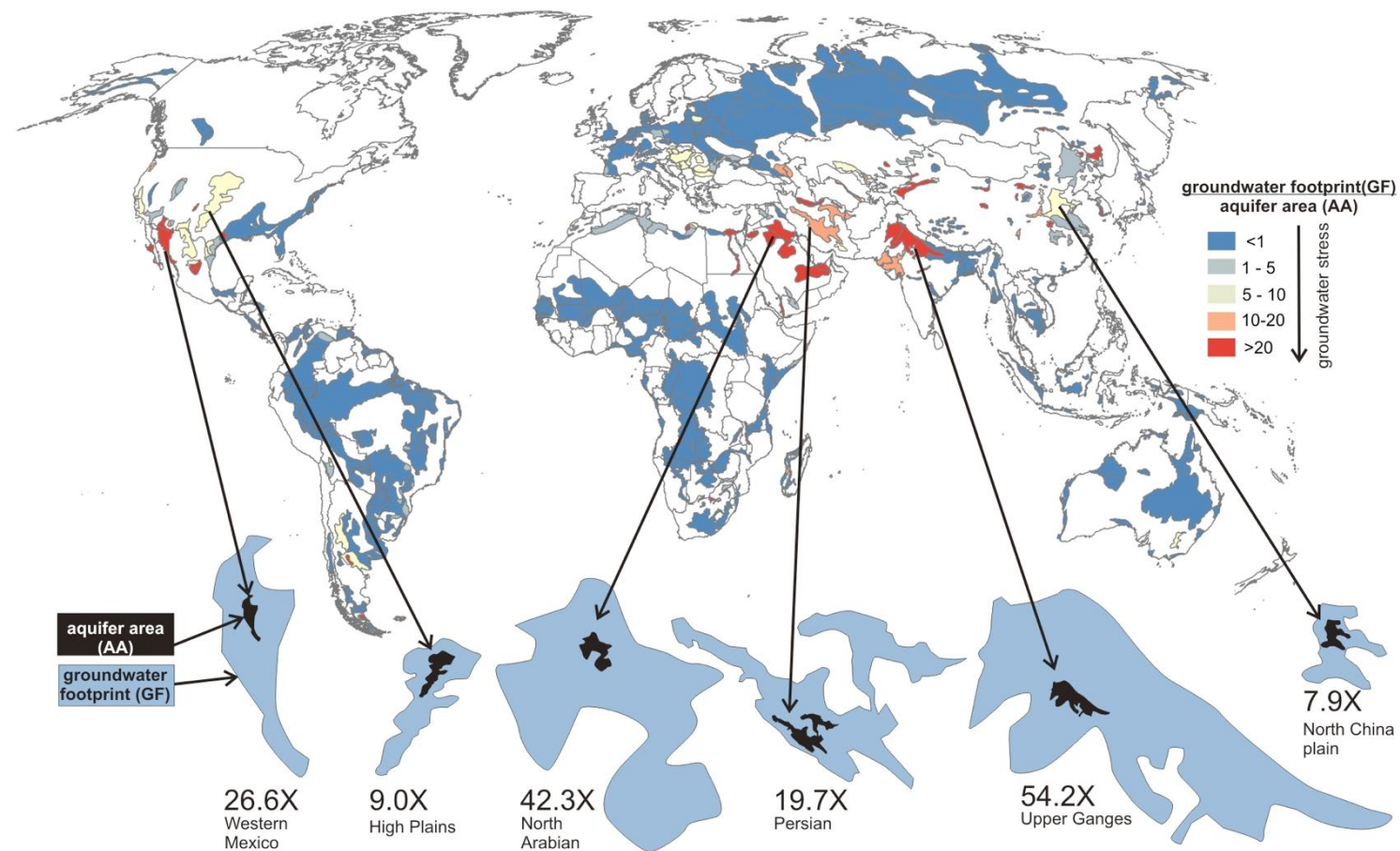


Water footprint





Groundwater footprint of major aquifers

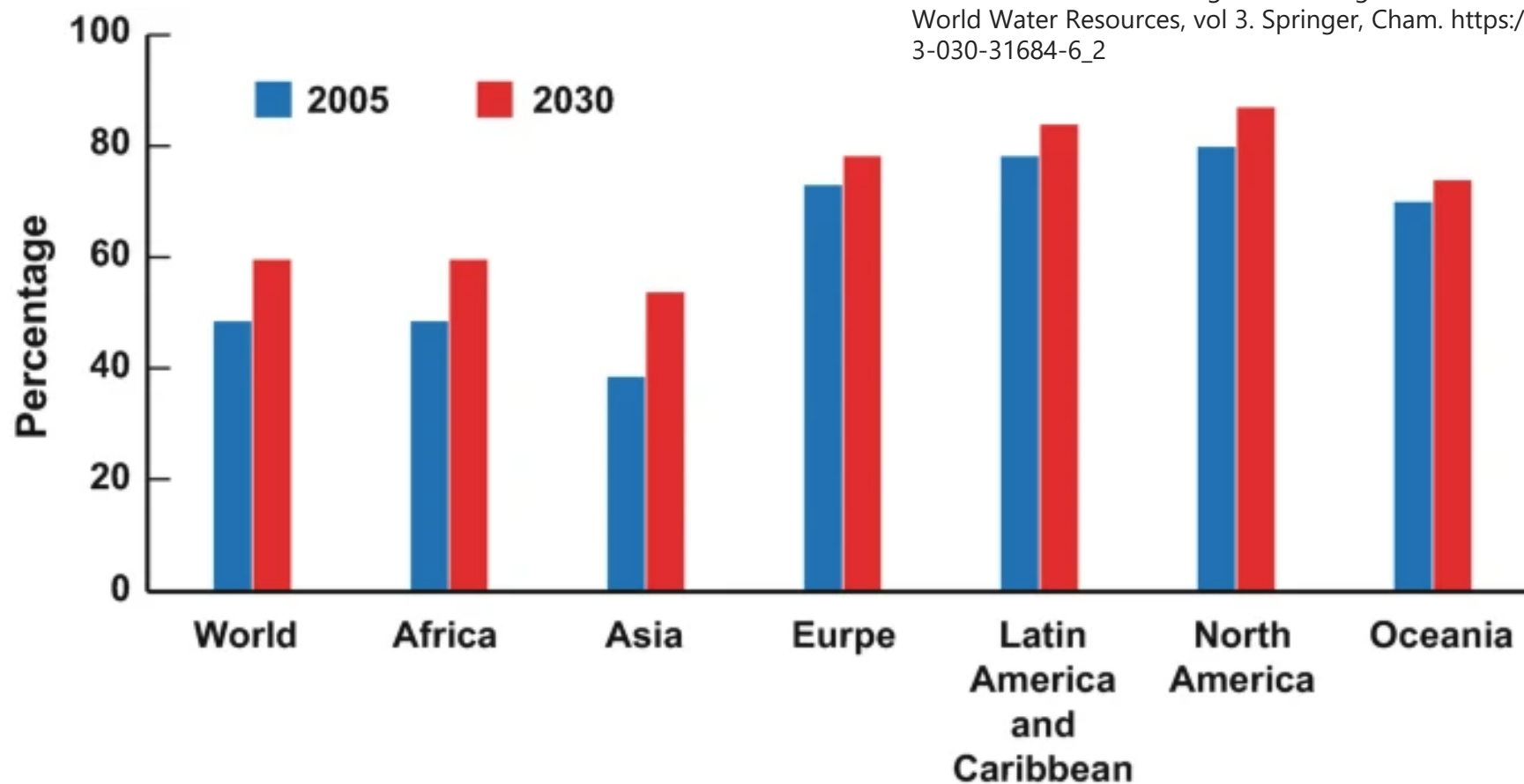


Gleeson, et al., Nature, 2012.

Global groundwater footprint = 3.5 times the global area of productive aquifers



Alsharhan, A.S., Rizk, Z.E. (2020). Overview on Global Water Resources. In: Water Resources and Integrated Management of the United Arab Emirates. World Water Resources, vol 3. Springer, Cham. https://doi.org/10.1007/978-3-030-31684-6_2



The growth of the percentage of people living in urban areas around the globe in 2005 and 2030. (After UN 2007)



Need for better assessment and management of Surface and groundwater resources

Pressure on groundwater resources

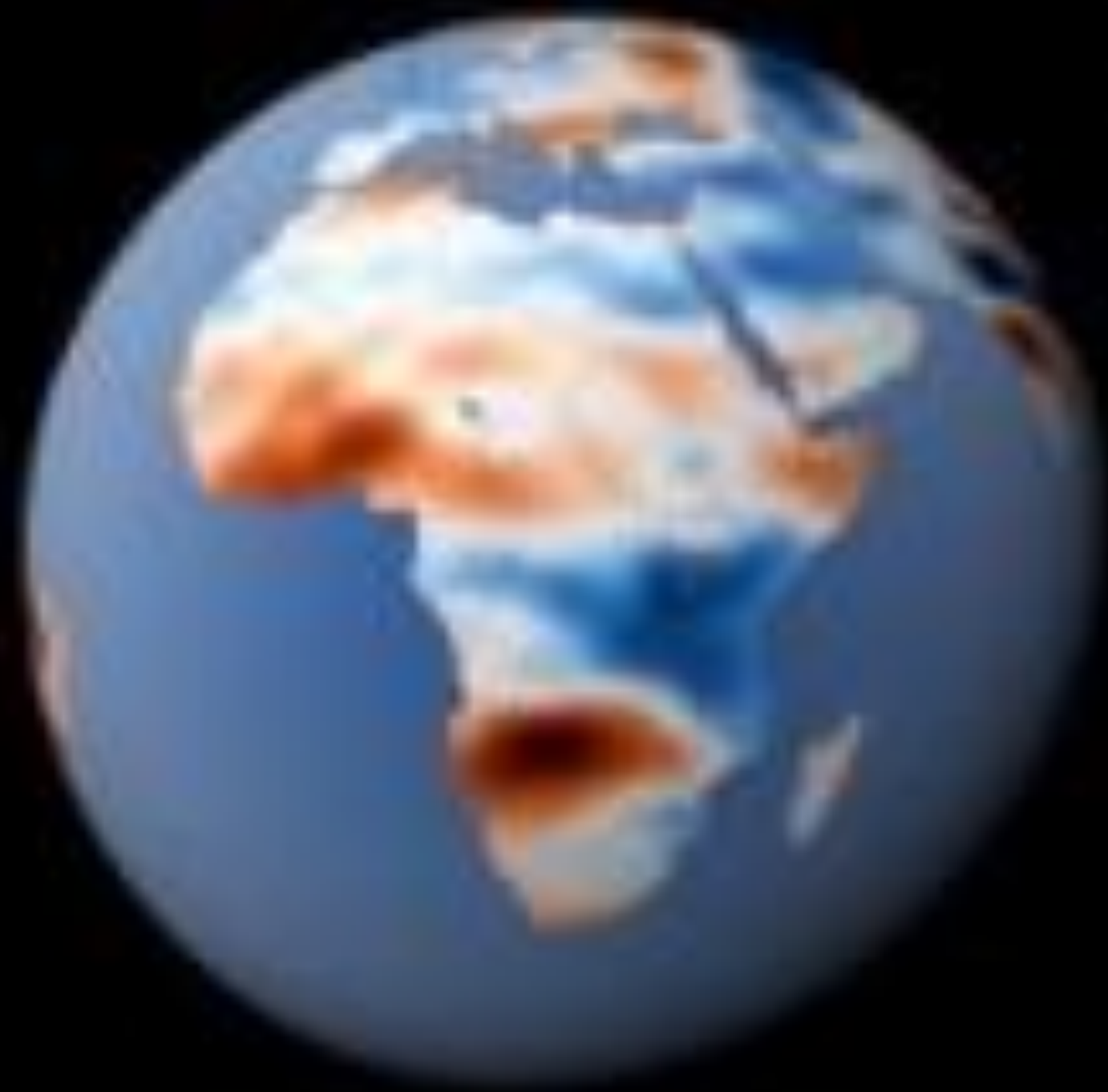
- ☐ **INCREASING DEMAND**
- ☐ **OVER-EXPLOITATION (impact on quality)**
- ☐ **POLLUTION OF SURFACE WATERS**
- ☐ **CLIMATE CHANGE**

Better management of groundwater resources means:

- ☐ **BETTER KNOWLEDGE OF AQUIFER SYSTEMS !!!**
- ☐ **GOOD LEGISLATION**
- ☐ **EFFICIENT ORGANIZATION**
- ☐ **FUNDING ...**

While the intensity of groundwater use, partly encouraged by subsidized rural electrification, has led to the emergence of many groundwater-dependent economies, their future is now threatened by aquifer depletion and pollution. Prospects for relaxing use of these key aquifers, remediating water quality and restoring groundwater services to ecosystems look remote unless alternative management approaches are developed.

Jun 21, 2002





PRECIPITACIONES EN EL MUNDO
SURAMÉRICA Y COLOMBIA

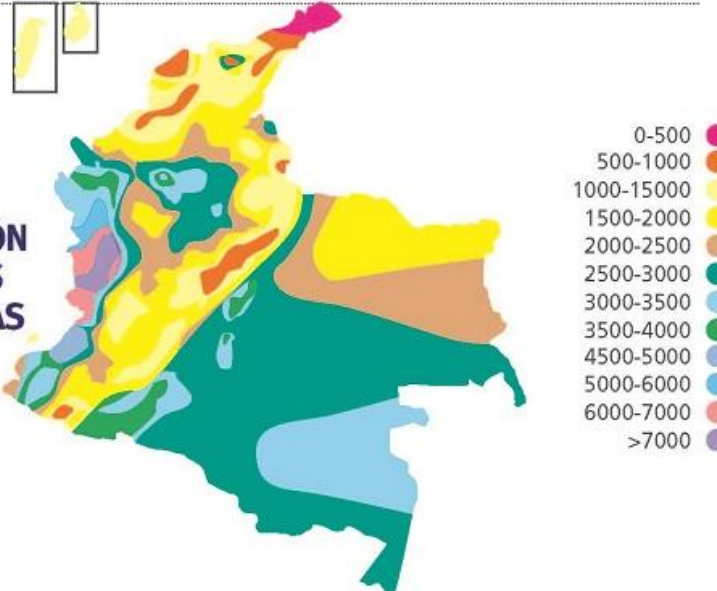
Fuente: Balance Hídrico Mundial y recursos hídricos de la tierra. Instituto de Hidrología UNE

BALANCE HÍDRICO GLOBAL

MUNDIAL	SUR AMÉRICA	COLOMBIA
Precipitación 900 mm / año	Precipitación 1.600 mm / año	Precipitación 3.000 mm / año
Escorrentía 47.000 Km ³ / año	Escorrentía 11.668 Km ³ / año	Escorrentía 2.112 Km ³ / año
64% Evaporación 580 mm / año	58% Evaporación 940 mm / año	38% Evaporación 1.150 mm / año
101/S Km ²	211/S Km ²	581/S Km ²

GRÁFICO 2.
VALORES DE
PRECIPITACIÓN
EN REGIONES
COLOMBIANAS

Fuente: IDEAM



- 0-500
- 500-1000
- 1000-1500
- 1500-2000
- 2000-2500
- 2500-3000
- 3000-3500
- 3500-4000
- 4500-5000
- 5000-6000
- 6000-7000
- >7000

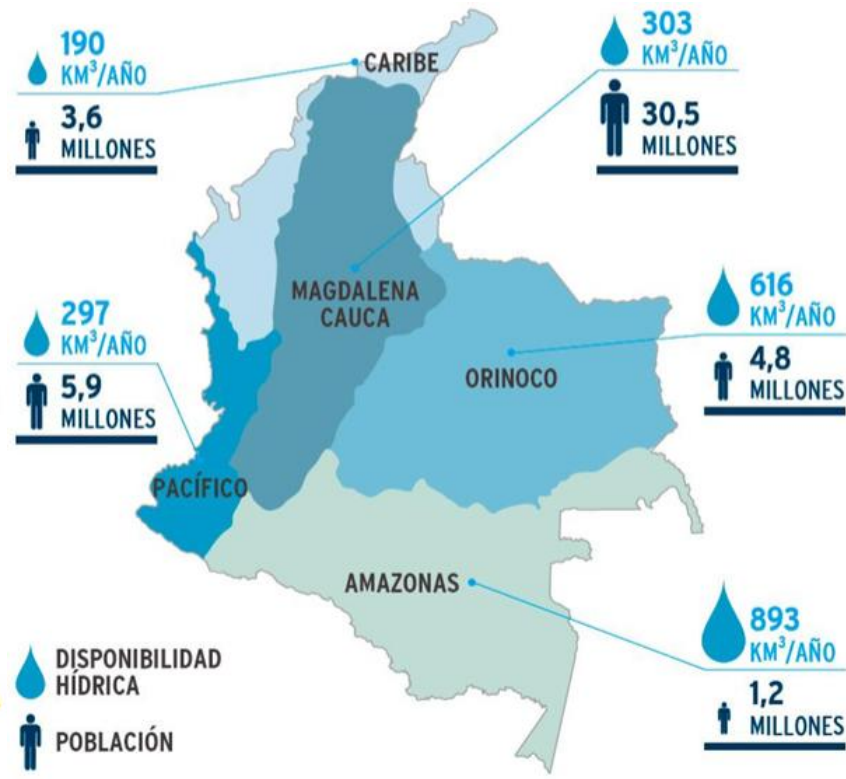


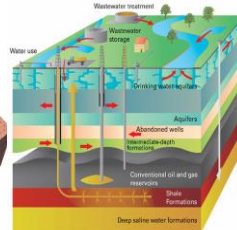
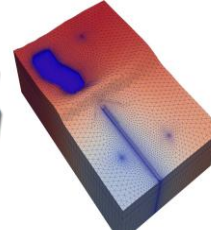
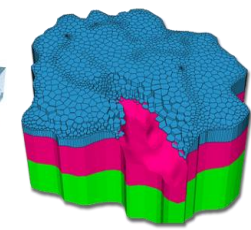
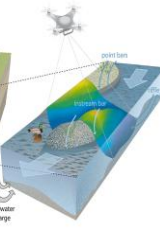
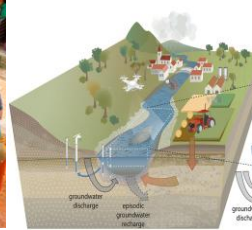
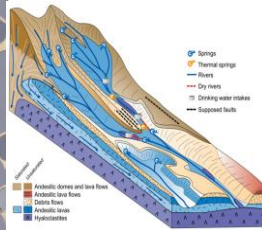
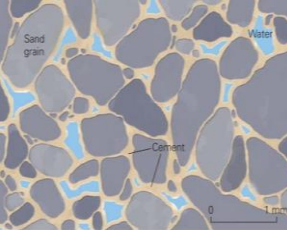
25%



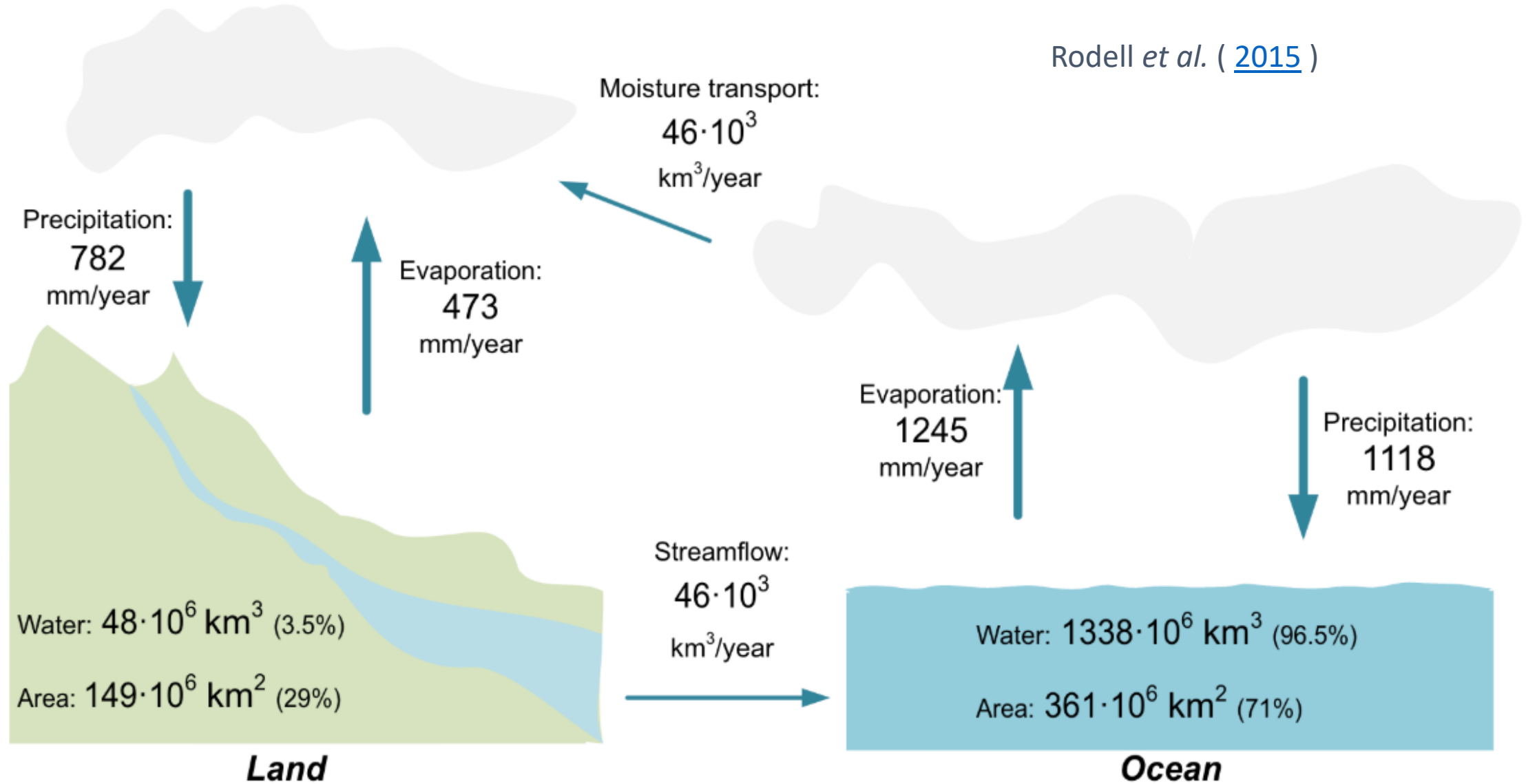
75%
5.848 kilómetros cúbicos
distribuidos en el 75%
del país

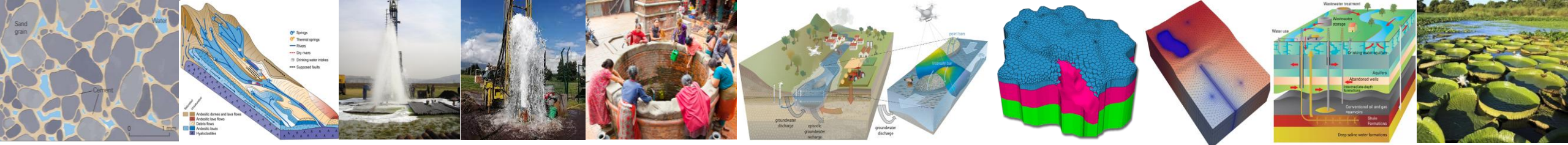
Disponibilidad hídrica en Colombia



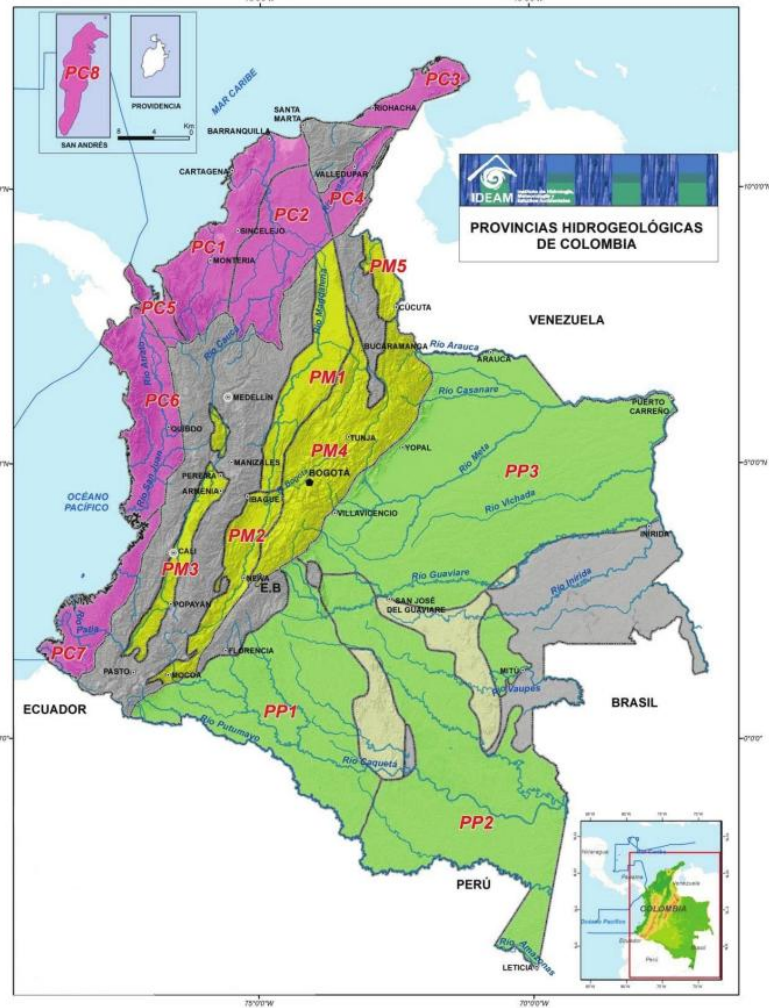


Rodell *et al.* ([2015](#))

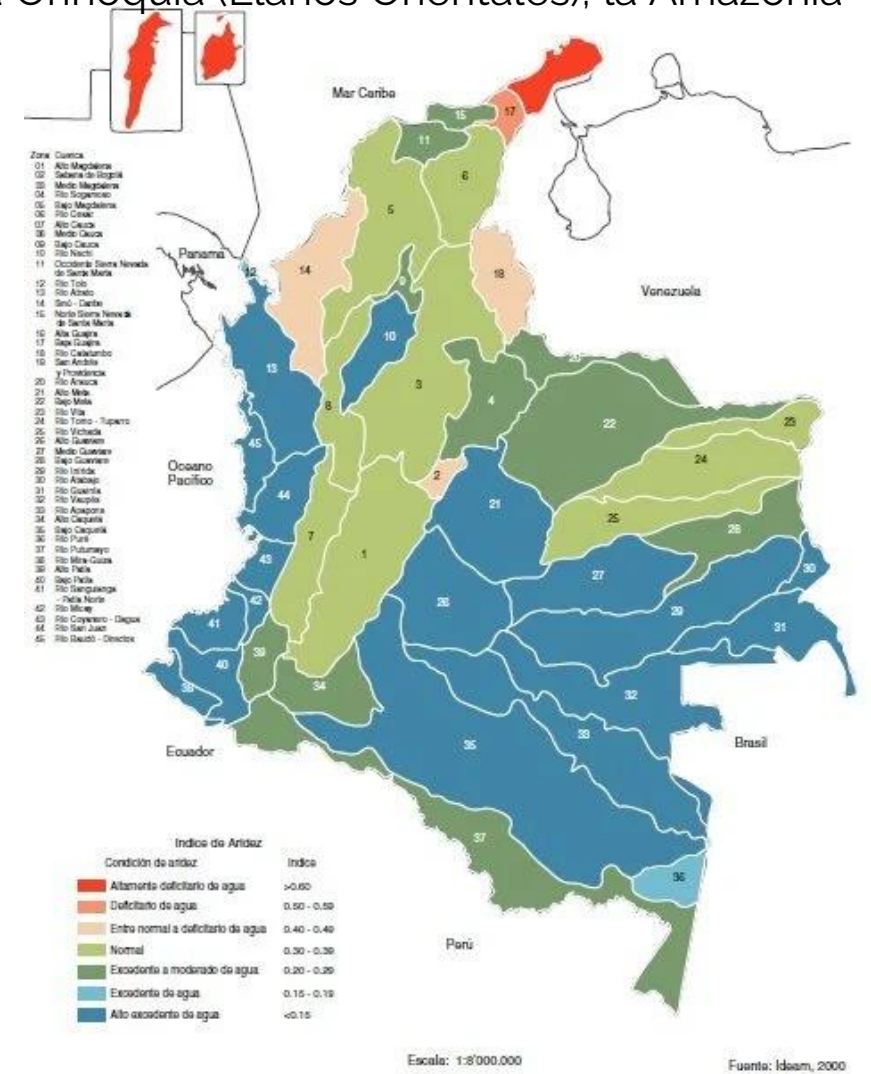




Se han identificado en total 16 provincias hidrogeológicas, las cuales cubren el 74% de la extensión total del territorio nacional. Sin embargo, el 51% de esta superficie corresponde a las cuencas de la Orinoquía (Llanos Orientales), la Amazonía (Vaupés-Amazonas y Caguán-Putumayo) y la costa pacífica (Tumaco y Chocó).



PROVINCIAS HIDROGEOLÓGICAS DE COLOMBIA	
MONTANAS E INTRAMONTANAS	
PM1	Valle Medio del Magdalena
PM2	Valle Alto del Magdalena
PM3	Cauca Patia
PM4	Cordillera Oriental
PM5	Catatumbo
COSTERAS	
PC1	Sinú - San Jacinto
PC2	Valle Bajo del Magdalena
PC3	Guajira
PC4	César - Ranchería
PC5	Urabá
PC6	Chocó
PC7	Tumaco
PC8	Isla de San Andrés
PERICRATÓNICAS	
PP1	Caguán - Putumayo
PP2	Vaupés - Amazonas
PP3	Llanos Orientales





THE KEY QUESTION:

WHY RIVERS FLOW WHEN IT DOES NOT RAIN?



El agua subterránea y el agua superficial interactúan a lo largo del paisaje, como se muestra en el dibujo adyacente. El paisaje conceptual muestra, de manera simplificada, la interacción del agua subterránea con todos los tipos de agua superficial, como arroyos, lagos y humedales, en muchos terrenos diferentes, desde las montañas hasta los océanos.



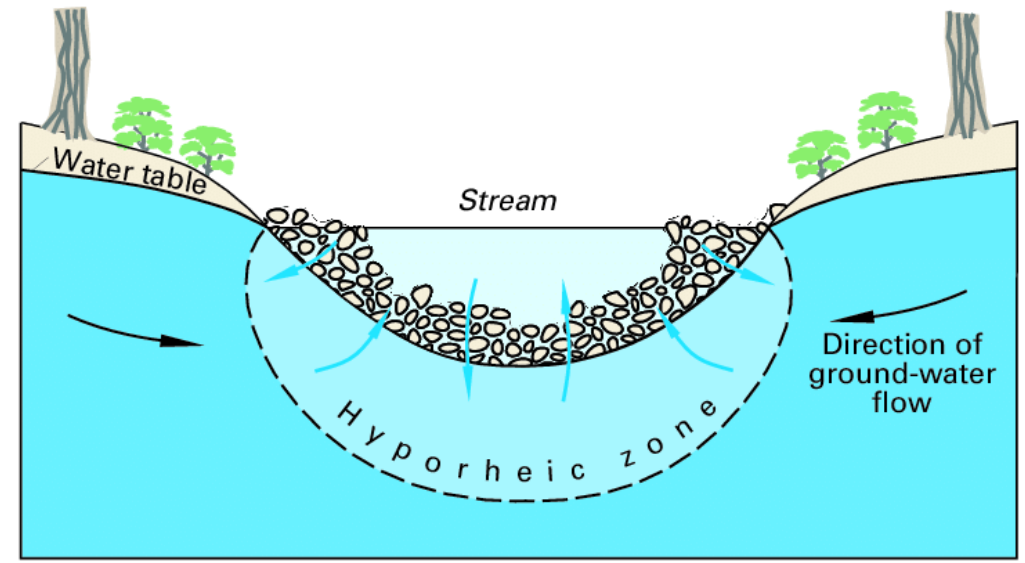
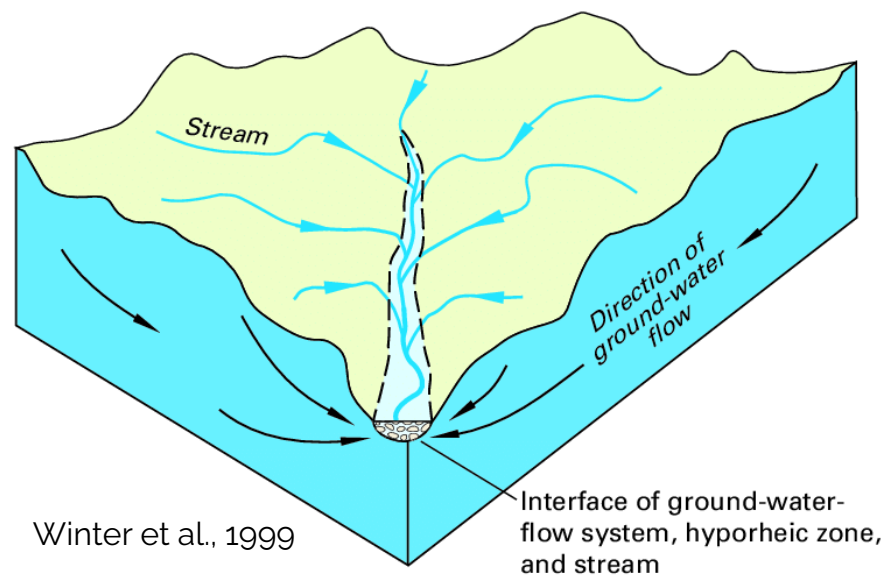
Alexander, L.C., 2015

Según Jackson et al. (2001) y Fleckenstein et al. (2010) tradicionalmente las aguas subterráneas y superficiales han sido tratadas como entes diferentes, tanto científica como legalmente, debido a la perspectiva espacial y temporal del flujo.

Entender la interacción GW-SW es importante para la sustentabilidad de los recursos hídricos ya que de ésta depende la recarga-descarga (Sophocleous, 2002), la regulación de la temperatura del río (Irvine et al., 2024), la biogeoquímica (Brunke y Gonser, 1997), ecología (Hayashi y Rosenberry, 2002), concentraciones de contaminantes (Kalbus et al., 2006), suministro de agua (Changming et al., 2001), salinidad (Jolly et al., 2008) y calidad del agua (Harvey y Bencala, 1993) de un sistema hidrológico.



Según Tóth (1999) y Sophocleous (2002) las interacciones GW-SW son complejas y para entenderlas hay que combinar diferentes factores como climáticos, edáficos, geológicos, topográficos, bióticos y de usos del suelo. Las características hidráulicas del intercambio de flujo GW-SW en un sistema dependen en gran medida del nivel freático, nivel de agua en el río, espesor y granulometría del lecho del río (Domenico, 1972; Winter et al., 1999; Kalbus et al., 2006; Jolly et al., 2008).





Los ríos interactúan con el agua subterránea de tres maneras:

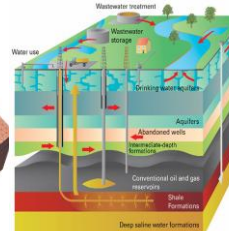
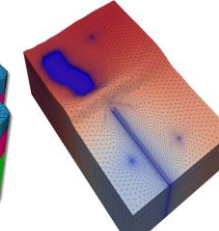
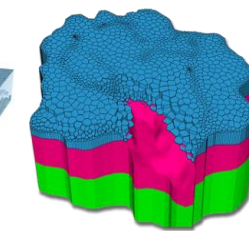
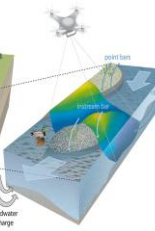
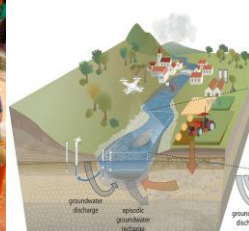
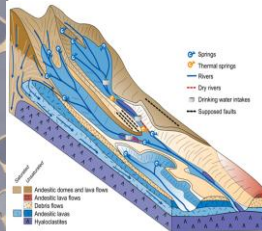
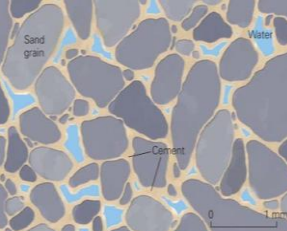
1. Los ríos ganan agua a través de la entrada de agua subterránea por el lecho del río (río ganador).
2. Pierden agua hacia el agua subterránea por flujo de salida a través del lecho del río (río perdedor)
3. Hacen ambas cosas, ganando en algunos tramos y perdiendo en otros.



Descarga Acuífero-Río
Río Ganador



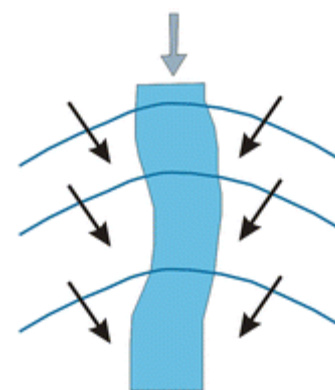
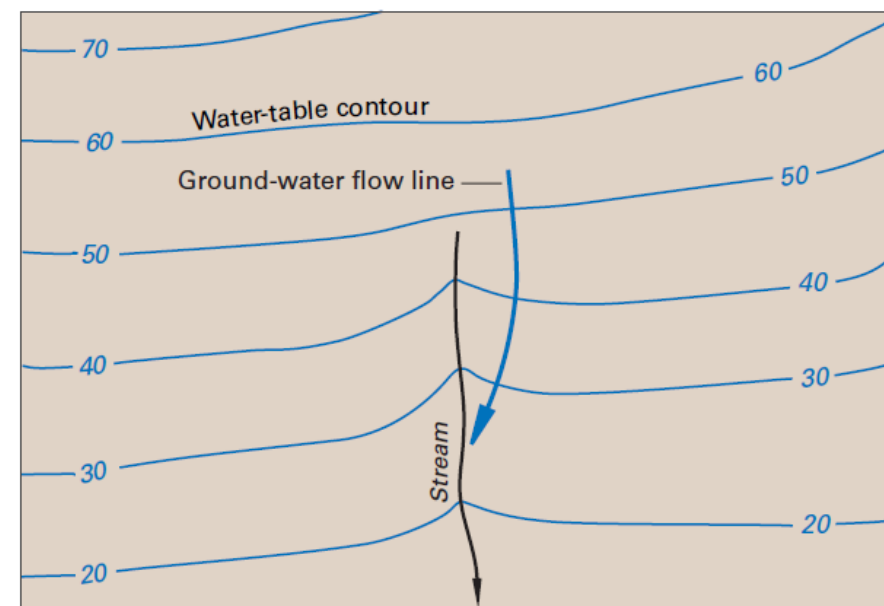
Recarga Río-Acuífero
Río Perdedor



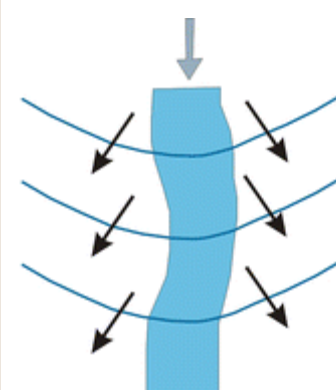
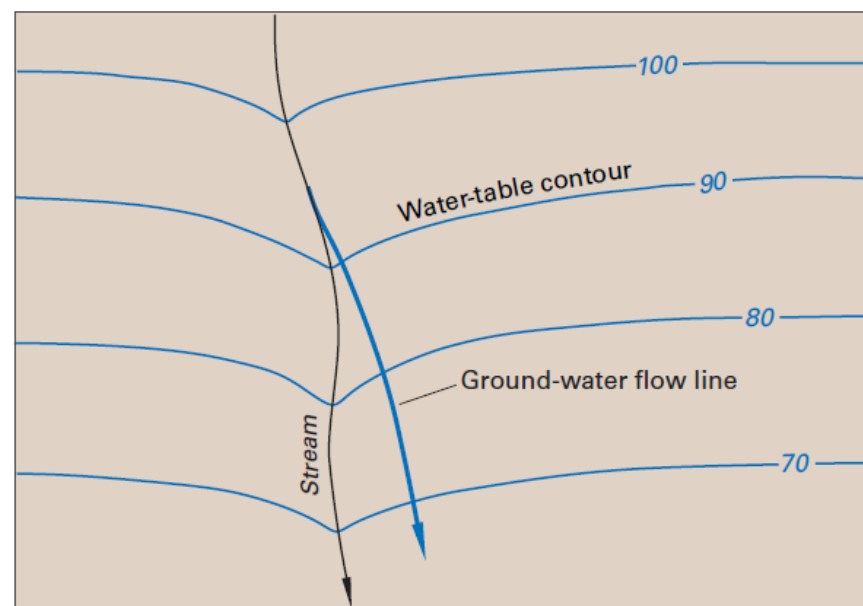
Descarga
Acuífero-Río
Río Ganador



Recarga Río-Acuífero
Río Perdedor

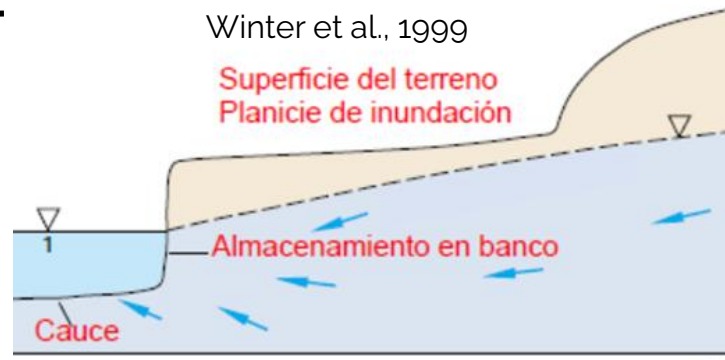


Winter et al., 1999

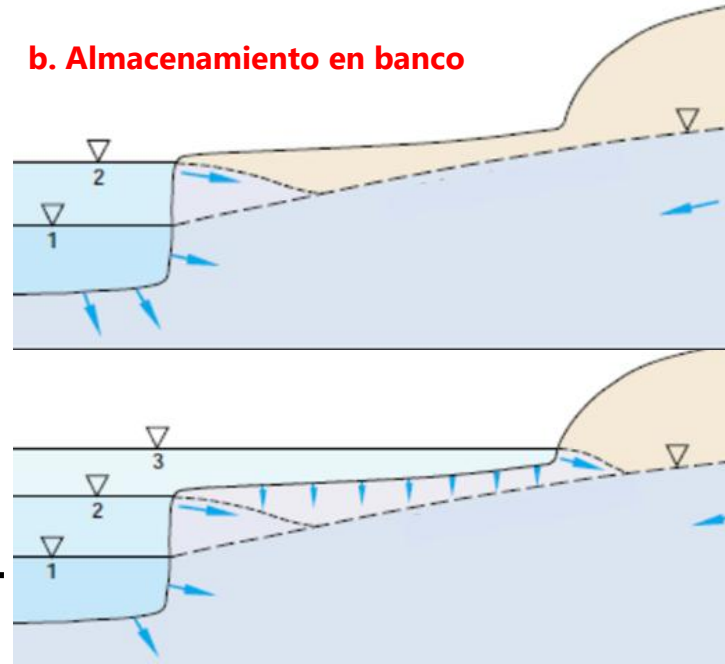




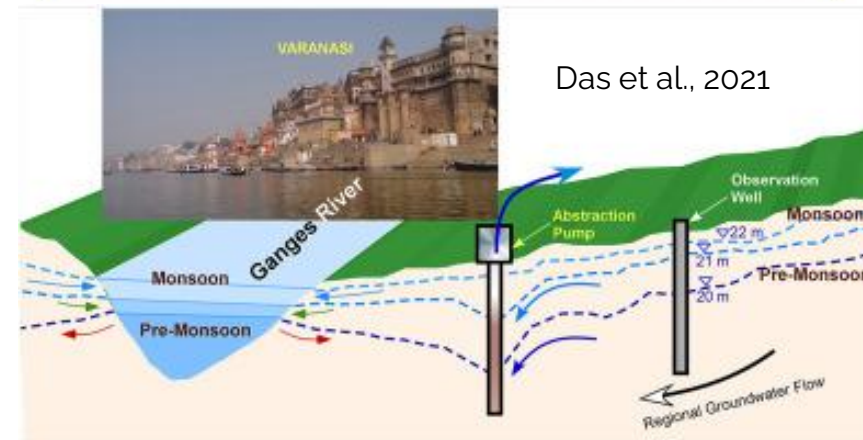
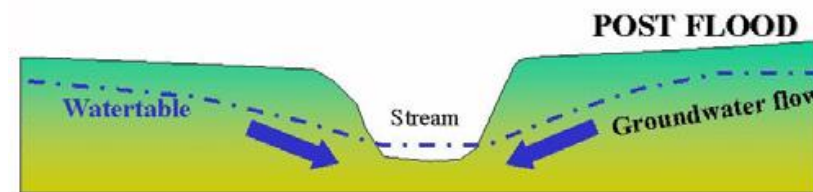
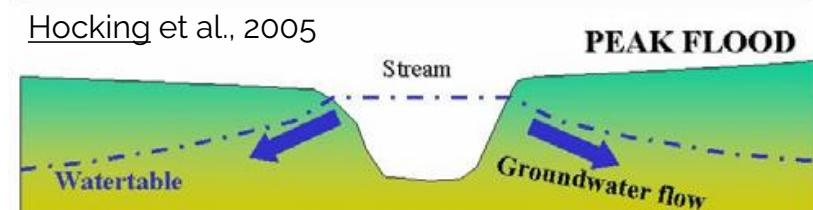
Unidad 1. Balance hídrico e introducción a las características del flujo entre las aguas subterráneas y superficiales (interacciones acuífero-río-planicie inundable)



b. Almacenamiento en banco

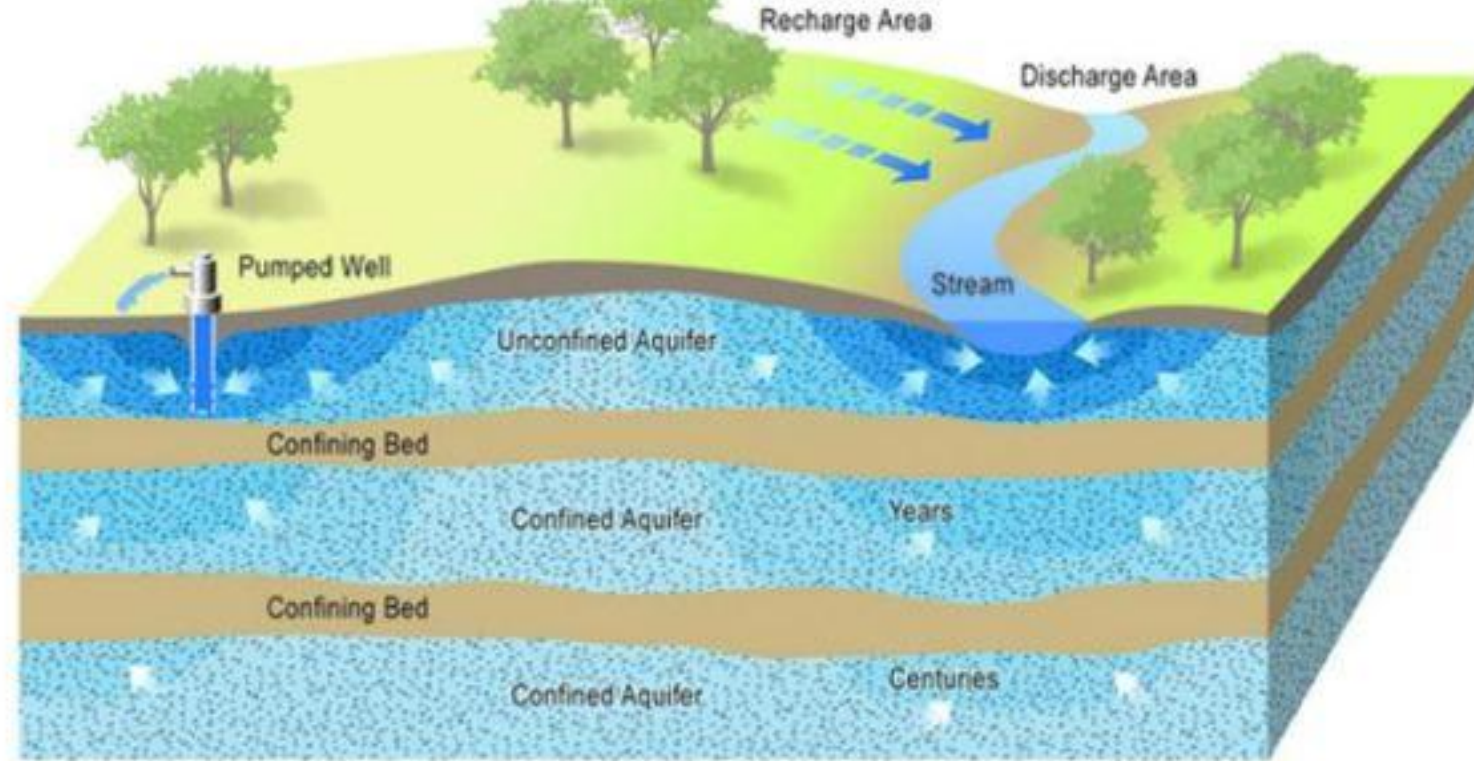


c. Almacenamiento sobre la planicie de inundación

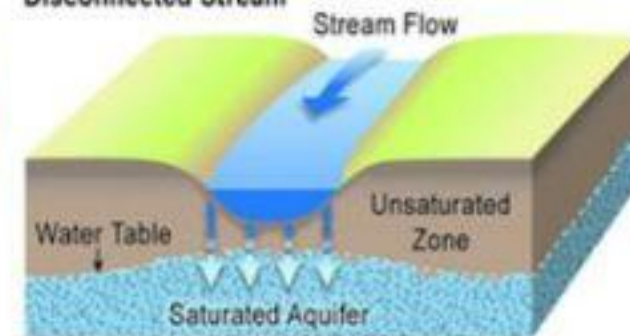


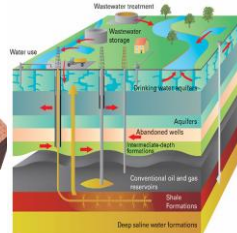
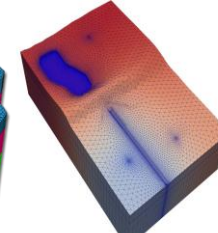
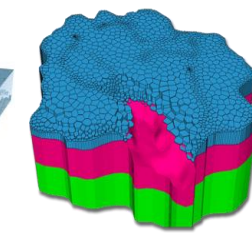
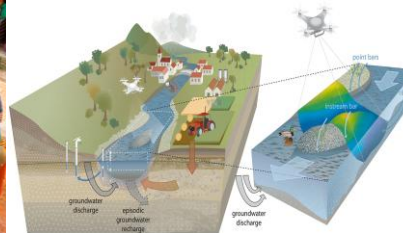
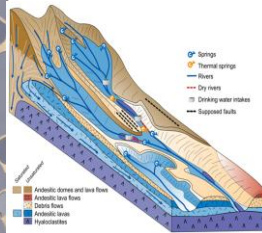
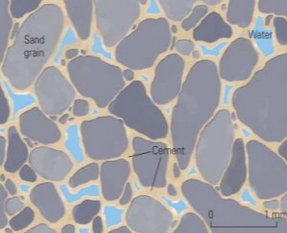
Das et al., 2021

Aumentos en el nivel del río pueden causar que la dirección del flujo a través del lecho del río se invierta y conduzca a almacenamiento en el banco (b) y a la recarga de agua subterránea en toda el área inundada o planicie de inundación (c).

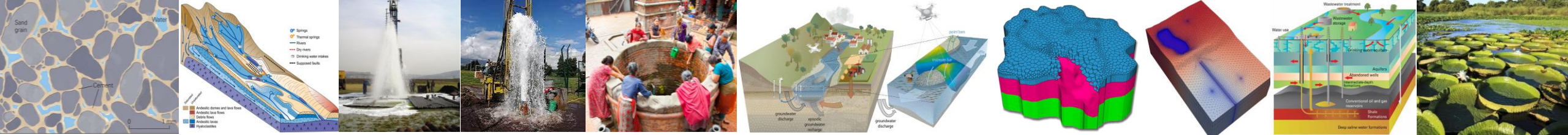


Gaining Stream

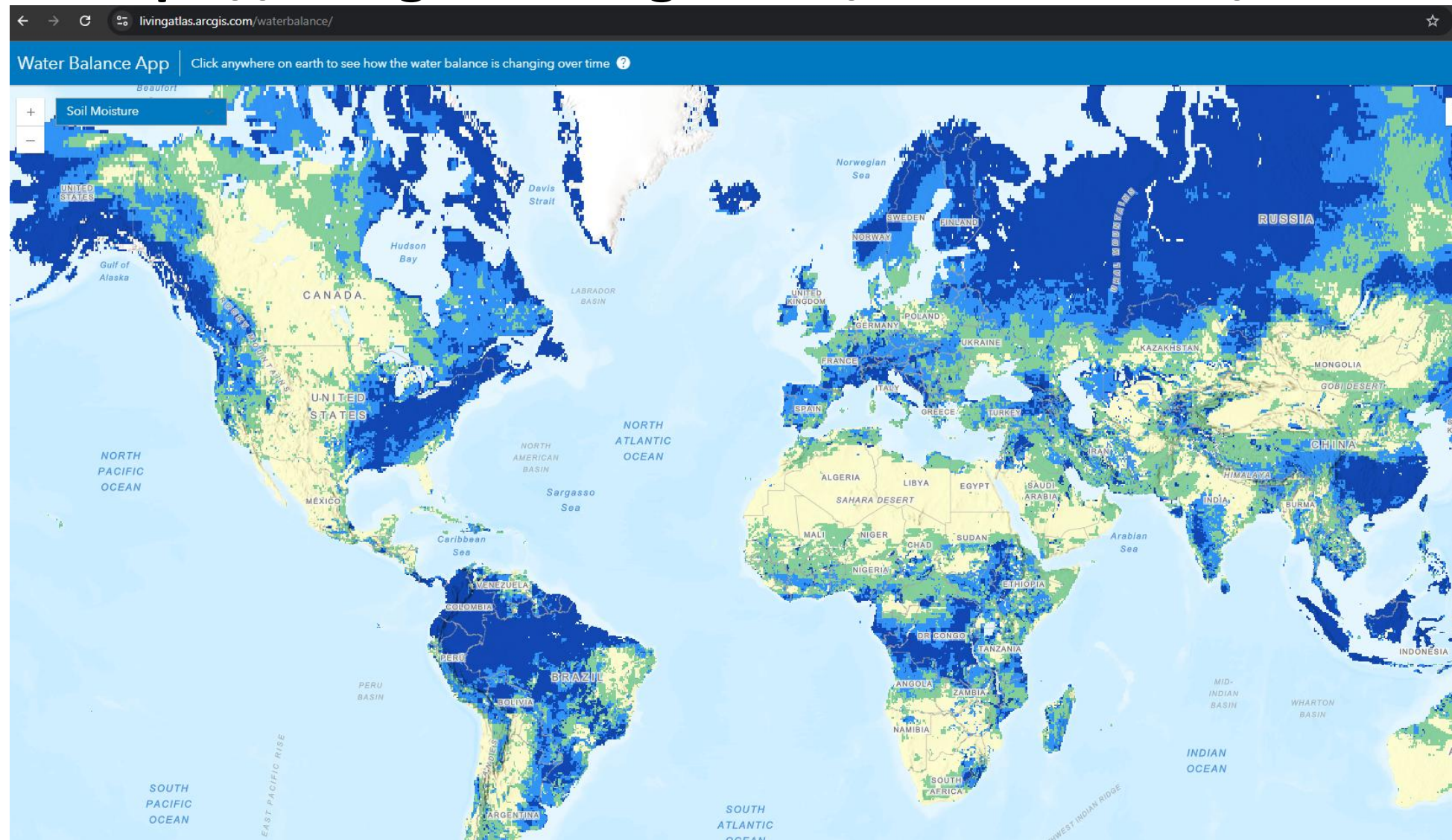




Caso práctico Real



<https://livingatlas.arcgis.com/waterbalance/>





Caso de estudio – Ejercicio práctico 1

Una pequeña cuenca hidrográfica (3.5 km^2) está formada por un arroyo con pendiente en ambos lados, y ambas laderas tienen la misma superficie.

El 45 % de la precipitación efectiva se escurre por la superficie de la ladera, y el resto se infiltra en el agua subterránea y fluye hacia el arroyo. Debido a una capa de arcilla en el subsuelo de la ladera derecha, solo el 30 % de la escorrentía infiltrada en dicha ladera fluye hacia el arroyo.

La precipitación anual es de 1.000 mm/año y la evaporación anual es de 900 mm/año .

- ¿Cuál es la escorrentía anual de la superficie de ambas laderas?
- ¿Cuál es la escorrentía anual total de la pendiente de la ladera derecha?
- ¿Cuál es la escorrentía anual total del arroyo?



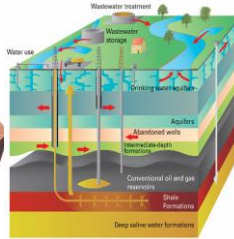
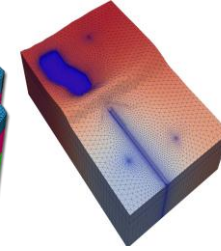
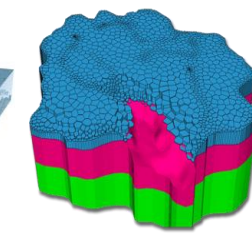
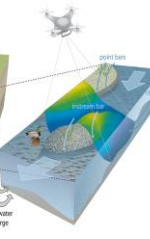
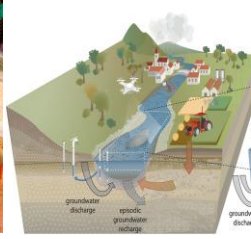
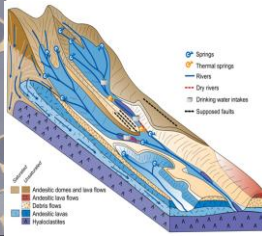
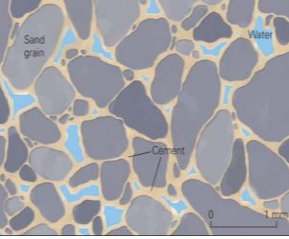
Datos

- Área de la cuenca: $A = 3,5 \text{ km}^2 = 3,5 \times 10^6 \text{ m}^2$.
- Dos laderas con la **misma** superficie $\Rightarrow A_{\text{ladera}} = 1,75 \text{ km}^2$ cada una.
- Precipitación anual: $P = 1000 \text{ mm/año}$.
- Evaporación anual: $E = 900 \text{ mm/año}$.
- Precipitación efectiva: $P_e = P - E = 100 \text{ mm/año}$.
- De P_e :
 - 45% es **escorrentía superficial** en ladera $\Rightarrow 45 \text{ mm}$.
 - 55% **infiltra** $\Rightarrow 55 \text{ mm}$.
 - Ladera derecha: **solo** 30% de lo infiltrado llega al arroyo.

¿Cuál es la escorrentía anual de la superficie de ambas laderas?

$$Q_{\text{surface}} = 0.45 \cdot A \cdot (P - E)$$

$$= 0.45 \cdot 3.5 \cdot 10^6 \cdot (1000 - 900) \cdot 10^{-3} = 157,500 \text{ m}^3/\text{year}$$



b) ¿Cuál es la escorrentía anual total de la pendiente de la derecha?

$$Q_{right} = 0.5 \cdot (Q_{surface} + 0.55 \cdot 0.3 \cdot A \cdot (P - E))$$

$$= 0.5 \cdot (157,500 + 0.55 \cdot 0.3 \cdot 3.5 \cdot 10^6 \cdot (1000 - 900) \cdot 10^{-3}) = 107,625 \text{ m}^3/\text{year}$$

¿Cuál es el caudal anual total del arroyo?

$$Q_{total} = Q_{right} + 0.5 \cdot A \cdot (P - E)$$

$$107,625 + 0.5 \cdot 3.5 \cdot 10^6 \cdot (1000 - 900) \cdot 10^{-3} = 282,625 \text{ m}^3/\text{year}$$



Caso de estudio – Ejercicio práctico 2

Se han recopilado los siguientes datos para una cuenca hidrográfica:

Precipitación media anual = 1000 mm/año = 1 m/año

Evaporación media anual = 500 mm/año = 0,5 m/año

Área = 80 km

a) ¿Cuál es el caudal anual total de la cuenca hidrográfica?

- En el período invernal (180 días), se produce la mitad de la precipitación anual y el 30 % de la evaporación. Durante este período, el almacenamiento aumentó en 10 cm.

b) ¿Cuál es el caudal medio en el periodo invernal?



a) ¿Cuál es el caudal anual total de la cuenca hidrográfica?

$$Q = A \cdot (P - E) = (80 \cdot 10^6) \cdot (1 - 0.5) = 40 \cdot 10^6 \text{ m}^3/\text{year}$$

b) ¿Cuál es el caudal medio en el periodo invernal?

$$P = 500 \text{ mm}/180 \text{ days} = 0.5 \text{ m}/180 \text{ days}$$

$$E = 0.3 \cdot 500 = 150 \text{ mm}/180 \text{ days} = 0.15 \text{ m}/180 \text{ days}$$

$$\frac{dS}{dt} = 10 \text{ cm}/180 \text{ days} = 0.10 \text{ m}/180 \text{ days}$$

Del balance hídrico se desprende que:

$$Q = A \cdot (P - E - dS/dt)$$

$$Q = (80 \cdot 10^6) \cdot (0.5 - 0.15 - 0.10) = 20 \cdot 10^6 \text{ m}^3/180 \text{ days}$$

$$Q = \frac{20 \cdot 10^6}{(180 \cdot 24 \cdot 3600)} = 1.29 \text{ m}^3/\text{s}$$

